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# PARAMETRIC ESTIMATION OF ROBOT MANIPULATORS ORIENTED TO MEDICAL ASSISTANCE

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*This work is dedicated to the loving memory of my dear friend **Beto** who showed me the true value of friendship and was also a huge fan of the UNAM soccer team.*

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# ABSTRACT

Robot manipulators are used in a wide variety of applications in many different industries. They efficiently handle all kinds of payload and improve productivity. To perform tasks with high precision, model based control can be employed, which required not only the robot parameters but also the payload ones. Most of the time, payload parameters are unknown and need to be estimated. Thus, mainly motivated by medical applications, the present work focuses on the estimation of payload parameters for robot manipulators. For that purpose, it is proposed to use a modification to the so-called Dynamic Regressor Extension and Mixing (DREM) technique, which arises for online estimation seeking to relax the hard to satisfy Persistence of Excitation (PE) condition. The key feature of the Modified DREM is that by design it increases a minimal amount of excitation to a desired level to achieve parameter convergence in fixed time. First, the Modified DREM technique is applied in simulation to a dynamic model derived from the Geomagic Touch haptic device, which was modified to include a payload and some assumptions were made to the model to obtain a reduced number of proper parameters (overparameterized). The robot parameters are also assumed to be unknown, so that the estimation of the payload parameters is done altogether with the robot parameters. Then, the parameter estimation scheme is verified experimentally with the actual Geomagic Touch. In addition, experiments are also carried out with the Franka Research 3 (FR3) robot to estimate only the payload parameters. Unlike the Geomagic Touch dynamic model, which was derived from the Euler-Lagrange formulation, the FR3 model used is derived from the Newton-Euler formulation. Therefore, the payload is associated to ten parameters, which are estimated.

# RESUMEN

Los robots manipuladores se utilizan en una amplia variedad de aplicaciones en diversas industrias. Manejan eficientemente todo tipo de cargas útiles y mejoran la productividad. Para realizar tareas con alta precisión, se puede emplear un control basado en modelo, que requiere no solo los parámetros del robot, sino también los de la carga útil. La mayoría de las veces, los parámetros de la carga útil son desconocidos y deben estimarse. Por lo tanto, principalmente motivado por aplicaciones médicas, el presente trabajo se centra en la estimación de los parámetros de la carga útil para robots manipuladores. Para ello, se propone utilizar una modificación de la técnica denominada Dynamic Regressor Extension and Mixing (DREM), que surge para la estimación en línea y que busca relajar la difícil condición de Excitación Persistente (PE, por sus siglas en inglés). La característica principal del DREM modificado es que, por diseño, aumenta una cantidad mínima de excitación hasta un nivel deseado para lograr la convergencia de parámetros en un tiempo fijo. La técnica DREM modificada se aplica primero en simulación a un modelo dinámico derivado del dispositivo háptico Geomagic Touch, el cual se modifica para incluir una carga útil. Se realizaron ciertas suposiciones en el modelo para obtener un número reducido de parámetros apropiados (sobreparametrizados). Se asume tambien que los parámetros del robot son desconocidos, de modo que la estimación de los parámetros de la carga útil se realiza en conjunto con los parámetros del robot. Posteriormente, el esquema de estimación de parámetros se verifica experimentalmente con el Geomagic Touch real. Además, se realizan experimentos con el robot Franka Research 3 (FR3) para estimar únicamente los parámetros de la carga útil. A diferencia del modelo dinámico de Geomagic Touch, derivado de la formulación de Euler-Lagrange, el modelo del robot FR3 se deriva de la formulación de Newton-Euler. Por lo tanto, la carga útil se asocia a diez parámetros, que se estiman.

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*Chapter 1*

# INTRODUCTION

## 1.1 Motivation

Robot manipulators have been widely used in medical applications in recent years. This is because of the many advantages they provide. They can make surgeons' movements more accurately, which is necessary in high-precision surgeries. They can help improving the comfort of surgeons as well as patients, and lastly they can reduce surgery time.

Some medical applications reported in Sandoval & Laribi (2022) and Koszulinski et al. (2022) are Doppler sonography, laparoscopic surgery, craniotomy, and spine surgery, just to mention some. In Doppler sonography, the manipulator is used with an ultrasound probe holder attached to the end effector. In laparoscopy surgery, the robot holds the laparoscope. Finally, in craniotomy and spine surgery, the robot manipulator holds a drilling tool. It is important to mention that a teleoperation scheme is employed in some of these and many other medical applications.

To perform the control task, is highly desirable to know the model and the exact parameters of the manipulator if we want the robot to perform as precise as required. Considering that the inertial parameters of the tool are generally unknown and that the tool attached to the end effector may change in weight and form depending on the application, this uncertainty could lead to imprecise movements of the manipulator and undesired errors. Therefore, it is critical to take the tool parameters into account and incorporate this information into the system controller.

One question we might ask ourselves is the following: How do we identify or estimate the inertial parameters of the payload? Currently, there exist robot manipulators equipped with joint torque sensors or a six-axis force sensor that are capable of identifying payload parameters using some integrated approach (Sandoval & Laribi 2022). However, not all manipulators are capable of doing this. They might not have this kind of sensor or this functionality. Consequently, the payload inertial parameters needs to be estimated in some way.

## 1.2 State of the art

As mentioned previously, knowledge of the inertial parameters of the manipulator payload is a priority in medical applications. Therefore, in this section, some work on parameter estimation and payload identification that has been done previously is reviewed. Inertial parameters can be estimated offline or online. First, some offline methods are briefly reviewed.

In the literature, methods can be found to identify the inertial parameters of the manipulator payload that are generally based on least squares. There exists a simple method in which the joint torques are measured for the same trajectory twice. First, without the tool attached, and second, with the tool attached. Then, the joint torque difference between each case can be used to compute the estimated parameters (Sandoval & Laribi 2022).

In Khalil et al. (2007), four different methods are described to identify offline the inertial parameters of the robot manipulator payloads. These methods are based on the weighted least-squares solution. The positions, velocities, and torques of the joints are assumed to be available. In three of the four methods presented, the dynamic parameters of the robot without the payload are assumed to be known, while the fourth method uses a global identification procedure to estimate both the link parameters and the payload inertial parameters.

In Gaz & De Luca (2017), the authors present a robust offline method to estimate the ten parameters (mass, center of mass, and inertia tensor) of an unknown manipulator payload without using additional external sensors. The robot parameters are assumed to have been previously identified. The concept of dynamic coefficients is used, which are linear combinations of the parameters that appear in the equations. The method calculates the difference between the dynamic coefficients of the loaded and unloaded robot, since they change linearly. A Jacobian of the payload is used to relate the ten physical parameters of the payload with the difference in the dynamic coefficients. Then the estimation is done through Least-Squares techniques. Another result of the same authors is reported in Gaz et al. (2019). However, for this other case the payload parameters are not identified, only the robot parameters for the Franka Emika Panda robot. Again, the dynamic coefficients are used to obtain a linear model, and they are first identified by Ordinary Least-Squares. Then, from the dynamic coefficients and through an optimization method, the robot's link parameters are identified, ensuring that they are physically consistent.

Alternatively to off-line estimation, adaptive control is widely used to deal with parameter uncertainty. In Ioannou & Fidan (2006), the following definition of adaptive control is given: *"Adaptive control is the combination of a parameter estimator, which generates parameter estimates online, with a control law in order to control classes of plants whose parameters are completely unknown and/or could change with time in an unpredictable manner"*. Therefore, we focus our work on the parameter estimator part of adaptive control.

Adaptive control has its beginning in the early 1950s. There were efforts to design high-performance aircraft autopilots. However, by then, research had to deal with a non-existing theory and lack of stability proofs, facing many challenges (Ioannou & Fidan 2006).

In Craig et al. (1986) the first adaptive control of the computed torque method for robot manipulators is presented. A formal proof of global stability is given, incorporating the complete nonlinear dynamic of the manipulator rather than the linear approximation or the slow variation. The adaptive law is based on parameter properties that are liner in the model, and it is driven by a filtered error. A parameter reset mechanism is introduced to maintain the estimations within known limits, which guaranties that the estimated inertia matrix is always invertible. Lastly, it formally establishes that the convergence of the parameter to their true values depends on the trajectory that satisfies the PE condition.

One of the most transcendent results on the adaptive control of robot manipulators is presented in Slotine & Li (1987). The unknown manipulator and payload parameters are estimated online, and a structural property of the dynamical model is exploited, which makes the algorithm simple. Furthermore, no joint accelerations are required. Nevertheless, a condition of persistency of excitation (PE) needs to be fulfilled in order to achieve convergence of the parameter estimation. Before the result of Slotine and Li, a similar one was presented in Horowitz & Tomizuka (1986), although with a slightly different approach. The authors proposed Model Reference Adaptive Control (MRAC) and made use of fundamental properties of the manipulator equations. The adaptive scheme compensates for a nonlinear term in the dynamics equations and decouples the dynamic interaction.

In Ortega & Spong (1988), a tutorial is given for the most relevant adaptive controllers for robot manipulators at that moment, which had been shown to be globally convergent. In the early adaptive control literature, adaptive controllers were classified into three categories, direct, indirect, and composite adaptive controllers. In Li & Slotine (1989), a brief description of the three controllers is given. Direct adaptive controllers use tracking errors of the joint motion to drive parameter adaptation. The main concern in this type of parameter estimator is to minimize the tracking errors. Indirect controllers use prediction errors on the filtered joint torques to produce the parameter estimates. The main concern in this type of parameter estimator is to obtain information about the actual parameters from the prediction errors. Composite adaptive controllers use both tracking errors in the joint motion and errors in the predicted filtered torques to drive the parameter adaptation. In this paper, an indirect adaptive controller is proposed and four parameter estimators are reviewed, gradient estimator, least-square estimator, gain-adjusted-forgetting estimator, and inherently-bounded-gain estimator. The global tracking convergence of the indirect adaptive controller is guaranteed.

In Slotine & Li (1989), a composite adaptive controller is obtained. A brief description of this type of controller was already made. The prediction error can be obtained by using a prediction of the filtered torque and the estimated parameters. The parameter estimator uses a time-varying gain matrix and it is obtained through three different techniques, i.e. the gradient method, the bounded-gain-forgetting method, and the cushioned-floor method. The proof of global asymptotic and exponential convergence of the tracking and parameter errors is shown for each of the three gain update techniques. Neither joint acceleration nor inversion of the inertia matrix is needed.

In Ham (1993), two non-adaptive controllers are derived using the Lyapunov stability theory. The inertial parameter of the robot's links and payload are assumed to be known, and global stability of the overall system is guaranteed. Secondly, assuming that the inertial parameters are partially or totally unknown, an adaptive controller is derived for the second control law obtained in the first part of the paper. Again, Lyapunov stability theory was used for this purpose. The design of the controllers takes advantage of a property of the dynamic model, i.e. it can be expressed linearly in terms of the inertial parameters of the links and payload.

In Tang & Arteaga (1994) an adaptive scheme is developed based on a passivity property. First, the passivity property of the control law and that of the adaptation algorithm is established. Next, a stability proof is carried out, and the exponential convergence of the tracking and parameter errors is shown under certain conditions. Moreover, there is no need for the regressor to be persistently exciting (PE), it is enough to satisfy a condition called sufficiency of excitation (SE), which is strictly weaker than PE. Lastly, this controller gives a smother performance of the tracking error and parameter convergence compared to other previous controllers.

In Swevers et al. (1997) the use of Fourier series was introduced to design trajectories to maximize the "excitation". This allowed the online parameter estimators to converge more quickly and with less sensitivity to sensor noise.

The poor transient response of most adaptive control algorithms is a well-known problem. Even if the PE condition is met, the convergence of the parameter error may take a long time to occur, thus reducing the transient performance of the tracking error. In Arteaga & Tang (2002), the authors introduce an adaptive control scheme based on robust control techniques to achieve a faster convergence of the parameter errors. The exponential convergence to zero of the parameter errors is guaranteed under the parameter-dependent persistent excitation condition, which is different from the PE condition (slightly relaxed). Unlike other adaptive control schemes which achieve improved transient performance by increasing control gains, this approach makes this happen by estimating the real parameters faster, which also improve control outputs.

Most adaptive control schemes for rigid robot manipulators assume that joint velocities are available. However, there may be some advantages to avoiding joint velocities measurements. In Arteaga (2003) a controller-observer is presented along with an adaptive scheme. The adaptive algorithm is based on Arteaga & Tang (2002). The observer used is linear and avoids the use of the inverse of the estimated inertia matrix. The whole scheme guarantees uniform ultimate boundedness of the tracking and observation errors and boundedness of the estimated parameters with arbitrary ultimate bound. Additionally, experiments are carried out where it is shown to have better results compared to obtaining joint velocities through numerical differentiation (assuming that the encoders are noise-free). A similar work is that of Arteaga–Pérez et al. (2020) where another adaptive control-observer scheme is proposed. It is shown to guarantee ultimate boundedness of the tracking and observation errors with arbitrary ultimate bounds, whereas parameter errors stay always bounded. Here, the improvement is that if a PE condition is satisfied, the tracking, observation, and parameter errors tend to zero as $t \to \infty$.

At some point, the convergence speed of the estimated parameters begins to have more relevance. For example, in Na et al. (2015) three parameter estimation algorithms are presented. The first of them achieves exponential convergence of the parameter estimation errors, while the other two are proved to yield finite-time convergence. This is accomplished by using a sliding mode technique. However, the persistence of excitation condition (PE) is still necessary. The introduced algorithms are then applied to robotics systems and joint accelerations are avoided. A similar work is presented in Yang et al. (2018) where the parameter estimator also produces finite-time convergence of the estimation errors. Nevertheless, the estimated parameters are only guaranteed to converge to a small neighborhood of the true values, and the PE condition is also needed.

In Aranovskiy et al. (2016), a new technique for designing parameter estimators is presented. The so-called Dynamic Regressor Extension and Mixing (DREM) technique establishes convergence without the usual requirement of persistency of excitation and improves performance. In Ortega et al. (2021) this technique is used to design a parameter estimator for Nonlinear Parameterized Regressions (NLPRE) that ensures convergence under a strictly weaker condition than PE. The estimator is then applied in an adaptive controller of Euler-Lagrange systems. In Romero et al. (2021), DREM is applied to the linear parameterization of Euler-Lagrange systems derived from the power balance equation, which is significantly simple, yet requires high excitation. Hence, this approach is used to successfully overcome this difficulty. In Ortega et al. (2022), the authors combine the DREM approach with the classical estimator based on Least-Squares (LS). The parameter estimator achieves global exponential convergence under the Interval Excitation condition, which is a weaker assumption than the PE condition. In addition, it is also applicable to NLPRE. Lastly, in Arteaga (2024) a modification is made to the DREM and it is applied to a robot manipulator system. This modification allows to have a way more relaxed condition than the Persistency of Excitation (PE) that, if it holds, it is guaranteed to achieve, theoretically, exact parameter estimation in fixed time.

### 1.3 Problem formulation

Robot manipulators can perform any kind of task and handle all types of payloads, from small to large. They can handle surgical tools, welding torches, car body parts, etc. Regardless of the manipulator payload, this can have an important impact on the stability of the operation trajectory, performance, and accuracy of the robot system. Focusing on the applications that motivate this research, in the medical field where precise robot movements are necessary, not knowing the inertial parameters of the surgical instrument could put the patient and the physician in danger or compromising the quality of medical practices. This is why knowledge of payload parameters is crucial in the control task of robot manipulators.

Parameter estimation often requires satisfying the PE condition, regardless of whether an online or offline estimation is used. It is well known that the PE condition brings many drawbacks in practice, for instance, actuator stress, increased wear, and pose risk, among others. Moreover, simple offline methods that use least squares, as one described briefly above, could be tedious. It is worth noting that although the PE condition can be relaxed, it is always present because it is related to the independence of the data set used for the estimation.

The main problem then is designing a parameter estimation scheme to be able to estimate the payload inertial parameters accurately, speeding up the procedure and making it easier. Considering that a robot manipulator could use different payloads as surgical tools in medical applications.

To tackle this challenge, the present work focuses on the study and application of the Modified Dynamic Regressor Extension and Mixing (MDREM) algorithm, which is a technique that arises initially for online parameter estimation. It takes advantage of the little excitation that a short movement of a robot manipulator could produce to estimate the parameters in fixed time. Thus, we avoid some of the disadvantages of the PE condition mentioned earlier. It is important to take a step back and mention some advantages, in general, that the DREM technique offers over the classical parameter estimators. First, it relaxes the convergence conditions; that is, the parameter convergence is ensured under a weaker condition than the PE. Second, it improves transient performance and ensures the element-wise transient monotonicity of parameter errors. Third, it decouples the estimated parameters, which allows one to design independent estimators for each parameter, making gain tuning easier. In addition, the technique can be applied to nonlinear parameter estimation scenarios.

## 1.4 Hypothesis

The approximation of the robot payload for the Geomagic Touch dynamic model is done by a homogeneous rectangular prism. In addition, the robot links are approximated by a hollow sphere and slender rods, respectively. It is expected that this allows us to estimate the robot's payload and link parameters with an acceptable margin of error for its control, significantly reducing the complexity of the mathematical model.

## 1.5 Contributions

The main contributions of this thesis are as follows:

- The MDREM algorithm based on adaptive control techniques is proposed to estimate the unknown inertial parameters of the payload. It is applied to two different robot manipulators. First, to a 3 DoF robot model and then verified in experimentation by the simplified 3 DoF Geomagic Touch haptic device, and second, to the 7 DoF Franka Research 3 (FR3) robot.
  - The linear parameterization model of the Geomagic Touch robot from Arteaga et al. (2022) is modified to include a payload. So, the estimation is done for all the robot parameters, including those of the payload. Simulations are carried out to validate the algorithm. Furthermore, experimental results are obtained.
  - Based on the work of a fellow student (Guajardo Benavides 2025), the experimental results for the estimation of the payload parameters of the FR3 robot are improved by adding a friction model to the dynamic model of the robot and with a better selection of the inertial parameters of the robot. The development of a ROS 2 (Robot Operator system 2) C++ node is done for the online implementation of the MDREM algorithm. It is worth noting that with this robot only payload estimation was involved, and nothing related to control law was done.

## 1.6 Thesis organization

The organization of this thesis is as follows. In Chapter 2, the preliminaries and mathematical tools are given. In Chapter 3, the dynamic model with the inclusion of the payload for a 3 DoF robot is derived, and the parameter estimation method (MDREM algorithm) is introduced. Chapter 4 presents the results of both the simulation for the 3 DoF robot and the validation experiment with the Geomagic Touch robot. In Chapter 5, the work done with the FR3 robot is developed and the experimental results are shown. Finally, Chapter 6 presents the conclusions and final comments.

*Chapter 2*

# PRELIMINARIES

## 2.1 Dynamic model

Knowledge of the dynamic model of robot manipulators can be useful for many purposes, for instance control-observer design, computer simulation, parameter estimation, robot design, and so on (Spong et al. 2020, Arteaga et al. 2022). As mentioned earlier, the task we are interested in is parameter estimation.

The dynamic model of robot manipulators presented in this section is derived from the Euler-Lagrange methodology, and the procedure is taken from Spong et al. (2020) and Arteaga et al. (2022). To see a brief description of how to obtain the dynamic model, see the Appendix A ; for further details, refer to the reference mentioned.

The model can be obtained by directly applying the Euler-Lagrange formulation for non-conservative systems; this is

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i}\right) - \frac{\partial \mathcal{L}}{\partial q_i} = \tau_i \quad i = 1, \ldots, n, \tag{2.1}$$

where $\mathcal{L}$ is the Lagrangian function and is equal to the kinetic energy ($\mathcal{K}$) – potential energy ($\mathcal{P}$); $\mathcal{K}$ is the total kinetic energy of the manipulator; $\mathcal{P}$ is the total potential energy of the manipulator; $q_i$ is the generalized coordinate of the manipulator; and $\tau_i$ is the generalized force/torque applied at the $i$-th joint to move the $i$-th link.

In the context of robot manipulators, the generalized coordinates refer to joint angles if the $i$-th joint is revolute, or displacements if it is prismatic.

In short, the kinetic energy can be expressed as

$$\mathcal{K} = \frac{1}{2}\dot{\boldsymbol{q}}^{\mathrm{T}}\boldsymbol{H}(\boldsymbol{q})\dot{\boldsymbol{q}}, \tag{2.2}$$

where

$$\boldsymbol{H}(\boldsymbol{q}) = \sum_{i=1}^{n}\left\{m_i \boldsymbol{J}_{\mathrm{v}ci}^{\mathrm{T}}(\boldsymbol{q})\boldsymbol{J}_{\mathrm{v}ci}(\boldsymbol{q}) + \boldsymbol{J}_{\omega \mathrm{c}i}^{\mathrm{T}}(\boldsymbol{q})\,{}^{0}\boldsymbol{R}_{\mathrm{c}i}(\boldsymbol{q})\boldsymbol{I}_i\,{}^{0}\boldsymbol{R}_{\mathrm{c}i}^{\mathrm{T}}(\boldsymbol{q})\boldsymbol{J}_{\omega ci}(\boldsymbol{q})\right\}, \tag{2.3}$$

and $\boldsymbol{H}(\boldsymbol{q})$ is the symmetric positive definite inertia matrix; $m_i$ is the mass of the $i$-th link; $\boldsymbol{J}_{\mathrm{v}ci}$ and $\boldsymbol{J}_{\omega ci}$ are the geometric Jacobian related to the linear and angular joint velocities, respectively; ${}^0\boldsymbol{R}_{\mathrm{c}i}$ is the rotation matrix with respect to the center of mass frame; and $\boldsymbol{I}_i \in \mathbb{R}^{3\times 3}$ is the inertia matrix.

Briefly, the potential energy can be expressed as

$$\mathcal{P} = \sum_{i=1}^{n} \mathcal{P}_i = -\sum_{i=1}^{n} m_i \overline{\boldsymbol{g}}^{\mathrm{T}}\,{}^0\overline{\boldsymbol{r}}_i, \tag{2.4}$$

where again $m_i$ is the mass of the $i$-th link; $\overline{\boldsymbol{g}}$ is a gravity vector expressed in the base coordinate frame; and ${}^0\overline{\boldsymbol{r}}_i$ is the position of the $i$-th center of mass.

Lastly, the dynamic model of a rigid robot manipulator is given by

$$\boldsymbol{H}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{D}\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{\tau} \tag{2.5}$$

where $\boldsymbol{D} \in \mathbb{R}^{n\times n}$ is a diagonal positive semidefinite matrix of viscous friction coefficients of the joint.

## 2.2 Some model properties

There exist some useful properties related to the model developed in the previous section that we are interested in. These are:

**Property 1** *By using the Christoffel symbols of the first kind to compute* $\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})$ *the matrix* $\dot{\boldsymbol{H}}(\boldsymbol{q}) - 2\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}}) \in \mathbb{R}^{n\times n}$ *is skew-symmetric.*

This property is based on the fact that the Christoffel symbols are used to compute $\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})$. In general, many matrices that may satisfy (A.7) could be found, yet do not fulfill this property. In contrast, it is known that the following equation always holds:

$$\dot{\boldsymbol{q}}^{\mathrm{T}}(\dot{\boldsymbol{H}}(\boldsymbol{q}) - 2\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}}))\dot{\boldsymbol{q}} = 0. \tag{2.6}$$

**Property 2** *The inertia matrix* $\boldsymbol{H}(\boldsymbol{q})$ *satisfies*

$$\lambda_{\min}(\boldsymbol{H}(\boldsymbol{q}))\|\boldsymbol{y}\|^2 \leq \boldsymbol{y}^{\mathrm{T}}\boldsymbol{H}(\boldsymbol{q})\boldsymbol{y} \leq \lambda_{\max}(\boldsymbol{H}(\boldsymbol{q}))\|\boldsymbol{y}\|^2 \quad \forall \boldsymbol{q} \in \mathbb{R}^n, \boldsymbol{y} \in \mathbb{R}^n, \tag{2.7}$$

with $0 < \lambda_{\min}(\boldsymbol{H}(\boldsymbol{q})) \leq \lambda_{\max}(\boldsymbol{H}(\boldsymbol{q})) < \infty$. $\lambda_{\min}(\cdot)$ and $\lambda_{\max}(\cdot)$ indicate the minimum and maximum eigenvalues, respectively, of a symmetric matrix (Slotine & Li 1991).

**Property 3** *The matrix $\boldsymbol{C}(\boldsymbol{q},\boldsymbol{y})$ satisfies $\|\boldsymbol{C}(\boldsymbol{q},\boldsymbol{y})\| \leq k_c\|\boldsymbol{y}\|$, with $k_c > 0$.*

**Property 4** *The vector of gravitational torques satisfies $\|\boldsymbol{g}(\boldsymbol{q})\| \leq k_g$, with $k_g > 0$.*

**Property 5** *With a proper choice of parameters, the model (2.5) can be rewritten as*

$$\boldsymbol{H}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{D}\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{\tau} = \boldsymbol{Y}(\boldsymbol{q},\dot{\boldsymbol{q}},\ddot{\boldsymbol{q}})\boldsymbol{\theta}, \tag{2.8}$$

*where $\boldsymbol{Y}(\boldsymbol{q},\dot{\boldsymbol{q}},\ddot{\boldsymbol{q}}) \in \mathbb{R}^{n\times p}$ is the so-called regressor; and $\boldsymbol{\theta} \in \mathbb{R}^p$ is a vector of constant parameters.*

For all the proofs of these properties, refer to Spong et al. (2020) or Slotine & Li (1991).

### 2.3 Adaptive control

One of the most transcendent results in adaptive control of robot manipulators was proposed in the well-known Slotine & Li (1987).

The control and adaptive laws that the authors first introduce produce a globally stable adaptive controller. However, even though they achieve zero joint velocity error in the steady-state, there is no guarantee that the steady-state position error is also zero. The authors solved this issue by restricting the error position and error velocity to lie on a sliding surface. Then, the tracking error tending to zero can be accomplished by replacing the desired trajectory $\boldsymbol{q}_{\mathrm{d}}(t)$ in the control and adaptive laws, by a reference trajectory given by

$$\mathbf{q}_{\mathrm{r}} = \mathbf{q}_{\mathrm{d}} - \boldsymbol{\Lambda}\int_0^t \boldsymbol{e}\,\mathrm{d}t, \tag{2.9}$$

where $\boldsymbol{\Lambda} \in \mathbb{R}^{n\times n}$ is a diagonal positive definite matrix; and in turn $\dot{\boldsymbol{q}}_{\mathrm{d}}$ and $\ddot{\boldsymbol{q}}_{\mathrm{d}}$ are changed by $\dot{\mathbf{q}}_{\mathrm{r}} = \dot{\mathbf{q}}_{\mathrm{d}} - \boldsymbol{\Lambda}\boldsymbol{e}$ and $\ddot{\mathbf{q}}_{\mathrm{r}} = \ddot{\mathbf{q}}_{\mathrm{d}} - \boldsymbol{\Lambda}\dot{\boldsymbol{e}}$, respectively. Now, the sliding surface is defined by

$$\boldsymbol{s} = \dot{\boldsymbol{q}} - \dot{\boldsymbol{q}}_{\mathrm{r}} = \dot{\boldsymbol{e}} + \boldsymbol{\Lambda}\boldsymbol{e}, \tag{2.10}$$

when $\boldsymbol{s}$=**0**. Thus, with the new control and adaptive law obtained from this, it can be shown that global convergence is maintained and that errors $\boldsymbol{e}$ and $\dot{\boldsymbol{e}}$ converge to the sliding surface $\boldsymbol{s} = \dot{\boldsymbol{e}} + \boldsymbol{\Lambda}\boldsymbol{e} = 0$.

It is important to highlight some features of the controller. The definition of $\boldsymbol{s}$ in (2.10) is equal to the stable linear filter given by

$$\dot{\boldsymbol{e}} = -\boldsymbol{\Lambda}\boldsymbol{e} + \boldsymbol{s}, \tag{2.11}$$

with $\boldsymbol{s}$ as the input, so that if $\boldsymbol{s}$ is bounded and tends to zero, so do $\boldsymbol{e}$ and $\dot{\boldsymbol{e}}$ as $t \to \infty$, hence $\boldsymbol{e}$, $\dot{\boldsymbol{e}}$, $\boldsymbol{q}$ and $\dot{\boldsymbol{q}}$ are bounded. Furthermore, the parameter estimation error $\tilde{\boldsymbol{\theta}} = \hat{\boldsymbol{\theta}} - \boldsymbol{\theta}$ and $\hat{\boldsymbol{\theta}}$ are also bounded. Finally, in order for the parameter estimation error to converge to zero, the regressor $\boldsymbol{Y}\left(\boldsymbol{q}, \dot{\boldsymbol{q}}, \dot{\boldsymbol{q}}_{\mathrm{d}}, \ddot{\boldsymbol{q}}_{\mathrm{d}}\right)$ must satisfy the condition of *persistency of excitation* (PE) defined by

$$\int_{t}^{t+T} \boldsymbol{Y}^{\mathrm{T}}(\vartheta)\boldsymbol{Y}(\vartheta)\, d\vartheta \geq \mu \boldsymbol{I}, \quad \forall t \geq t_0. \tag{2.12}$$

## 2.4 Dynamic Regressor Extension and Mixing (DREM) approach

The *Dynamic Regressor Extension and Mixing* (DREM) is a technique to design parameter estimators that exhibit some significant improvements compared to classical estimators. The main properties of this technique are as follows:

- It guaranties parameter convergence under strictly weaker conditions than the PE (Persistency of Excitation), which is less restrictive. Moreover, if the PE condition is satisfied, then the convergence is exponential.
- It improves the transient performance. It guaranties the element-wise monotonicity of the individual parameter errors, which means that these either decrease or remain, but not increase. Thus, we avoid oscillations and peaking phenomena.
- The DREM procedure decouples the estimated parameters, giving one the ability to estimate each parameter independently. As a result of this,it simplifies gain tuning.

- Furthermore, the DREM technique can be applied to nonlinear parameterizations without having to fulfill the PE condition.

The DREM was first proposed in Aranovskiy et al. (2017). This approach arose from efforts to relax the PE condition, since it is well-known to be hard to satisfy. The procedure described here was taken from Arteaga (2024) and is a general procedure for designing the adaptive law of an adaptive controller based on the DREM technique. The first step of the DREM has several ways to go. In this case, it is done by applying a first-order filter; hence, the filtered regressor $\boldsymbol{Y}_{\varphi 1}(\boldsymbol{q}, \dot{\boldsymbol{q}}) \in \mathbb{R}^{n\times p}$ is defined as

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{Y}_{\varphi 1} = -\lambda_{\varphi}\boldsymbol{Y}_{\varphi 1} + \lambda_{\varphi}\boldsymbol{Y}(\boldsymbol{q}, \dot{\boldsymbol{q}}, \ddot{\boldsymbol{q}}); \quad \boldsymbol{Y}_{\boldsymbol{\varphi}1}\,(t_0) = \boldsymbol{O}, \tag{2.13}$$

where $\lambda_{\varphi} > 0$ and the regressor $\boldsymbol{Y}(\boldsymbol{q}, \dot{\boldsymbol{q}}, \ddot{\boldsymbol{q}})$ comes from the dynamic model 2.5 and Property 5. By filtering the regressor, acceleration measurements can be avoided. Alternatively, another way to avoid joint acceleration was first introduced in Slotine & Li (1989) and later exploited in Romero et al. (2021) to design DREM based parameter estimator. The key difference lies in the fact that the linear parameterization model is derived from the power balance equation, and the resulting model does not depend on the joint acceleration. In other words, the acceleration avoidance comes from the linear parameterization based on the power balance equation, whereas in the present work it comes from taking advantage of the filtering step of the DREM. Then, continuing with the DREM algorithm of this work, this is possible by performing an integration by parts before the implementation of (2.13). Similarly, the filtered input $\boldsymbol{\tau}_{\varphi} \in \mathbb{R}^n$ is defined as

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\tau}_{\varphi} = -\lambda_{\varphi}\boldsymbol{\tau}_{\varphi} + \lambda_{\varphi}\boldsymbol{\tau}; \quad \boldsymbol{\tau}_{\varphi}(t_0) = \boldsymbol{0}. \tag{2.14}$$

Now, applying the LTI filters from (2.13), (2.14) of the form $\frac{v}{p+v}$ where $p = \frac{\mathrm{d}}{\mathrm{d}t}$, to $\boldsymbol{\tau} = \boldsymbol{Y}\boldsymbol{\theta}$ we get

$$\frac{v}{p+v}(\boldsymbol{\tau}) = \frac{v}{p+v}(\boldsymbol{Y}\boldsymbol{\theta}) \tag{2.15}$$

Then, using the Swapping Lemma C.3 on the right-hand side of (2.15) we have

$$\frac{v}{p+v}\,(\boldsymbol{Y}\boldsymbol{\theta}) = \left(\frac{v}{p+v}\boldsymbol{Y}\right)\boldsymbol{\theta} - \frac{1}{p+v}\left[\left(\frac{v}{p+v}\boldsymbol{Y}\right)\dot{\boldsymbol{\theta}}\right], \tag{2.16}$$

since the vector of parameters is constant, it implies $\dot{\boldsymbol{\theta}} = 0$, which in turn means that the second term of (2.16) is zero. Thus, from this result and from (2.13) and (2.14) we obtain

$$\boldsymbol{\tau}_{\varphi} = \boldsymbol{Y}_{\varphi 1}(\boldsymbol{q}, \dot{\boldsymbol{q}})\boldsymbol{\theta}. \tag{2.17}$$

The next step is to compute an expanded regressor $\boldsymbol{Y}_{\mathrm{f}} \in \mathbb{R}^{p\times p}$, which is a square matrix, and an expanded input $\boldsymbol{\tau}_{\mathrm{f}} \in \mathbb{R}^{p}$. For this task, the filtered regressor $\boldsymbol{Y}_{\varphi 1}(\boldsymbol{q}, \dot{\boldsymbol{q}})$ and the filtered input $\boldsymbol{\tau}_{\varphi}$ can be used. Without loss of generality, it is assumed that $n < p$. There exists more than one way to compute the expanded regressor and expanded input, here, it is presented just one way of doing it:

1. The $n$ rows of $\boldsymbol{Y}_{\varphi 1}$ are defined as the first $n$ rows of $\boldsymbol{Y}_{\mathrm{f}}$. Hence, $p - n$ rows still need to be defined.

2. Denote by $\tilde{n}$ the (integer) total number of matrices $\boldsymbol{Y}_{\varphi 1}, \dots, \boldsymbol{Y}_{\varphi \tilde{n}}$ required to compute $\boldsymbol{Y}_{\mathrm{f}}$, so that $\tilde{n} - 1$ additional matrices are necessary to form $\boldsymbol{Y}_{\mathrm{f}}$ as

$$\boldsymbol{Y}_{\mathrm{f}} = \begin{bmatrix} \boldsymbol{Y}_{\varphi 1} \\ \vdots \\ \boldsymbol{Y}_{\varphi \tilde{n}} \end{bmatrix} \in \mathbb{R}^{p\times p}. \tag{2.18}$$

3. If $\tilde{n} = \frac{p}{n}$ is an integer number, then the $\tilde{n} - 1$ matrices $\boldsymbol{Y}_{\varphi 2}, \dots, \boldsymbol{Y}_{\varphi \tilde{n}} \in \mathbb{R}^{n\times p}$ must be created by using $(\tilde{n} - 1)$ linear $\mathcal{L}_\infty$ stable operators $G_j : \mathcal{L}_\infty \to \mathcal{L}_\infty$ for $j = 2, \dots, \tilde{n}$ to get (2.18).

4. If the quotient $\tilde{n} = \frac{p}{n}$ is not an integer, the DIV operator can be used since it delivers the integer part of a division (e.g., $\frac{6}{4} = 1.5$ but 6 DIV 4 = 1). This means that $\tilde{n} = (p \text{ DIV } n) + 1$. In that case, only the last matrix $\boldsymbol{Y}_{\varphi \tilde{n}}$ in (2.18) is of different dimension given by $(p \text{ MOD } n) \times p$, where the MOD operator gives the remainder of an integer division (e.g., 6 MOD 4 = 2).

5. Keeping in mind the previous subdivisions and dimensions, for $j = 2, \dots, \tilde{n}$ the linear $\mathcal{L}_\infty$ stable operator $G_j : \mathcal{L}_\infty \to \mathcal{L}_\infty$ are chosen as linear time invariant filters of the form

$$\dot{\boldsymbol{Y}}_{\varphi j} = -b_j \boldsymbol{Y}_{\varphi j} + a_j \boldsymbol{Y}_{\varphi 1} \quad \boldsymbol{Y}_{\varphi j}(t_0) = \boldsymbol{O}, \tag{2.19}$$

where $a_j \neq 0,\ b_i \neq b_j > 0$ for $i, j = 2, \dots, \tilde{n}$.

It is important to note that if the gain $b_j$ is selected large enough, the extended regressor $\boldsymbol{Y}_{\mathrm{f}}$ could lose the linear independency, so that $\phi$ never becomes different than zero regardless of the excitation level. Therefore, it is preferable for $b_j$ to be chosen small enough to act as pure time delays at low frequencies.

After finishing constructing the expanded regressor $\boldsymbol{Y}_{\mathrm{f}}$, in a similar fashion, the exact $(\tilde{n}-1)$ linear $\mathcal{L}_\infty$ stable operators $G_j : \mathcal{L}_\infty \to \mathcal{L}_\infty$ that were applied to calculate $\boldsymbol{Y}_{\mathrm{f}}$, now are applied to the filtered input $\boldsymbol{\tau}_\varphi$ in the same order to compute the expanded input vector $\boldsymbol{\tau}_{\mathrm{f}}$. Hence, having $(\tilde{n}-1)$ additional filtered input vectors $\boldsymbol{\tau}_{\mathrm{f}2}, \ldots, \boldsymbol{\tau}_{\mathrm{f}\tilde{n}}$ that make up the expanded input vector given by

$$\boldsymbol{\tau}_{\mathrm{f}} = \begin{bmatrix} \boldsymbol{\tau}_{\mathrm{f}1}^{\mathrm{T}} & \cdots & \boldsymbol{\tau}_{\mathrm{f}\tilde{n}}^{\mathrm{T}} \end{bmatrix}^{\mathrm{T}}, \tag{2.20}$$

where $\boldsymbol{\tau}_{\mathrm{f}1} = \boldsymbol{\tau}_\varphi$. Keeping in mind that if $\tilde{n} = \frac{p}{n}$ is not an integer, not all rows of $\boldsymbol{Y}_{\varphi\tilde{n}}$ and elements of $\boldsymbol{\tau}_{\mathrm{f}\tilde{n}}$ will be used to construct $\boldsymbol{Y}_{\mathrm{f}}$ and $\boldsymbol{\tau}_{\mathrm{f}}$, respectively. Although for this particular case differential operators were chosen to compute the extended regressor $\boldsymbol{Y}_{\mathrm{f}}$, other option could have been using pure delay filters. However, there is no reason to think that such operators will work better since they cannot influence the original response. Following (2.17), the next relationship still holds

$$\boldsymbol{\tau}_{\mathrm{f}} = \boldsymbol{Y}_{\mathrm{f}}\boldsymbol{\theta}. \tag{2.21}$$

Now, consider the following

$$\mathrm{adj}\left(\boldsymbol{Y}_{\mathrm{f}}\right)\boldsymbol{Y}_{\mathrm{f}} = \boldsymbol{Y}_{\mathrm{f}}\,\mathrm{adj}\left(\boldsymbol{Y}_{\mathrm{f}}\right) = \phi\boldsymbol{I}, \tag{2.22}$$

where

$$\phi = \det\left(\boldsymbol{Y}_{\mathrm{f}}\right), \tag{2.23}$$

and adj$(\cdot)$ means the adjoint matrix of $(\cdot)$. By multiplying both sides of (2.21) by adj$\left(\boldsymbol{Y}_{\mathrm{f}}\right)$ the adjoint matrix of the expanded regressor $\boldsymbol{Y}_{\mathrm{f}}$, the following is obtained

$$\boldsymbol{\tau}_{\mathrm{e}} = \mathrm{adj}\left(\boldsymbol{Y}_{\mathrm{f}}\right)\boldsymbol{\tau}_{\mathrm{f}} = \phi\boldsymbol{\theta}, \tag{2.24}$$

that is, (2.21) has been decoupled, now having $p$ equations of the form

$$\tau_{ei} = \phi \theta_i, \tag{2.25}$$

for $i = 1, \ldots, p$, where $\theta_i$ and $\tau_{ei}$ are the $i$-th element of $\boldsymbol{\theta} \in \mathbb{R}^p$ and $\boldsymbol{\tau}_\mathrm{e} \in \mathbb{R}^p$, respectively, and, moreover, $\boldsymbol{\tau}_\mathrm{e}$ is known. Consequently, having $p$ equations as (2.25) means that $p$ independent estimators can be designed. A simple and effective choice, known as the gradient algorithm, was the one proposed in Aranovskiy et al. (2017), this is

$$\dot{\hat{\theta}}_i = -\gamma_i \phi(\phi\, \hat{\theta}_i - \tau_{ei}), \tag{2.26}$$

with $\gamma_i > 0$. Since $\boldsymbol{\theta}$ is a constant vector, the dynamics of the parameter estimation error is the same as (2.26), however, it can be reduced to

$$\dot{\tilde{\theta}}_i = -\gamma_i \phi^2 \tilde{\theta}_i, \tag{2.27}$$

where $\tilde{\theta}_i$ denotes the parameter estimation error for $i = 1, \ldots, p$, whose is given by

$$\tilde{\boldsymbol{\theta}} = \hat{\boldsymbol{\theta}} - \boldsymbol{\theta} \in \mathbb{R}^p. \tag{2.28}$$

The convergence condition given in Aranovskiy et al. (2017) is defined by

$$\phi \notin \mathcal{L}_2 \quad \Rightarrow \quad \lim_{t \to \infty} \tilde{\theta}_i = 0 \tag{2.29}$$

for $i = 1, ..., p$. There are a few things worth mentioning. First, research has shown that if the regressor $\boldsymbol{Y}(\boldsymbol{q}, \dot{\boldsymbol{q}}, \ddot{\boldsymbol{q}})$ is PE then so is $\phi$ and, hence, for exact parameter estimation, it is always a better choice to use an approach based on the DREM algorithm. Secondly, the convergence condition (2.29) is strongly related to the performance of the determinant $\phi$ which, in sequence, depends on the selection of the $(\tilde{n} - 1)$ linear $\mathcal{L}_\infty$ stable operators $G_j : \mathcal{L}_\infty \to \mathcal{L}_\infty$ for $j = 2, ..., \tilde{n}$. In other words, for the stable operators chosen here (2.19), a good election of $a_j$ and $b_j$ is crucial to avoid a poor performance and $\phi$ turning zero. In addition, the performance of $\phi$ is also related to the condition of PE in some way, since its formal definition establishes that the regressor is of full rank over a period. In the present approach, if the original regressor stops being full rank, so does the extended regressor, and $\phi$ turns into zero. Finally, if a DREM based algorithm is used in combination with a control law, in which case the main goal might be trajectory tracking, and the determinant $\phi$ simply becomes zero, the adaptation process can just stop and therefore not achieve $\boldsymbol{e}, \dot{\boldsymbol{e}} \rightarrow \mathbf{0}$.

*Chapter 3*

# PAYLOAD PARAMETER ESTIMATION FOR THE GEOMAGIC TOUCH

## 3.1 Proposed adaptive algorithm: *Modified Dynamic Regressor Extension and Mixing* (MDREM)

To pursue the objective of estimating the parameters of the robot payload, a modification of the DREM algorithm, described in Section 2.4, is proposed. This proposal is taken from Arteaga (2024) and is called *Modified Dynamic Regressor Extension and Mixing* (MDREM).

The central feature of the MDREM approach is that it takes a predefined minimal amount of excitation and increases it, by design, to a level to achieve exact parameter estimation in fixed time. So, the first thing that needs to be addressed is how to define a minimal amount of excitation.

The main goal of the MDREM approach is to design an adaptive algorithm capable of performing exact parameter estimation when

$$\phi^2 \geq \eta_{\mathrm{m}} \quad \text{for} \quad T_{\mathrm{m}}\ [\mathrm{s}], \tag{3.1}$$

which represents a convergence condition, and where $\eta_{\mathrm{m}}$ and $T_{\mathrm{m}}$ are positive constants that can be assumed to be arbitrarily small. Then, $\eta_{\mathrm{m}}$ can be viewed as the predefined minimal amount of excitation. The condition (3.1) also indicates that $\phi \neq 0$ for $T_{\mathrm{m}}$. More formally, this could be expressed as that for a $T_{\mathrm{m}}$, $\phi^2 \notin \mathcal{L}_2$. Now, define

$$\zeta = \begin{cases} 1 & \text{if} \quad \phi^2 < \eta_{\mathrm{m}} \\ \left(\frac{\phi_{\mathrm{d}}}{|\phi|}\right)^{\frac{1}{p}} & \text{if} \quad \phi^2 \geq \eta_{\mathrm{m}} \end{cases} \tag{3.2}$$

with $\phi_{\mathrm{d}}^2 > \eta_{\mathrm{m}}$. Now, the *Modified Dynamic Regressor Extension and Mixing* (MDREM) process can be introduced by defining

$$\boldsymbol{Y}_{\mathrm{m}} = \zeta \boldsymbol{Y}_{\mathrm{f}}, \tag{3.3}$$

and

$$\boldsymbol{\tau}_{\mathrm{m}} = \zeta \boldsymbol{\tau}_{\mathrm{f}}. \tag{3.4}$$

A similar relationship as in (2.21) can be obtained from (3.3) and (3.4), that is,

$$\boldsymbol{\tau}_{\mathrm{m}} = \boldsymbol{Y}_{\mathrm{m}} \boldsymbol{\theta} \tag{3.5}$$

By multiplying both sides by the adjoint matrix of $\boldsymbol{Y}_{\mathrm{m}}$, a new vector $\boldsymbol{\tau}_{\epsilon}$ is obtained and defined by

$$\boldsymbol{\tau}_{\epsilon} = \mathrm{adj}\left(\boldsymbol{Y}_{\mathrm{m}}\right) \boldsymbol{\tau}_{\mathrm{m}} = \phi_{\mathrm{m}} \boldsymbol{\theta}, \tag{3.6}$$

with

$$\phi_{\mathrm{m}} = \det\left(\boldsymbol{Y}_{\mathrm{m}}\right). \tag{3.7}$$

This means (3.5) has been decoupled, now having $p$ equations of the form

$$\tau_{\epsilon i} = \phi_{\mathrm{m}} \theta_i, \tag{3.8}$$

for $i = 1, \ldots, p$, where $\theta_i$ and $\tau_{\epsilon i}$ are the $i$th element of $\boldsymbol{\theta} \in \mathbb{R}^p$ and $\boldsymbol{\tau}_{\epsilon} \in \mathbb{R}^p$, respectively, and, moreover, $\boldsymbol{\tau}_{\epsilon}$ is known. Having decoupled these equations means that we can choose any adaptive law to estimate the parameters. One more particularity of the MDREM algorithm is that it uses an adaptive law that allows to estimate the parameters in fixed time. Thus, taking advantage of the short time that the excitation could last. Initially, the fixed-time adaptive law presented in Arteaga (2024) is described by

$$\dot{\hat{\theta}}_i = -\gamma_i \, \mathrm{sign}\left(e_{\theta i}\right) \left|e_{\theta i}\right|^{\frac{\lambda_{\theta i} e_{\theta i}^2}{1+\alpha_{\theta i} e_{\theta i}^2}}, \tag{3.9}$$

where $e_{\theta i}$ is defined as in (3.12), $\gamma_i$, $\lambda_{\theta i}$ and $\alpha_{\theta i}$ are adaptive gains that must be tuned. Nevertheless, the adaptive law used here is slightly different. The difference lies in the exponent; here $\frac{\lambda_{\theta i} e_{\theta i}^2}{1+\alpha_{\theta i} e_{\theta i}^2}$ will be replaced by $\lambda_{\theta i} \tanh(e_{\theta i}^2)$. This idea was taken from Arteaga et al. (2025). Then, the adaptive law is given by

$$\dot{\hat{\theta}}_i = -\gamma_i \,\mathrm{sign}\,(e_{\theta i})\, |e_{\theta i}|^{\lambda_{\theta i}\tanh(e_{\theta i}^2)} \equiv -\gamma_i \,\mathrm{sign}\,(e_{\theta i})\, f_{\theta i}(e_{\theta i}), \tag{3.10}$$

with

$$f_{\theta i}(e_{\theta i}) = |e_{\theta i}|^{\lambda_{\theta i}\tanh(e_{\theta i}^2)}\,, \tag{3.11}$$

and

$$e_{\theta i} = \phi_{\mathrm{m}} \left(\phi_{\mathrm{m}}\hat{\theta}_i - \tau_{\epsilon i}\right) = \phi_{\mathrm{m}}^2 \tilde{\theta}_i, \tag{3.12}$$

where $\tilde{\theta}_i$ is the $i$-th element of $\tilde{\boldsymbol{\theta}} \in \mathbb{R}^p$ and (2.28) has been used; $\gamma_i$ denotes the $i$-th element of a diagonal positive definite matrix $\boldsymbol{\Gamma} \in \mathbb{R}^{p\times p}$; lastly, $\lambda_{\theta i}$ is a positive constant that satisfies

$$\lambda_{\theta i}\tanh(1) = \beta_{\theta i} > 1, \quad \lambda_{\theta i} > \frac{\alpha_{\theta i}}{\tanh(1)}, \quad \alpha_{\theta i} > 1, \tag{3.13}$$

while the sign function for $x \in \mathbb{R}$ is defined as

$$\mathrm{sign}(x) = \begin{cases} 1 & \text{if} \quad x > 0 \\ 0 & \text{if} \quad x = 0 \\ -1 & \text{if} \quad x < 0 \end{cases} \tag{3.14}$$

**Theorem 1.** *Consider the adaptive law* (3.10) *with* $\lambda_{\theta i} > 0$ *satisfying* (3.13)*. Then, for i = 1, ... ,p the following holds*

1. $\tilde{\theta}_i \in \mathcal{L}_\infty$

2. *If condition* (3.1) *holds, that is, if* $\phi^2 \geq \eta_{\mathrm{m}}$ *for* $T_{\mathrm{m}}$ *seconds from some time* $t_i \geq t_0$*, where both* $\eta_{\mathrm{m}}$ *and* $T_{\mathrm{m}}$ *can be assumed arbitrarily small, then for i = 1, ... ,p it holds* $\tilde{\theta}_i(t) \equiv 0$ *for all* $t \geq t_i + T_{\mathrm{m}}$ *as long as it is set*

$$\left(\frac{\mu_{\theta i} + (\beta_{\theta i} - 1)}{T_{\mathrm{m}}\gamma_i\mu_{\theta i}\,(\beta_{\theta i} - 1)}\right)^{\frac{1}{2}} \leq \phi_{\mathrm{d}}, \tag{3.15}$$

with $\mu_{\theta i}$ as defined in (3.29) and $\beta_{\theta i}$ as in (3.13).

*Proof.* The proof is almost identical to that in Arteaga (2024) with just slight differences. First, it will be shown that (3.11) is continuous at $e_{\theta i} = \phi_{\mathrm{m}}^2 \tilde{\theta}_i = 0$. Then, consider that it is rewritten as

$$f_{\theta i}(x) = |x|^{\lambda_{\theta i} \tanh(x)} = e^{(\lambda_{\theta i} \tanh(x) \ln(|x|))}, \tag{3.16}$$

for $x \in \mathbb{R}$. Note that

$$\begin{aligned} \lim_{x \to 0} x \ln(|x|) &= \lim_{x \to 0} \frac{\ln\left((x^2)^{\frac{1}{2}}\right)}{\frac{1}{x}} = \lim_{x \to 0} \frac{\frac{1}{|x|}\frac{1}{2}(x^2)^{-\frac{1}{2}} 2x}{-\frac{1}{x^2}} \\ &= -\lim_{x \to 0} \frac{\frac{1}{|x|}\frac{1}{|x|} x}{\frac{1}{x^2}} = -\lim_{x \to 0} \frac{\frac{1}{x}}{\frac{1}{x^2}} = -\lim_{x \to 0} \frac{1}{\frac{1}{x}} = 0, \end{aligned} \tag{3.17}$$

where the L'Hôpital rule and the identity $|x| = (x^2)^{\frac{1}{2}}$ have been used. Based on (3.17), it can be shown that

$$\lim_{x \to 0} f_{\theta i}(x) = \lim_{x \to 0} e^{\left(\lambda_{\theta i} \tanh(x^2) \ln(|x|)\right)} = e^{(0)} = 1, \tag{3.18}$$

where basic limit properties were used. Therefore, $f_{\theta i}(e_{\theta i}) = f_{\theta i}(\phi_{\mathrm{m}}^2 \tilde{\theta}_i) = \ldots$ $= |\phi_{\mathrm{m}}^2 \tilde{\theta}_i|^{\lambda_{\theta i} \tanh(\phi_{\mathrm{m}}^2 \tilde{\theta}_i)}$ is continuous in $e_{\theta i} = \phi_{\mathrm{m}}^2 \tilde{\theta}_i = 0$ with $f_{\theta i}(0) = 1$. In other words, the right-hand side of (3.10) is locally bounded. Now, the next steps of the proof are divided into two: First, the boundedness of $\tilde{\theta} i$ is shown, and secondly, the fixed-time convergence is proven.

1. To show that $\tilde{\theta}_i$ is bounded, consider the positive definite function

   $$V_{\theta i} = \phi_i^4 \tilde{\theta}_i^2, \tag{3.19}$$

   and recall that since the vector of parameters $\boldsymbol{\theta}$ is constant, then $\dot{\hat{\theta}}_i = \dot{\tilde{\theta}}_i$. Consequently, the derivative of (3.19) along (3.10) is given by

   $$\dot{V}_{\theta i} = 2\phi_{\mathrm{d}}^4 \tilde{\theta}_i \dot{\theta}_i = -2\gamma_i \phi_{\mathrm{d}}^4 \tilde{\theta}_i \operatorname{sign}\left(\phi_{\mathrm{m}}^2 \tilde{\theta}_i\right) \left|\phi_{\mathrm{m}}^2 \tilde{\theta}_i\right|^{\lambda_{\theta i} \tanh(\phi_{\mathrm{m}}^2 \tilde{\theta}_i)} \le 0, \tag{3.20}$$

where (3.10) has been used. Note that based on (3.14) $\tilde{\theta}_i \operatorname{sign}\left(\phi_{\mathrm{m}}^2 \tilde{\theta}_i\right) > 0$ for $\phi_{\mathrm{m}}^2 > 0$, $\tilde{\theta}_i \operatorname{sign}\left(\phi_{\mathrm{m}}^2 \tilde{\theta}_i\right) \equiv 0$ for $\phi_{\mathrm{m}}^2 = 0$, so that the previous inequality holds. This shows that $\tilde{\theta}_i \in L_\infty$ and, therefore, $\hat{\theta}_i \in L_\infty$.

2. Next, to show fixed-time convergence to zero of the parameter error $\tilde{\theta}_i$, assume that condition (3.1) holds for $t \in [t_i, t_i + T_{\mathrm{m}}]$ for some $t_i \geq t_0$, and recall that

$$\phi_{\mathrm{m}} = \det\left(\boldsymbol{Y}_{\mathrm{m}}\right) = \det\left(\zeta \boldsymbol{Y}_{\mathrm{f}}\right) = \zeta^p \det\left(\boldsymbol{Y}_{\mathrm{f}}\right), \tag{3.21}$$

and in view of (3.2), this implies that

$$\phi_{\mathrm{m}} = \begin{cases} \phi & \text{if} \quad \phi^2 < \eta_{\mathrm{m}} \\ \phi_{\mathrm{d}} \operatorname{sign}(\phi) & \text{if} \quad \phi^2 \geq \eta_{\mathrm{m}} \end{cases} \tag{3.22}$$

then the derivative (3.20) becomes

$$\dot{V}_{\theta i} = -2\gamma_i \phi_{\mathrm{d}}^4 \tilde{\theta}_i \operatorname{sign}\left(\phi_{\mathrm{d}}^2 \tilde{\theta}_i\right) \left|\phi_{\mathrm{d}}^2 \tilde{\theta}_i\right|^{\lambda_{\theta i} \tanh(\phi_{\mathrm{m}}^2 \tilde{\theta}_i)} = -2\gamma_i \phi_{\mathrm{d}}^2 \left|\phi_{\mathrm{d}}^2 \tilde{\theta}_i\right|^{\lambda_{\theta i} \tanh(\phi_{\mathrm{m}}^2 \tilde{\theta}_i)+1} \leq 0. \tag{3.23}$$

Now, assume that $V_{\theta i}(t_{\mathrm{i}}) = \phi_{\mathrm{d}}^4 \tilde{\theta}_i^2 > 1$, which in turn means that $|\phi_{\mathrm{d}}^2 \tilde{\theta}_i| > 1$. In that instance, the following applies

$$\lambda_{\theta i} \tanh(\phi_{\mathrm{d}}^4 \tilde{\theta}_i^2) + 1 > \lambda_{\theta i} \tanh(1) + 1 = \beta_{\theta i} + 1 > 2 \tag{3.24}$$

because $\beta_{\theta i} > 1$ based on (3.13), so that (3.23) becomes

$$\dot{V}_{\theta i} \leq -2\gamma_i \phi_{\mathrm{d}}^2 \left|\phi_{\mathrm{d}}^2 \tilde{\theta}_i\right|^{\beta_{\theta i}+1} = -2\gamma_i \phi_{\mathrm{d}}^2 \left(\left(\phi_{\mathrm{d}}^2 \tilde{\theta}_i\right)^2\right)^{\frac{\beta_{\theta i}+1}{2}} = -2\gamma_i \phi_{\mathrm{d}}^2 V_{\theta i}^{\frac{\beta_{\theta i}+1}{2}} < 0. \tag{3.25}$$

Hence, according to Lemma C.1, $V_{\theta i}(t_{\mathrm{i}}) > 1$ becomes $V_{\theta i}(t_{\mathrm{i}}) \leq 1$ in a fixed-time no greater than

$$t \geq t_{\mathrm{i}} + \frac{1}{2\gamma_i \phi_{\mathrm{d}}^2 \left(\frac{\beta_{\theta i}+1}{2} - 1\right)} = t_{\mathrm{i}} + \underbrace{\frac{1}{\gamma_i \phi_{\mathrm{d}}^2 \left(\beta_{\theta i} - 1\right)}}_{T_{\theta 1}} = t_{\mathrm{i}} + T_{\theta 1}. \tag{3.26}$$

In a similar way, once $V_{\theta i}(t)$ has become $V_{\theta i}(t_{\mathrm{i}} + T_{\theta 1}) = \phi_{\mathrm{d}}^4 \tilde{\theta}_i^2 \leq 1$, which also implies $|\phi_{\mathrm{d}}^2 \tilde{\theta}_i| \leq 1$, (3.23) can be rewritten as

$$\dot{V}_{\theta i} = -2\gamma_i \phi_{\mathrm{d}}^2 \left|\phi_{\mathrm{d}}^2 \tilde{\theta}_i\right| \left|\phi_{\mathrm{d}}^2 \tilde{\theta}_i\right|^{\lambda_{\theta i} \tanh(\phi_{\mathrm{d}}^4 \tilde{\theta}_i^2)} \leq 0. \tag{3.27}$$

For the sake of simplicity $x = \phi_{\mathrm{d}}^2 \tilde{\theta}_i$ will be used. To find the minimum value of $|x|^{\lambda_{\theta i} \tanh(x^2)} = e^{\lambda_{\theta i} \tanh(x^2) \ln(|x|)}$, the derivative must be calculated and equaled to zero

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}x}\left(e^{\lambda_{\theta i} \tanh(x^2)\ln(|x|)}\right) &= e^{\lambda_{\theta i} \tanh(x^2)\ln(|x|)} \frac{\mathrm{d}}{\mathrm{d}x}\left(\lambda_{\theta i} \tanh(x^2)\ln(|x|)\right) \\ &= |x|^{\lambda_{\theta i} \tanh(x^2)} \lambda_{\theta i} \left(\frac{2x \ln(|x|)}{\cosh^2(x^2)} + \frac{\tanh(x^2)}{x}\right) \equiv 0 \end{aligned} \tag{3.28}$$

where $\frac{\mathrm{d}}{\mathrm{d}x}|x| = \frac{\mathrm{d}}{\mathrm{d}x}\left(x^2\right)^{\frac{1}{2}} = \frac{1}{|x|}x = \operatorname{sign}(x)$ has been used. It is known from (3.18) that $|x|^{\lambda_{\theta i} \tanh(x^2)}|_{x=0} = 1$, so that $|x|^{\lambda_{\theta i} \tanh(x^2)} > 0$. For this reason, the second term of the product must be zero; that happens at $x = x_{\mathrm{m}} \approx 0.5829$. This means that

$$\mu_{\theta i} = x_{\mathrm{m}}^{\lambda_{\theta i} \tanh(x_{\mathrm{m}}^2)} \leq |x|^{\lambda_{\theta i} \tanh(x^2)} \leq 1 \tag{3.29}$$

Therefore, following (3.27) $\dot{V}_{\theta i}$ satisfies

$$\dot{V}_{\theta i} \leq -2\gamma_i \phi_{\mathrm{d}}^2 \left|\phi_{\mathrm{d}}^2 \tilde{\theta}_i\right| \mu_{\theta i} = -2\gamma_i \mu_{\theta i} \phi_{\mathrm{d}}^2 \left(\left(\phi_{\mathrm{d}}^2 \tilde{\theta}_i\right)^2\right)^{\frac{1}{2}} = -2\gamma_i \mu_{\theta i} \phi_{\mathrm{d}}^2 V_{\theta i}^{\frac{1}{2}}. \tag{3.30}$$

Thus, in accordance with Lemma C.2, $V_{\theta i}(t_{\mathrm{i}} + T_{\theta 1}) = \phi_{\mathrm{d}}^4 \tilde{\theta}_i^2 \leq 1$ becomes $V_{\theta i}(t) \equiv 0$ in a fixed-time no greater than

$$T_{\theta 2} = \frac{1}{2\gamma_i \mu_{\theta i} \phi_{\mathrm{d}}^2 \left(1 - \frac{1}{2}\right)} = \frac{1}{\gamma_i \mu_{\theta i} \phi_{\mathrm{d}}^2} \tag{3.31}$$

The previous analysis shows that regardless the initial conditions of the estimated parameters $\hat{\theta}_i$, $V_{\theta i}(t_{\mathrm{i}})$ becomes zero, and as a result the parameter error $\tilde{\theta}_i$ converges to zero in a time no larger than

$$T_{\mathrm{m}} = T_{\theta 1} + T_{\theta 2} = \frac{1}{\gamma_i \phi_{\mathrm{d}}^2 (\beta_{\theta i} - 1)} + \frac{1}{\gamma_i \mu_{\theta i} \phi_{\mathrm{d}}^2} = \frac{\mu_{\theta i} + (\beta_{\theta i} - 1)}{\gamma_i \mu_{\theta i} \phi_{\mathrm{d}}^2 (\beta_{\theta i} - 1)}, \tag{3.32}$$

according to (3.26) and (3.31). Finally, parameter estimation errors can be guaranteed to converge to zero with predefined minimal amount of excitation since condition (3.15) can always be satisfied.

■

Since the structure of the adaptive law (3.10) is based on robust fixed-time stability theory, this suggests that it could have a certain degree of robustness against noise, although no formal analysis is given in the original source (Arteaga 2024) or considered here. For this matter, an interesting result is presented in Ortega et al. (2022), where a Least-Squares parameter estimator based on DREM (LS+DREAM) is proposed. Formal robustness analysis is derived, where additive perturbations are considered for a Nonlinearly parameterized regressor equation, yielding to bounded-input-bounded-state (BIBS) stability under the assumption of Interval Excitation (IE), which is weaker than the PE condition. Similarly, the approach proposed by the authors relaxes the convergence conditions compared to PE. The LS+DREM estimator achieves global exponential convergence of parameter errors with the assumption of Interval Excitation (IE) (Tao 2003, Definition 3.1), which states that input signals must only be sufficiently rich over a finite time window to ensure convergence.

Finally, a few last comments on the MDREM procedure. First, the parameters $\eta_\mathrm{m}$ and $T_\mathrm{m}$ are not design parameters that can be chosen a priori; rather, they are determined by the dynamics of the system and the desired trajectories. Consequently, they should not be confused with tunable adaptive gains. For the purpose of this analysis, it can be assumed that $\eta_\mathrm{m}$ and $T_\mathrm{m}$ are arbitrarily small. Furthermore, $T_\mathrm{m}$ is, in a certain way, dependent on the selection of $a_j$ and $b_j$. In scenarios where the input signal becomes static, what is desired is to take advantage of the transient response of the filters (2.19) within the extended regressor. Consequently, $T_\mathrm{m}$ should be limited by the settling time of the filters. Otherwise, condition (3.1) could not hold. Increasing $b_j$ (for $j = 2, \dots \tilde{n}$) shortens this transient response, and for values high enough, the filtered signals would tend to the original one too fast. This would result in a loss of linear independence in the extended regressor and, in turn $\phi = 0$. Secondly, the standard DREM approach has an interesting characteristic, it ensures that the individual parameter errors $\tilde{\theta}_i$ are monotonically non-increasing. This means that if the excitation stops, the parameter errors will not diverge and, in fact, (3.1) could be satisfied by intervals and not necessarily at once. This property is also maintained by the MDREM procedure. Lastly, it is necessary to point out that working with a very small threshold $\eta_{\mathrm{m}}$ allows $\phi$ to be considerably close to zero, indicating that it is almost singular. This could mean that the regressor is poorly numerically conditioned, which can affect the calculation of the adjoint, causing numerical errors while computing $\phi$.

## 3.2 *Geomagic touch* haptic device Robot

The present work is developed based on the dynamic model of the *Geomagic touch* haptic device (see Figure 3.1). Therefore, in this section, the dynamic model of such a robot is presented. For more details, see Arteaga et al. (2022) Section 14.2.


Figure 3.1: *Geomagic touch* haptic device

### 3.2.1 Simplified 3 DoF

Originally, the *Geomagic touch* is a sub-actuated robot manipulator with five degrees of freedom (DoF). Nevertheless, a totally actuated three-DoF configuration can be obtained by just blocking the last two non-actuated joints. In Figure 3.2 the Denavit-Hartenberg configuration can be seen and its corresponding parameters in Table 3.1.

### 3.2.2 Dynamic model of the 3-DoF configuration

The dynamic model presented here is derived from the Euler-Lagrange equations of motions as described in Section 2.1 so that it has the well-known form of equation

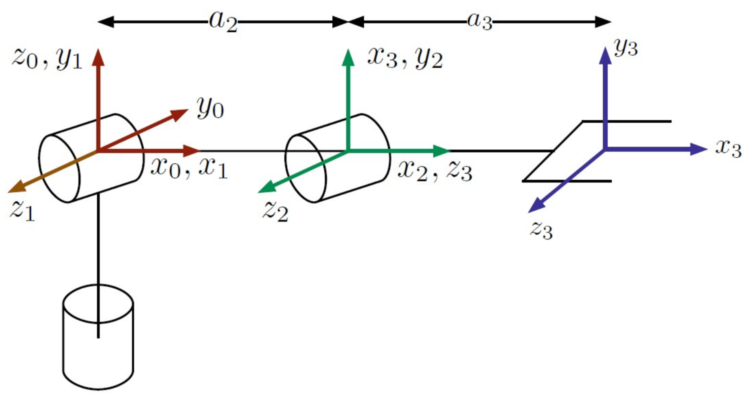


Figure 3.2: Denavit-Hartenberg configuration of the simplified 3 DoF *Geomagic touch*

Table 3.1: Denavit-Hartenberg parameters of the simplified 3 DoF Geomagic Touch

| **Joint** | $a_i$[m] | $d_i$[m] | $\alpha_i$[rad] | $q_i$[rad] |
|---|---|---|---|---|
| **1** | 0 | 0 | $\pi/2$ | ${q_1}^*$ |
| **2** | 0.145 | 0 | 0 | ${q_2}^*$ |
| **3** | 0.1738 | 0 | 0 | ${q_3}^*$ |

* variable quantity

(2.5). Thus, the elements of the inertia matrix $\boldsymbol{H}(\boldsymbol{q})$ are given by

$$
\begin{aligned}
h_{11} &= m_2 c_2^2 \ell_{c2}^2 + m_3 \left(a_2 c_2 + c_{23}\ell_{c3}\right)^2 + c_2^2 I_{yy2} + c_{23}^2 I_{yy3} + s_2^2 I_{xx2} + s_{23}^2 I_{xx3} + I_{yy1} \\
h_{12} &= 0 \\
h_{13} &= 0 \\
h_{22} &= m_2 \ell_{c2}^2 + m_3 \left(2 a_2 c_3 \ell_{c3} + a_2^2 + \ell_{c3}^2\right) + I_{zz2} + I_{zz3} \\
h_{23} &= m_3 \ell_{c3} \left(a_2 c_3 + \ell_{c3}\right) + I_{zz3} \\
h_{33} &= m_3 \ell_{c3}^2 + I_{zz3},
\end{aligned}
\tag{3.33}
$$

where $m_i$ is the mass of the $i$-th link for $i = 1, 2, 3$; $\ell_{ci}$ is the distance from the center of mass of the $i$-th link to the origin of the frame $O_{x_{i-1}, y_{i-1}, z_{i-1}}$ for $i = 2, 3$; $c_2 = cos(q_2)$, $c_3 = cos(q_3)$, $c_{23} = cos(q_2 + q_3)$, $s_2 = sin(q_2)$ and $s_{23} = sin(q_2 + q_3)$; $\boldsymbol{I}_{xxi}$, $\boldsymbol{I}_{yyi}$ and $\boldsymbol{I}_{zzi}$ are moments of inertia for $i = 1, 2, 3$; and $a_2$ is a Denavit-Hartenberg parameter given in Table 3.1.

The elements of the matrix $\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})$ are given by

$$
\begin{aligned}
c_{\mathrm{e}11} =& c_2\dot{q}_2 s_2 I_{\mathrm{xx}2} + c_{23}\dot{q}_2 s_{23} I_{\mathrm{xx}3} + c_{23}\dot{q}_3 s_{23} I_{\mathrm{xx}3} - c_2\dot{q}_2 s_2 I_{\mathrm{yy}2} - c_{23}\dot{q}_2 s_{23} I_{\mathrm{yy}3} \\
& - c_{23}\dot{q}_3 s_{23} I_{\mathrm{yy}3} - m_2 c_2 \dot{q}_2 s_2 \ell_{\mathrm{c}2}^2 + m_3\left(-a_2 c_{23}\dot{q}_2 s_2 \ell_{\mathrm{c}3} - a_2 c_2 \dot{q}_2 s_{23}\ell_{\mathrm{c}3}\right. \\
& \left. -a_2 c_2 \dot{q}_3 s_{23}\ell_{\mathrm{c}3} - a_2^2 c_2 \dot{q}_2 s_2 - c_{23}\dot{q}_2 s_{23}\ell_{\mathrm{c}3}^2 - c_{23}\dot{q}_3 s_{23}\ell_{\mathrm{c}3}^2\right) \\
c_{\mathrm{e}12} =& c_2\dot{q}_1 s_2 I_{\mathrm{xx}2} + c_{23}\dot{q}_1 s_{23} I_{\mathrm{xx}3} - c_2\dot{q}_1 s_2 I_{\mathrm{yy}2} - c_{23}\dot{q}_1 s_{23} I_{\mathrm{yy}3} - c_2\dot{q}_1 m_2 s_2 \ell_{\mathrm{c}2}^2 \\
& + m_3\left(-2a_2 c_2 c_3 \dot{q}_1 s_2 \ell_{\mathrm{c}3} - a_2 c_2^2 \dot{q}_1 s_3 \ell_{\mathrm{c}3} - a_2^2 c_2 \dot{q}_1 s_2 + a_2 \dot{q}_1 s_2^2 s_3 \ell_{\mathrm{c}3} + c_2 \dot{q}_1 s_2 s_3^2 \ell_{\mathrm{c}3}^2\right. \\
& \left. -c_2 c_3^2 \dot{q}_1 s_2 \ell_{\mathrm{c}3}^2 - c_2^2 c_3 \dot{q}_1 s_3 \ell_{\mathrm{c}3}^2 + c_3 \dot{q}_1 s_2^2 s_3 \ell_{\mathrm{c}3}^2\right) \\
c_{\mathrm{e}13} =& c_{23}\dot{q}_1 s_{23} I_{\mathrm{xx}3} - c_{23}\dot{q}_1 s_{23} I_{\mathrm{yy}3} + m_3\left(-a_2 c_2 \dot{q}_1 s_{23}\ell_{\mathrm{c}3} - c_{23}\dot{q}_1 s_{23}\ell_{\mathrm{c}3}^2\right) \\
c_{\mathrm{e}21} =& - c_2\dot{q}_1 s_2 I_{\mathrm{xx}2} - c_{23}\dot{q}_1 s_{23} I_{\mathrm{xx}3} + c_2\dot{q}_1 s_2 I_{\mathrm{yy}2} + c_{23}\dot{q}_1 s_{23} I_{\mathrm{yy}3} + m_2 c_2 \dot{q}_1 s_2 \ell_{\mathrm{c}2}^2 \\
& + m_3\left(2a_2 c_2 c_3 \dot{q}_1 s_2 \ell_{\mathrm{c}3} + a_2 c_2^2 \dot{q}_1 s_3 \ell_{\mathrm{c}3} + a_2^2 c_2 \dot{q}_1 s_2 - a_2 \dot{q}_1 s_2^2 s_3 \ell_{\mathrm{c}3} - c_2 \dot{q}_1 s_2 s_3^2 \ell_{\mathrm{c}3}^2\right. \\
& \left. +c_2 c_3^2 \dot{q}_1 s_2 \ell_{\mathrm{c}3}^2 + c_2^2 c_3 \dot{q}_1 s_3 \ell_{\mathrm{c}3}^2 - c_3 \dot{q}_1 s_2^2 s_3 \ell_{\mathrm{c}3}^2\right) \\
c_{\mathrm{e}22} =& - m_3 a_2 \dot{q}_3 s_3 \ell_{\mathrm{c}3} \\
c_{\mathrm{e}23} =& - m_3\left(a_2 \dot{q}_2 s_3 \ell_{\mathrm{c}3} + a_2 \dot{q}_3 s_3 \ell_{\mathrm{c}3}\right) \\
c_{\mathrm{e}31} =& c_{23}\dot{q}_1 s_{23} I_{\mathrm{yy}3} - c_{23}\dot{q}_1 s_{23} I_{\mathrm{xx}3} + m_3 \dot{q}_1 s_{23}\ell_{\mathrm{c}3}\left(a_2 c_2 + c_{23}\ell_{\mathrm{c}3}\right) \\
c_{\mathrm{e}32} =& m_3 a_2 \dot{q}_2 s_3 \ell_{\mathrm{c}3} \\
c_{\mathrm{e}33} =& 0,
\end{aligned}
\tag{3.34}
$$

where $s_3 = sin(q_3)$. The gravity torque vector $\boldsymbol{g}(\boldsymbol{q})$ is given by

$$
\boldsymbol{g}(\boldsymbol{q}) = \begin{bmatrix} 0 \\ g c_2 \ell_{\mathrm{c}2} m_2 + g\left(a_2 c_2 + c_{23}\ell_{\mathrm{c}3}\right) m_3 \\ g c_{23}\ell_{\mathrm{c}3} m_3 \end{bmatrix}, \tag{3.35}
$$

where $g \approx 9.81 m/s$ is gravity.

To compute the inertia tensor, some considerations are taken into account. The inertia products, that is, the elements outside the diagonal of the inertia tensor, are assumed to be zero. The first link is a hollow sphere of radius $r_1$ and mass $m_1$, and its main moments of inertia are given by

$$
I_{xx1} = I_{yy1} = I_{zz1} = \frac{2}{3} m_1 r_1^2. \tag{3.36}
$$

The second and third bodies are approximated by rectangular prisms with sides $b_2$, $h_2$, $b_3$, $h_3$, and lengths $a_2$ and $a_3$, respectively. The main moments of inertia of these links are given by

$$\begin{aligned} I_{xxi} &= \frac{m_i}{12}\left(b_i^2 + h_i^2\right) \\ I_{yyi} &= \frac{m_i}{12}\left(4a_i^2 + h_i^2\right) \\ I_{zzi} &= \frac{m_i}{12}\left(4a_i^2 + b_i^2\right), \end{aligned} \tag{3.37}$$

for $i = 2, 3$; and where Steiner's theorem (parallel axis theorem) has been used.

## 3.3 Modeling and inclusion of the payload in the dynamic model

Consider the model (2.5) of the robot Geomagic touch presented in the previous section, which is a simplified fully-actuated model of three degrees of freedom (DoF) and is described by equations (3.33)-(3.35). The main goal of this work is to estimate the inertial parameters of the payload. This will be done in conjunction with the entire set of parameters of the robot manipulator. Therefore, the payload must be included in the robot's dynamic model. Thus, the last body is considered to be made up of two elements, the actual robot link and the payload itself. The third (last) link and the payload, each one, are then approximated by a rectangular prism. Figure 3.3 shows the diagram of the third compound body of the robot. Now, the total mass of the compound body is given by $m_3 = m_{31} + m_{32}$, where $m_{31}$ is the link mass, $m_{32}$ is the payload mass. The center of mass is considered to be along the $x_3$ axis for simplicity, which is an assumption already made in the reference model (3.33)-(3.35). The inertia tensor of this body is $I_3 = I_{31} + I_{32}$ where $I_{31}$ is the inertia of the link and $I_{32}$ is the inertia tensor of the payload, which are diagonal matrices. The inertia tensor is then determined by

$$\begin{aligned} I_{xx3} &= \frac{m_{31}}{12}\left(b_{31}^2 + h_{31}^2\right) + \frac{m_{32}}{12}\left(b_{32}^2 + \ell_{32}^2\right) \\ I_{yy3} &= \frac{m_{31}}{12}\left(4a_3^2 + b_{31}^2\right) + \frac{m_{32}}{12}\left(b_{32}^2 + h_{32}^2\right) + m_{32}a_3^2 \\ I_{zz3} &= \frac{m_{31}}{12}\left(4a_3^2 + h_{31}^2\right) + \frac{m_{32}}{12}\left(h_{32}^2 + \ell_{32}^2\right) + m_{32}a_3^2, \end{aligned} \tag{3.38}$$

where Steiner's theorem or parallel axis theorem has been used; $I_{xx3}$, $I_{yy3}$ and $I_{zz3}$ are the moments of inertia about the $x$, $y$ and $z$ axis of the compound third body; $b_{31}$, $h_{31}$, $b_{32}$ and $h_{32}$ are the sides of the link and payload, respectively; $a_3$ is a D-H parameter and can be consulted in Table 3.1; and and $\ell_{32}$ is the length of the payload.

In order to continue with the goal of the work, it is necessary to obtain a linear parameterization of the model. However, since it is not possible to obtain a linear relation using the actual physical parameters, the vector of unknown parameters must be overparameterized using the actual physical ones (Romero et al. 2021, Remark 2.3).

In Arteaga et al. (2022) Section 14.2.5, the authors make some assumptions on robot links to obtain a simplified and compact model with a reduced number of parameters to estimate. It is assumed that the second and third bodies of the robot are slender rods instead of rectangular prisms. In fact, the Geomagic touch has a pen (not actuated) at the end of the third link, which here will be modified to account for the robot payload. For the sake of simplifying the analysis and as stated above in the hypothesis in Section 1.4, the same assumptions mentioned are made in this work. Nevertheless, in agreement with the hypothesis, the robot payload is still assumed as a rectangular prism, so the compound third body is made up of a slender rod (link) and a rectangular prism (payload). Therefore, this implies that the radius $r_1$ of the body of the first link and the sides $b_2, h_2, b_{31}, h_{31}$ of the second and third links are assumed to be zero, which appear in the inertia formulas (3.36), (3.37) and (3.38); and in turn this means that as in (3.33)-(3.34) the moments of inertia $I_{xx1} = I_{yy1} = I_{zz1} = I_{xx2} = 0$, and $I_{yy2} = I_{zz2} \triangleq I_2$.

Considering the above simplifications and assumptions, the geometric structure of the robot does not change, but many parameter values do because the actual robot links are not slender rods or rectangular prisms. Therefore, the starting point for deriving the linear parameterized model is the reference dynamic model (3.33)-(3.35). It was found that the model can be expressed in terms of a set of ten parameters, which are the following:

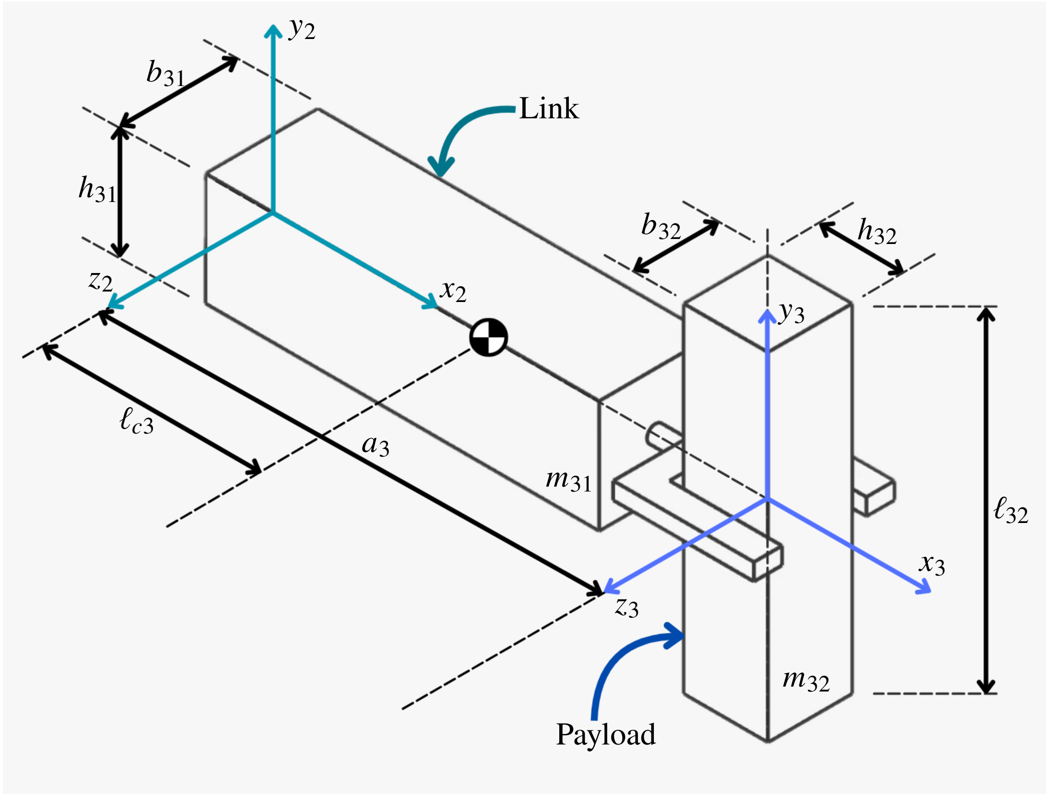


Figure 3.3: Coordinate frames, geometry of the third link and payload of the 3 DoF robot

$$\boldsymbol{\theta} = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \\ \theta_4 \\ \theta_5 \\ \theta_6 \\ \theta_7 \\ \theta_8 \\ \theta_9 \\ \theta_{10} \end{bmatrix} = \begin{bmatrix} m_2\ell_{c2}^2 + m_3a_2^2 + I_2 \\ m_3a_2\ell_{c3} \\ m_3\ell_{c3}^2 + I_{yy3} \\ I_{xx3} \\ m_3\ell_{c3}^2 + I_{zz3} \\ c_{f1} \\ c_{f2} \\ c_{f3} \\ m_2\ell_{c2} + m_3a_2 \\ m_3\ell_{c3} \end{bmatrix}, \tag{3.39}$$

where $m_2$ and $I_2 = I_{yy2} = I_{zz2}$ are the mass and the moment of inertia of the second body, respectively; $\ell_{ci}$ is the distance from the origin of the frame $O_{x_{i-1},y_{i-1},z_{i-1}}$ to the center of mass of the $i$-th body for $i = 2, 3$; $c_{f1}$, $c_{f2}$ and $c_{f3}$ are the friction coefficients of joints 1, 2, and 3, respectively.

Using these parameters and considering the terms that cancel out, i.e. the inertia elements that become zero, the dynamic model (3.33)-(3.35) can be rewritten in a more compact form. The inertia matrix $\boldsymbol{H}(\boldsymbol{q})$ is now given by

$$\boldsymbol{H}(\boldsymbol{q}) = \begin{bmatrix} c_2^2\theta_1 + 2c_2c_{23}\theta_2 + c_{23}^2\theta_3 + s_{23}^2\theta_4 & 0 & 0 \\ 0 & \theta_1 + 2c_3\theta_2 + \theta_5 & c_3\theta_2 + \theta_5 \\ 0 & c_3\theta_2 + \theta_5 & \theta_5 \end{bmatrix}. \tag{3.40}$$

The matrix $\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})$ is given by

$$\begin{aligned}
c_{e11} &= -c_2s_2\dot{q}_2\theta_1 - \frac{1}{2}\left(c_2s_{23}\left(\dot{q}_2+\dot{q}_3\right) + s_2c_{23}\dot{q}_2\right)\theta_2 - c_{23}s_{23}\left(\dot{q}_2+\dot{q}_3\right)\theta_3 + c_{23}s_{23}\left(\dot{q}_2+\dot{q}_3\right)\theta_4 \\
c_{e12} &= -c_2s_2\dot{q}_1\theta_1 - \frac{1}{2}\left(s_2c_{23} + c_2s_{23}\right)\dot{q}_1\theta_2 - s_{23}c_{23}\dot{q}_1\theta_3 + s_{23}c_{23}\dot{q}_1\theta_4 \\
c_{e13} &= -\frac{1}{2}c_2s_{23}\dot{q}_1\theta_2 - c_{23}s_{23}\dot{q}_1\theta_3 + c_{23}s_{23}\dot{q}_1\theta_4 \\
c_{e21} &= c_2s_2\dot{q}_1\theta_1 + \frac{1}{2}\left(s_2c_{23} + c_2s_{23}\right)\dot{q}_1\theta_2 + s_{23}c_{23}\dot{q}_1\theta_3 - s_{23}c_{23}\dot{q}_1\theta_4 \\
c_{e22} &= -s_3\dot{q}_3\theta_2 \\
c_{e23} &= -s_3\left(\dot{q}_2+\dot{q}_3\right)\theta_2 \\
c_{e31} &= \frac{1}{2}c_2s_{23}\dot{q}_1\theta_2 + c_{23}s_{23}\dot{q}_1\theta_3 - c_{23}s_{23}\dot{q}_1\theta_4 \\
c_{e32} &= s_3\dot{q}_2\theta_2 \\
c_{e33} &= 0.
\end{aligned} \tag{3.41}$$

The symmetric positive semidefinite matrix of joint viscous friction coefficients $\boldsymbol{D}$ is described by

$$\boldsymbol{D} = \begin{bmatrix} \theta_6 & 0 & 0 \\ 0 & \theta_7 & 0 \\ 0 & 0 & \theta_9 \end{bmatrix}, \tag{3.42}$$

and lastly, the gravity torque vector $\boldsymbol{g}(\boldsymbol{q})$ is given by

$$\boldsymbol{g}(\boldsymbol{q}) = \begin{bmatrix} 0 \\ g\left(c_2\theta_9 + c_{23}\theta_{10}\right) \\ gc_{23}\theta_{10} \end{bmatrix}, \tag{3.43}$$

where $q_i$ and $\dot{q}_i$ are the joint positions and velocities of the $i$-th joint for $i = 1, 2$ and 3; $c_2 = \cos(q_2)$, $c_3 = \cos(q_3)$, $c_{23} = \cos(q_2 + q_3)$, $s_2 = \sin(q_2)$, $s_3 = \sin(q_3)$ and $s_{23} = \sin(q_2 + q_3)$.

Finally, recalling Property 5, the left-hand side of (2.5) can be rewritten as the product of the regressor $\boldsymbol{Y}(\boldsymbol{q}, \dot{\boldsymbol{q}}, \ddot{\boldsymbol{q}}) \in \mathbb{R}^{n \times p}$ by a vector of constant parameters $\boldsymbol{\theta} \in \mathbb{R}^p$, that is,

$$\boldsymbol{H}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{D}\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{Y}(\boldsymbol{q}, \dot{\boldsymbol{q}}, \ddot{\boldsymbol{q}})\boldsymbol{\theta}.$$

As a result, the elements of the regressor $\boldsymbol{Y}(\boldsymbol{q}, \dot{\boldsymbol{q}}, \ddot{\boldsymbol{q}})$ are then given by

$$\begin{aligned}
y_{11} &= c_2^2 \ddot{q}_1 - 2c_2 s_2 \dot{q}_1 \dot{q}_2 \\
y_{12} &= 2c_2 c_{23} \ddot{q}_1 - \left(c_2 s_{23} \left(\dot{q}_2 + \dot{q}_3\right) + s_2 c_{23} \dot{q}_2\right) \dot{q}_1 \\
y_{13} &= c_{23}^2 \ddot{q}_1 - 2c_{23} s_{23} \left(\dot{q}_2 + \dot{q}_3\right) \dot{q}_1 \\
y_{14} &= s_{23}^2 \ddot{q}_1 + 2c_{23} s_{23} \left(\dot{q}_2 + \dot{q}_3\right) \dot{q}_1 \\
y_{15} &= 0 \\
y_{16} &= \dot{q}_1 \\
y_{17} &= y_{18} = y_{19} = y_{110} = 0 \\
y_{21} &= \ddot{q}_2 + c_2 s_2 \dot{q}_1^2 \\
y_{22} &= 2c_3 \ddot{q}_2 + c_3 \ddot{q}_3 + \frac{1}{2}\left(s_2 c_{23} + c_2 s_{23}\right) \dot{q}_1^2 - 2s_3 \dot{q}_2 \dot{q}_3 - s_3 \dot{q}_3^2 \\
y_{23} &= s_{23} c_{23} \dot{q}_1^2 \\
y_{24} &= -s_{23} c_{23} \dot{q}_1^2 \\
y_{25} &= \ddot{q}_2 + \ddot{q}_3 \\
y_{26} &= 0 \\
y_{27} &= \dot{q}_2 \\
y_{28} &= 0 \\
y_{29} &= g c_2 \\
y_{210} &= g c_{23} \\
y_{31} &= 0 \\
y_{32} &= c_3 \ddot{q}_2 + \frac{1}{2} c_2 s_{23} \dot{q}_1^2 + s_3 \dot{q}_2^2 \\
y_{33} &= c_{23} s_{23} \dot{q}_1^2
\end{aligned} \tag{3.44}$$

$$y_{34} = -c_{23}s_{23}\dot{q}_1^2$$
$$y_{35} = \ddot{q}_2 + \ddot{q}_3$$
$$y_{36} = y_{37} = 0$$
$$y_{38} = \dot{q}_3$$
$$y_{39} = 0$$
$$y_{310} = gc_{23}.$$

Taking into account that some assumptions are made to simplify the analysis, it is necessary to keep in mind that we aimed at still reaching a good enough approximation of the parameter estimation. Although the robot payload could take any shape, it is assumed that it could be approximated by a rectangular prism, which is a very strong hypothesis, though. The practical goal of parameter estimation in this work is to be able to apply it to different payloads that may vary in shape and weight. For simulation and experimental purposes, here the robot payload is considered with a defined shape (rectangular prism) and weight.

### 3.4 Control law

In this section, the control law that is used in simulation and experimentation is introduced. The control law was taken from the same paper on which the MDREM is based. However, two changes are made to the original one. First, in a similar way to the change made to the adaptive law (3.10), the exponent is replaced by $\lambda_s \tanh(s^2)$. Secondly, instead of the original sliding surface $s$, a new one is used taken from Arteaga et al. (2025).

To introduce the control law, it is necessary to define the control problem as achieving the tracking error given by

$$\boldsymbol{e} = \boldsymbol{q} - \boldsymbol{q}_{\mathrm{d}}, \tag{3.45}$$

tends to zero under the assumption that the parameter vector $\boldsymbol{\theta}$ is uncertain or unknown; where $\boldsymbol{q}_{\mathrm{d}} \in \mathbb{R}^n$ is a given bounded desired trajectory with at least bounded first and second derivatives. As seen in Section 2.3, this can be done by defining the reference velocity

$$\dot{\boldsymbol{q}}_{\mathrm{r}} = \dot{\boldsymbol{q}}_{\mathrm{d}} - \boldsymbol{\Lambda}\boldsymbol{e} - \boldsymbol{K}_{\mathrm{e}}\tau_{\mathrm{e}}, \tag{3.46}$$

where $\boldsymbol{\Lambda}$, $\boldsymbol{K}_{\mathrm{e}} \in \mathbb{R}^{n\times n}$ are diagonal positive definite matrices and the last term is added due to the new sliding surface $\boldsymbol{s}$. The $i$th element of the vector $\boldsymbol{\tau}_{\mathrm{e}} \in \mathbb{R}^{n}$ for $i = 1, ..., n$ is defined as

$$\tau_{\mathrm{e}i} = \operatorname{sign}(e_i)\,\frac{\tanh\left(|e_i|^{\frac{1}{2}}\right)}{\tanh(1)}\,|e_i|^{\frac{1}{2}\left(1-2\frac{\tanh\left(e_i^2\right)}{\tanh(1)}\right)}\,|e_i|^{\lambda_{\mathrm{e}i}\tanh\left(e_i^2\right)}, \tag{3.47}$$

with $\lambda_{\mathrm{e}i}$, the $i$th positive element of the vector $\boldsymbol{\lambda}_{\mathrm{e}} \in \mathbb{R}^{n}$, satisfying

$$\lambda_{\mathrm{e}i} \geq \frac{\frac{1}{2}\theta_{\mathrm{e}i} + 1}{\tanh(1)}, \quad \theta_{\mathrm{e}i} > 1. \tag{3.48}$$

Taking into account (3.46), it is possible to define the new sliding surface as follows

$$\boldsymbol{s} = \dot{\boldsymbol{q}} - \dot{\boldsymbol{q}}_{\mathrm{r}} = \dot{\boldsymbol{e}} + \boldsymbol{\Lambda}\boldsymbol{e} + \boldsymbol{K}_{\mathrm{e}}\tau_{\mathrm{e}}. \tag{3.49}$$

If (3.49) is compared to (2.10) introduced in Slotine & Li (1987), the difference is the addition of the term $\boldsymbol{K}_{\mathrm{e}}\boldsymbol{\tau}_{\mathrm{e}}$ here . The design of term $\boldsymbol{\tau}_{\mathrm{e}}$ is based on fixed-time stability techniques of Sliding Mode Control. Hence, its effect is such that once the system reaches the sliding surface ($\boldsymbol{s} = 0$), the error dynamics is given by $\dot{\boldsymbol{e}} = -\boldsymbol{\Lambda}\boldsymbol{e} - \boldsymbol{K}_{\mathrm{e}}\tau_{\mathrm{e}}$, so that the position tracking error $\boldsymbol{e}$ becomes zero in fixed-time, as proved in Arteaga et al. (2025) in Section 3, Theorem 1. In addition, unlike other fixed-time control schemes, the sliding surface (3.49) avoids the classic singularities of the origin. Then, the control law is given by

$$\begin{aligned}\boldsymbol{\tau} &= \hat{\boldsymbol{H}}(\boldsymbol{q})\ddot{\boldsymbol{q}}_{\mathrm{r}} + \hat{\boldsymbol{C}}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}}_{\mathrm{r}} + \hat{\boldsymbol{D}}\dot{\boldsymbol{q}}_{\mathrm{r}} + \hat{\boldsymbol{g}}(\boldsymbol{q}) - \boldsymbol{K}_{\mathrm{v}}\underbrace{\operatorname{sign}(\boldsymbol{s})|\boldsymbol{s}|^{\lambda_{\mathrm{s}}\tanh\left(s^2\right)}}_{\boldsymbol{\tau}_{\mathrm{s}}} - \boldsymbol{K}_{\mathrm{p}}\|\boldsymbol{s}\|\boldsymbol{s}, \\ &= \boldsymbol{Y}_{\mathrm{a}}\hat{\boldsymbol{\theta}} - \boldsymbol{K}_{\mathrm{v}}\boldsymbol{\tau}_{\mathrm{s}} - \boldsymbol{K}_{\mathrm{p}}\|\boldsymbol{s}\|\boldsymbol{s},\end{aligned} \tag{3.50}$$

where Property 5 of Section 2.2 and $\boldsymbol{Y}_{\mathrm{a}} = \boldsymbol{Y}(t, \boldsymbol{q}, \dot{\boldsymbol{q}}, \dot{\boldsymbol{q}}_{\mathrm{r}}, \ddot{\boldsymbol{q}}_{\mathrm{r}})$ has been used for simplicity; $\boldsymbol{K}_{\mathrm{v}}, \boldsymbol{K}_{\mathrm{p}} \in \mathbb{R}^{n\times n}$ are diagonal positive definite matrices, and $\boldsymbol{\lambda}_{\mathrm{s}} \in \mathbb{R}^{n}$ is a vector of tuning parameters. In addition, the $i$th element $\tau_{\mathrm{s}i}$ of $\boldsymbol{\tau}_{\mathrm{s}} \in \mathbb{R}^{n}$ is defined as

$$\tau_{\mathrm{s}i} = \operatorname{sign}(s_i)\,|s_i|^{\lambda_{\mathrm{s}i}\tanh(s_i^2)}, \tag{3.51}$$

for $i = 1, \ldots , n$, and with $\lambda_{\mathrm{s}i}$ the $i$th positive element of $\boldsymbol{\lambda}_{\mathrm{s}}$, which satisfies

$$\lambda_{si} \geq \frac{\theta_{si}}{\tanh(1)}, \quad \theta_{si} > 1. \tag{3.52}$$

*Chapter 4*

# SIMULATION AND EXPERIMENTAL VALIDATION WITH THE GEOMAGIC TOUCH

In this chapter, the adaptive algorithm developed in the previous chapter, Chapter 3, is applied in simulation to the 3 DoF robot, verified in experimentation with the *Geomagic touch haptic device*, and the results are presented. The model used for parameter estimation is derived in Section 3.3 and is given by equations (3.40)-(3.43) or otherwise by the regressor $\boldsymbol{Y}(\boldsymbol{q}, \dot{\boldsymbol{q}}, \ddot{\boldsymbol{q}})$ given in (3.44) of the linear parameterization model. The simulations were carried out using MATLAB and SIMULINK, while the experiment was carried out using the actual *Geomagic touch* with the adaptive control scheme implemented in a *Visual Studio/C++* application.

## 4.1 Simulation/experiment description

The goal is to estimate the set of parameters of the entire robot with a payload attached. Recalling that the payload of the robot was model as part of the robot's last link, the inertial parameters associated with this link should view an increase in their values. In order to make this change more evident, two scenarios are proposed:

1. Scenario one: parameter estimation without payload.
2. Scenario two: parameter estimation with attached payload.

First, the simulation/experiment is run without payload. Next, a payload is added to the end effector, and the simulation/experiment is run again. Therefore, in the first scenario the algorithm should notice the absence of the payload, consequently, in the second one it should estimate the addition of the payload.

## 4.2 Desired trajectory

To implement the control law (3.50) a desired trajectory is necessary. This one is generated by a fifth-order polynomial for each of the $n$ joints and is given by

$$q_{j,\mathrm{d}} = \alpha_{j,0} + \alpha_{j,1}t + \alpha_{j,2}t^2 + \alpha_{j,3}t^3 + \alpha_{j,4}t^4 + \alpha_{j,5}t^5 \tag{4.1}$$

for $j = 1, 2, \ldots, n$ joints, and where $\alpha_{j,0}, \alpha_{j,1}, ..., \alpha_{j,5}$ are the polynomial coefficients corresponding to their respective joints; $t$ is the time and is the independent variable. Given a constant desired initial and final joint position, a final time need to be set for which the trajectory will be executed. After this final time, the desired position is constant. The shorter the final time, the higher the transitory-state of the desired joint velocity and acceleration. So, to maintain reasonable values, it is not recommended to set the final time too small. To obtain the desired velocity and acceleration, respectively, derivation of (4.1) must be carried out as follows

$$\dot{q}_{j,\mathrm{d}} = \alpha_{j,1} + 2\alpha_{j,2}t + 3\alpha_{j,3}t^2 + 4\alpha_{j,4}t^3 + 5\alpha_{j,5}t^4 \tag{4.2}$$

$$\ddot{q}_{j,\mathrm{d}} = 2\alpha_{j,2} + 6\alpha_{j,3}t + 12\alpha_{j,4}t^2 + 20\alpha_{j,5}t^3 \tag{4.3}$$

The coefficients $\alpha_{j,i}$ for $j$ = 1, 2, ..., $n$ and $i$ = 0, 1, ... ,5, depend on the initial and final time and the desired initial and final position, velocity and acceleration of the corresponding joint $j$. To find these coefficients, first, substitute the initial time ($t_0 = 0$), the desired initial joint position, velocity, and acceleration into (4.1), (4.2) and (4.3), respectively, to directly obtain $\alpha_{j,0}$, $\alpha_{j,1}$ and $\alpha_{j,2}$. Then, substitute these three coefficients, the final time ($t_f$), the desired final joint position, velocity, and acceleration, again in (4.1), (4.2), and (4.3), respectively; a system of three equations and three unknown is obtained and thus can be solved to determine $\alpha_{j,3}$, $\alpha_{j,4}$, and $\alpha_{j,5}$ for $j$ = 1, 2, ..., $n$.

The purpose of having such a trajectory is for a smooth motion of the robot manipulator, although it needs to produce the minimum amount of excitation.

## 4.3 Tuning guidelines

To tune the MDREM parameters, some tuning rules are given in the original MDREM paper, Arteaga (2024) in Section 4.1. These are summarized as follows:

- Choose $\lambda_\varphi$ in (2.13) (and (2.14)) according to your own experience but small enough to filter high frequencies.
- Choose the values $b_j$ for $j = 2, \ldots, \tilde{n}$ in (2.19) small enough to work as pure time delays at low frequencies.
- Choose the values $a_j$ for $j = 2, \ldots, \tilde{n}$ (2.19) to change the magnitude of $\phi^2$. The larger the coefficients $a_j$ the larger $\phi^2$ becomes and vise versa.

Regarding $\eta_{\mathrm{m}}$, it is a good idea first to see the behavior of $\phi^2$ and based on it adjust its value depending on the desired performance and duration of $\phi_{\mathrm{m}}$. Recall that the values of $a_j$ and $b_j$ (for $j = 2, \ldots, \tilde{n}$) also affect the shape of $\phi^2$ so that in conjunction with $\eta_{\mathrm{m}}$ they might be tuned together. In general, this is done by trial and error.

For the adaptive law (3.10) the corresponding gains in the present work are tuned following these rules:

- The gains $\boldsymbol{\lambda}_{\theta}$ are tuned according to (3.13).
- The elements $\gamma_i$ for $i = 1, \ldots, p$ are set very small first and then gradually increase or decrease until the desired performance is achieved.

For tuning the gains of the control law (3.50), the paper Arteaga (2024) also states some tuning rules, and they are complemented with those of Arteaga et al. (2025) Remark 7. These are summarized as:

- Set $\boldsymbol{K}_{\mathrm{e}} = \boldsymbol{O}$, where $\boldsymbol{O} \in \mathbb{R}^{n \times n}$ is the null matrix, and choose $\boldsymbol{\Lambda}$ according to his/her own experience for first order linear systems in (3.46).
- The parameter $\lambda_{\mathrm{s}i}$ in (3.51) has to be chosen in accordance with (3.52).
- The gain $\boldsymbol{K}_{\mathrm{v}}$ in (3.50) can be set according to his/her own experience with derivative gains.
- The gain $\boldsymbol{K}_{\mathrm{p}}$ in (3.50) allows to yield the sliding surface faster to zero in a predefined fixed time, so to avoid chattering it should not be set too large.
- Choose $\lambda_{\mathrm{e}i}$ in (3.47) as given in (3.48).
- Slowly increase $\boldsymbol{K}_{\mathrm{e}}$ to yield $\boldsymbol{e}$ and $\dot{\boldsymbol{e}}$ to zero.

It should be noted that large values of $\boldsymbol{K}_{\mathrm{e}}$ improve precision, but also create larger input peaks.

## 4.4 Simulation results

### 4.4.1 First simulation (scenario one): parameter estimation without payload

The corresponding inertial parameters for the first simulation are taken from Guajardo Benavides & Arteaga (2022) and are shown in Table 4.1, where the parameters $\theta_3 = \theta_5$ and $\theta_4 = 0$ due to the absence of the payload and the assumptions made on the robot links. Thus, for the no payload case, the system employs a twofold linear overparameterization. First, the parameters are defined specifically to make the value increments evident when estimating with the payload. Second, this formulation respects the structural constraints discussed in Section 3.3, where an inherent overparameterization is required because the true physical parameters cannot be estimated independently.

Table 4.1: Parameters for simulation without payload. Source: Guajardo Benavides & Arteaga (2022)

| Parameter | Expression | Value |
|---|---|---|
| $\theta_1$ | $m_2\ell_{c2}^2 + m_3a_2^2 + I_2$ | $0.00251729\ \mathrm{kg \cdot m^2}$ |
| $\theta_2$ | $m_3a_2\ell_{c3}$ | $0.00108246\ \mathrm{kg \cdot m^2}$ |
| $\theta_3$ | $m_3\ell_{c3}^2 + I_{yy3}$ | $0.00137408\ \mathrm{kg \cdot m^2}$ |
| $\theta_4$ | $I_{xx3}$ | $0\ \mathrm{kg \cdot m^2}$ |
| $\theta_5$ | $m_3\ell_{c3}^2 + I_{zz3}$ | $0.00137408\ \mathrm{kg \cdot m^2}$ |
| $\theta_6$ | $c_{f1}$ | $0.00076823$ kg/s |
| $\theta_7$ | $c_{f2}$ | $0.03526735$ kg/s |
| $\theta_8$ | $c_{f3}$ | $0.00744473$ kg/s |
| $\theta_9$ | $m_2\ell_{c2} + m_3a_2$ | $0.00449158\ \mathrm{kg \cdot m}$ |
| $\theta_{10}$ | $m_3\ell_{c3}$ | $0.00534505\ \mathrm{kg \cdot m}$ |

The MDREM algorithm is intended to achieve exact parameter estimation with very little excitation. Therefore, the robot performs only a brief movement, from a constant initial position to a desired constant final position. The desired joint position, velocity, and acceleration are given by

$$\begin{aligned} \boldsymbol{q}_\mathrm{d} &= [45^\circ \ \ -30^\circ \ \ -100^\circ]^\mathrm{T} \\ \dot{\boldsymbol{q}}_\mathrm{d} &= [0\ 0\ 0]^\mathrm{T}\ {}^\circ/\mathrm{s} \\ \ddot{\boldsymbol{q}}_\mathrm{d} &= [0\ 0\ 0]^\mathrm{T}\ {}^\circ/\mathrm{s}^2, \end{aligned} \tag{4.4}$$

while the corresponding initial joint position, velocity, and acceleration, respectively, are given by

$$
\begin{aligned}
\boldsymbol{q}(t_0) &= [-45^\circ\ 60^\circ\ -10^\circ]^{\mathrm{T}} \\
\dot{\boldsymbol{q}}(t_0) &= [0\ 0\ 0]^{\mathrm{T}}\ {}^\circ/\mathrm{s} \\
\ddot{\boldsymbol{q}}(t_0) &= [0\ 0\ 0]^{\mathrm{T}}\ {}^\circ/\mathrm{s}^2.
\end{aligned}
\tag{4.5}
$$

To provide a smooth transition, the trajectory from the initial position to the final position is given by the fifth-order polynomial (4.1). To compute the extended regressor $\boldsymbol{Y}_{\mathrm{f}}$ as in (2.18), the corresponding parameters are chosen as $\lambda_\varphi = 1$, $b_2 = 1.35$, $b_3 = 3.46$, $b_4 = 5$ and $a_2 = a_3 = a_4 = 9$ for (2.13) and (2.19), respectively. The gains of the adaptive law (3.10) are given in Table 4.2, while $\phi_{\mathrm{d}} = 0.2$ and $\eta_{\mathrm{m}} = 0.0001$. Since $p$ independent estimators are designed, there will be one $T_{\mathrm{m}}$ for each estimated parameter. The time $T_{\mathrm{m}}$ is obtained from Equation (3.15) by solving for $T_{\mathrm{m}}$ given the other tuning parameters. Although $T_{\mathrm{m}}$ could have been first proposed to determine the necessary $\phi_{\mathrm{d}}$, that approach was not followed here, instead $\phi_{\mathrm{d}}$ is treated as another gain to be tuned. Therefore, the $T_{\mathrm{m}}$ values for the given tuning parameters are

$$
T_{\mathrm{m}} = [5955.5\quad 8337.7\quad 1413.2\quad 189494.1\quad 4904.5\quad 21656.5\quad 591.3\quad 2689.6\quad 1029.3\quad 1199.7\,]^{\mathrm{T}}\ \mathrm{s}
$$

Recall that the value $T_{\mathrm{m}}$ represents the minimum time required for (3.1) to hold and ensure convergence of the parameters independently of the initial conditions. Nevertheless, convergence could happen before this time. The simulation sample time is $T = 1 \times 10^{-5}$.

These parameters were tuned according to the rules stated in Section 4.3. Still, it is difficult to tell what the first approach should be for $\phi_d$ and $\eta_{\mathrm{m}}$, so they were tuned mainly by trial and error. Increasing $\phi_d$ can help achieve the fixed-time convergence of parameters faster; however, as will be seen, with a relatively small value of $\phi_d$, good results were obtained. In general, the whole tuning process is complemented by trial and error.

Table 4.2: Gains of the adaptive law for the simulation

| Gain | Value |
|---|---|
| $\boldsymbol{\Gamma}$ | diag $\{0.014, 0.01, 0.059, 0.00044, 0.017, 0.00385, 0.141, 0.031, 0.081, 0.0695\}$ |
| $\boldsymbol{\lambda}_\theta$ | $[2\quad 2\quad 2\quad 2\quad 2\quad 2\quad 2\quad 2\quad 2\quad 2\,]^{\mathrm{T}}$ |

The gains of the control law (3.50) and those of (3.46)-(3.47) are given in Table 4.3. To tune these gains, the guidelines in Section 4.3 were also followed. In addition, the trail-and-error approach was still needed to complement the tuning.

Table 4.3: Gains of the control law for the simulation

| **Gain** | **Value** |
|---|---|
| $\boldsymbol{\Lambda}$ | $\mathrm{diag}\,\{20, 20, 20\}$ |
| $\boldsymbol{K}_{\mathrm{e}}$ | $\mathrm{diag}\,\{0.8, 0.8, 0.8\}$ |
| $\boldsymbol{K}_{\mathrm{v}}$ | $\mathrm{diag}\,\{0.01, 0.04, 0.01\}$ |
| $\boldsymbol{K}_{\mathrm{p}}$ | $\mathrm{diag}\,\{0.01, 0.01, 0.01\}$ |
| $\boldsymbol{\lambda}_{\mathrm{s}}$ | $[2 \quad 2 \quad 2]^{\mathrm{T}}$ |
| $\boldsymbol{\lambda}_{\mathrm{e}}$ | $[2.813 \quad 2.813 \quad 2.813]^{\mathrm{T}}$ |

Figure 4.1 shows the results for position regulation. As can be seen in this figure, all the joints achieve zero position error very quickly. Regardless of whether exact parameter estimation is accomplished, this is expected, since a robust control law is employed. However, in Figure 4.2, it can be confirmed that 10 of 10 estimated parameters reach zero estimated error. This is done before second 0.5 as can be verified in this figure. From Table 4.1, the parameter $\theta_4$ is zero, so since the initial conditions of the estimated parameters are zero, the parameter estimation error $\tilde{\theta}_4$ begins and remains zero. Figure 4.3 shows the behavior of the determinant $\phi_{\mathrm{m}}^2$ for the MDREM (top plot) and the determinant $\phi^2$ for the DREM (bottom plot). Thus, we can see that according to (3.2), $\phi_{\mathrm{m}}^2$ increases to $\phi_{\mathrm{d}}^2 = 0.04$ for approximately 0.63 seconds.

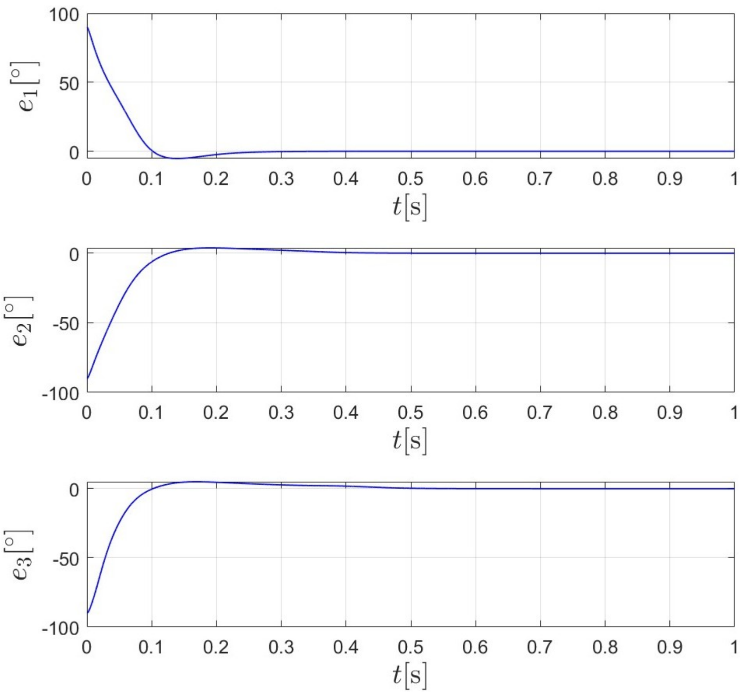


Figure 4.1: First simulation. Joint position errors.

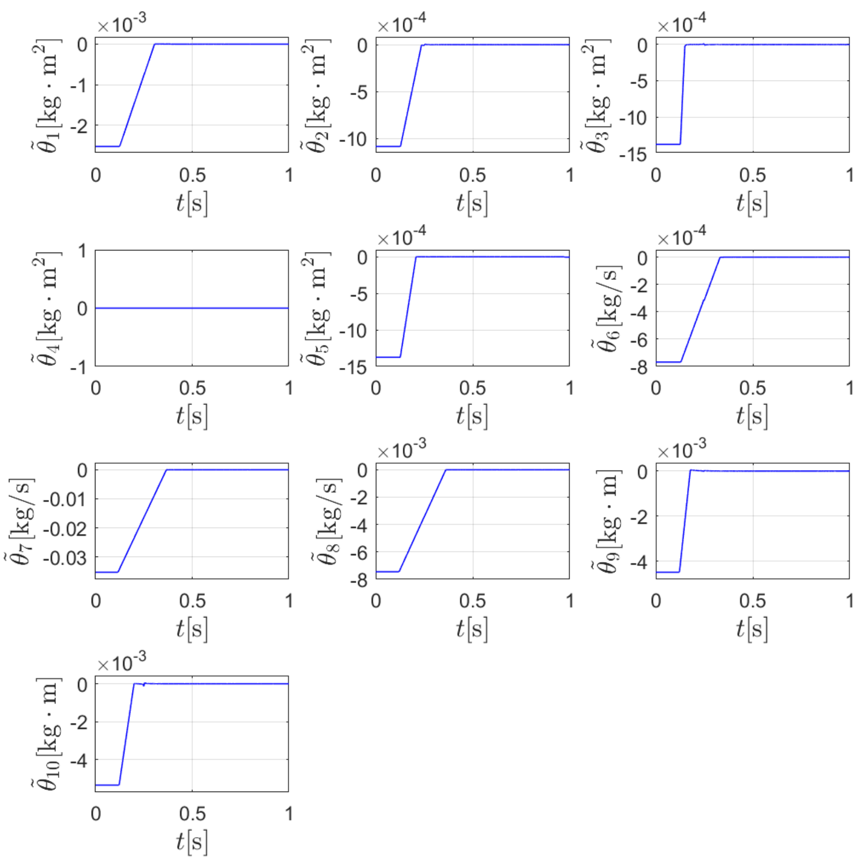


Figure 4.2: First simulation. Parameter estimation errors with initial conditions of $\hat{\boldsymbol{\theta}}$ equal to zero.

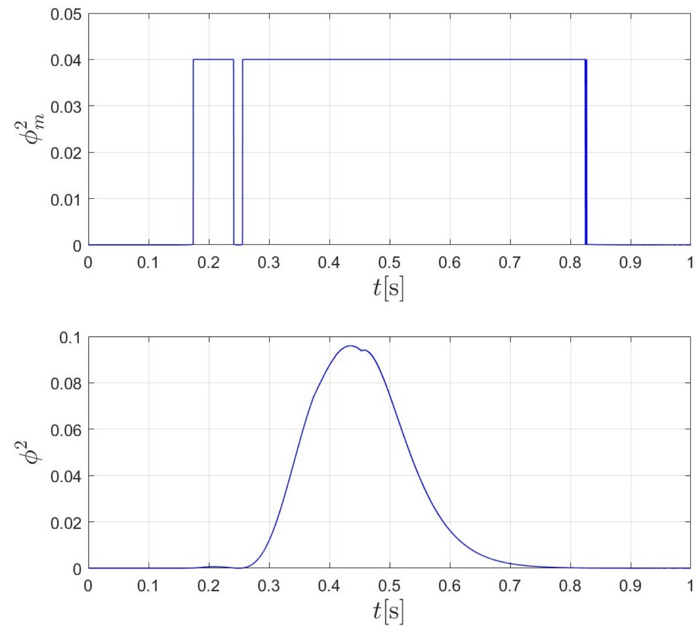


Figure 4.3: First simulation. Square determinant $\phi_{\mathrm{m}}^2$ and $\phi^2$.

### 4.4.2 Second simulation (scenario two): parameter estimation with payload

For the second simulation, the 3 DoF robot virtually holds a payload represented by a rectangular prism, so a change in some values of the parameters in Table 4.1 is perceived. The resulting parameter values are shown in Table 4.4.

Table 4.4: Parameters for the simulation with payload.

| Parameter | Expression | Value |
|---|---|---|
| $\theta_1$ | $m_2\ell_{c2}^2 + m_3a_2^2 + I_2$ | $0.00349359$ kg · m$^2$ |
| $\theta_2$ | $m_3a_2\ell_{c3}$ | $0.00248272$ kg · m$^2$ |
| $\theta_3$ | $m_3\ell_{c3}^2 + I_{yy3}$ | $0.00459359$ kg · m$^2$ |
| $\theta_4$ | $I_{xx3}$ | $0.00006993$ kg · m$^2$ |
| $\theta_5$ | $m_3\ell_{c3}^2 + I_{zz3}$ | $0.00465868$ kg · m$^2$ |
| $\theta_6$ | $c_{f1}$ | $0.00076823$ kg/s |
| $\theta_7$ | $c_{f2}$ | $0.03526735$ kg/s |
| $\theta_8$ | $c_{f3}$ | $0.00744473$ kg/s |
| $\theta_9$ | $m_2\ell_{c2} + m_3a_2$ | $0.02191225$ kg · m |
| $\theta_{10}$ | $m_3\ell_{c3}$ | $0.01712223$ kg · m |

The desired and initial joint position, velocity and acceleration were the same as in the first simulation given by (4.4) and (4.5), respectively. The tuning parameters to compute $\boldsymbol{Y}_{\mathrm{f}}$ are $\lambda_\varphi = 1$, $b_2 = 1.35$, $b_3 = 3.46$, $b_4 = 5$ and $a_2 = a_3 = a_4 = 9$

for (2.13) and (2.19), respectively. Also, $\phi_{\mathrm{d}} = 0.2$ and $\eta_{\mathrm{m}} = 0.0001$. These are the same as for the first simulation, so the tuning process is as described earlier in Section 4.4.1. Similarly, the gains of the adaptive law (3.10) and those of the implementation of the control law (3.50) are the same as for the first simulation and are given in tables 4.2 and 4.3, respectively. Again, the simulation sample time is $T = 1 \times 10^{-5}$.

Figure 4.4 shows the joint position errors for the second simulation. It can be seen that all three joints achieve zero position error with a higher overshoot compared to that of the first simulation in Figure 4.1. In Figure 4.5 the parameter estimation errors are shown, where it can be seen that the set of ten parameters achieve zero estimation error before the second 0.5. The initial conditions of the estimated parameters $\hat{\boldsymbol{\theta}}$ are set to zero. All parameter adaptations have a similar behavior, yet parameter 3 is the fastest. Finally, Figure 4.6 shows the behavior of the determinant $\phi_{\mathrm{m}}^2$ for the MDREM (top plot) and the determinant $\phi^2$ for the DREM (bottom plot). It can be seen that the values of $\phi^2$ have been relatively higher than $\phi_{\mathrm{m}}^2$ for some time. However, $\phi_{\mathrm{m}}^2$ remain constant in accordance with (3.2) for about 0.61 seconds, which is enough to achieve the exact parameter estimation. This exhibits a property of the MDREM, it increases the amount of excitation but also decreases it, if it is too high, to a desired constant value.

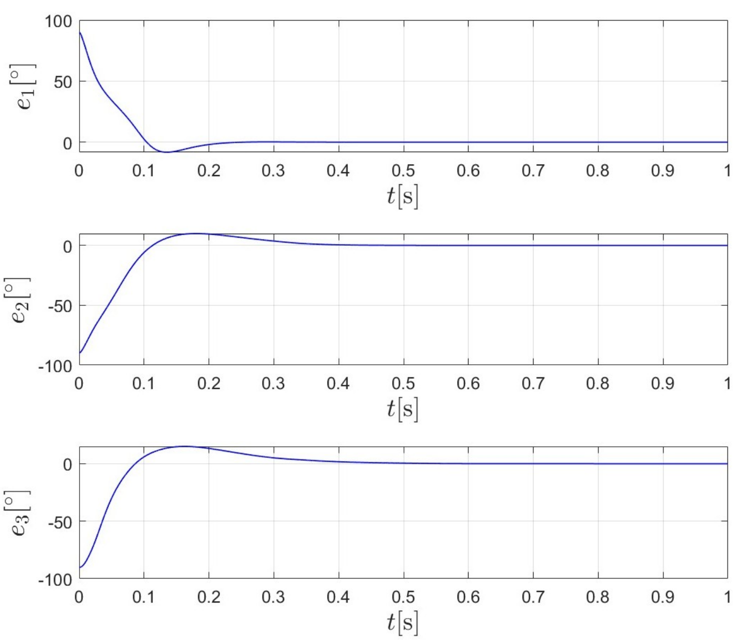


Figure 4.4: Second simulation. Joint position errors.

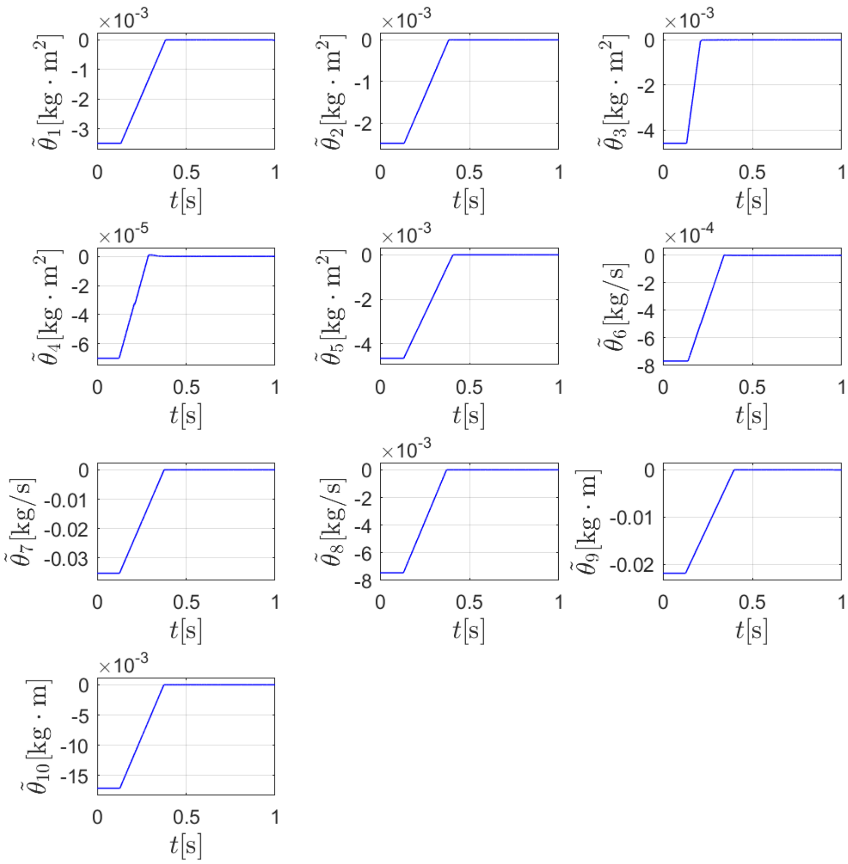


Figure 4.5: Second simulation. Parameter estimation errors with initial conditions of $\hat{\boldsymbol{\theta}}$ equal to zero.

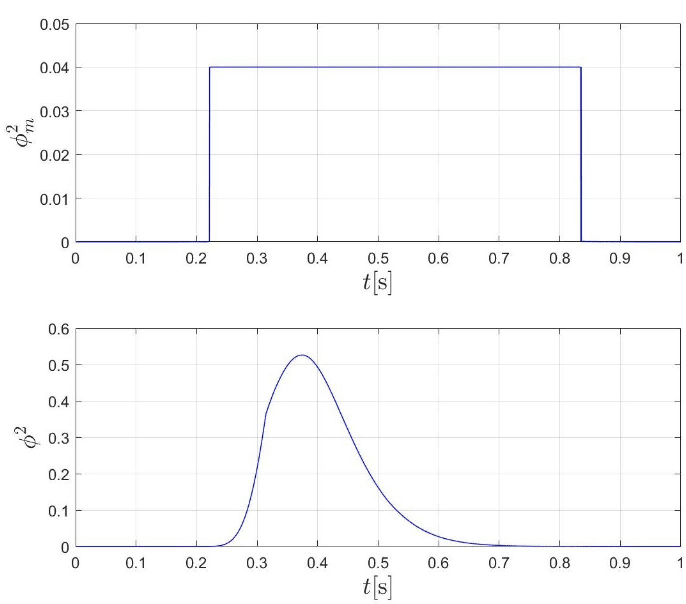


Figure 4.6: Second simulation. Square determinant $\phi_{\mathrm{m}}^2$ and $\phi^2$.

### 4.5 Experimentation results

For experimentation, the trajectory is different from the one used in the simulation. The desired joint position, velocity, and acceleration are given by

$$\begin{aligned} \boldsymbol{q}_{\mathrm{d}} &= [50^\circ\ 55^\circ\ \ -70^\circ]^{\mathrm{T}} \\ \dot{\boldsymbol{q}}_{\mathrm{d}} &= [0\ 0\ 0]^{\mathrm{T}}\ {}^\circ/\mathrm{s} \\ \ddot{\boldsymbol{q}}_{\mathrm{d}} &= [0\ 0\ 0]^{\mathrm{T}}\ {}^\circ/\mathrm{s}^2, \end{aligned} \tag{4.6}$$

and are planned to be reached in 1 second by the polynomial trajectory given by (4.1)-(4.3). The initial joint position, velocity, and acceleration are as follows

$$\begin{aligned} \boldsymbol{q}(t_0) &= [-50^\circ\ 10^\circ\ \ -90^\circ]^{\mathrm{T}} \\ \dot{\boldsymbol{q}}(t_0) &= [0\ 0\ 0]^{\mathrm{T}}\ {}^\circ/\mathrm{s} \\ \ddot{\boldsymbol{q}}(t_0) &= [0\ 0\ 0]^{\mathrm{T}}\ {}^\circ/\mathrm{s}^2. \end{aligned} \tag{4.7}$$

The tuning gains to calculate the extended regressor $\boldsymbol{Y}_{\mathrm{f}}$ as in (2.18) are shown in Table 4.5 together with other MDREM gains. Moreover, the gains of the adaptive law (3.10) are given in Table 4.6. In general, the same tuning rules mentioned before and stated in Section 4.3 were used to tune the adaptive gains for experimentation. However, recall that some assumptions were made to the dynamic model to simplify it, so some uncertainties were introduced. In simulation, this does not have an impact since the model used for the robot is the same as that for the MDREM algorithm. In contrast, in experimentation, the dynamics of the actual robot might not be perfectly described by the obtained model, which might make the tuning process more difficult. Making $\phi_d$ large could mathematically achieve parameter convergence faster; nevertheless, in case of modeling uncertainty the parameter error is scaled by this parameter, as seen in equation (3.12) since $\phi_{\mathrm{m}}^2$ becomes $\phi_{\mathrm{d}}^2$ when $\phi^2 \geq \eta_{\mathrm{m}}$. Therefore, the convergence accuracy can be degraded, so it is recommended to be relatively small. Again, the overall tuning process is complemented by trial and error.

Table 4.5: MDREM gains for experimentation.

| Gain | Equation | Value |
|---|---|---|
| $\lambda_\varphi$ | (2.13) | 1 |
| $a_2 = a_3 = a_4$ | (2.19) | 6 |
| $b_2, b_3, b_4$ | (2.19) | $\{1.2, 4.4, 4.8\}$ |
| $\phi_\mathrm{d}$ | (3.2) | 0.2 |
| $\eta_\mathrm{m}$ | (3.2) | $1 \times 10^{-9}$ |

Table 4.6: Gains of the adaptive law (3.10) for experimentation.

| Gain | Value |
|---|---|
| $\boldsymbol{\Gamma}$ | $\mathrm{diag}\ \{0.4,\ 0.6,\ 0.45,\ 0.2,\ 0.05,\ 0.1,\ 0.25,\ 0.05,\ 0.052,\ 0.13\}$ |
| $\boldsymbol{\lambda}_\theta$ | $[3 \quad 3 \quad 3 \quad 3 \quad 3 \quad 3 \quad 3 \quad 3 \quad 3 \quad 3\,]^\mathrm{T}$ |

In the simulation, the control law (3.50) explicitly uses the estimated parameters. In other words, the control law uses the estimated parameters that are being updated in "real time". In experimentation, since we are dealing with unknown parameters, doing so could be dangerous because there is no certainty of what the behavior of the robot will be. Therefore, in experimentation, the estimated parameters that the control law uses will not be updated in real time. Nevertheless, the control law still needs a set of acceptable parameters to operate. For this purpose, the parameters shown in Table 4.1 will be used for both scenarios, with and without payload. Thus, each experiment consists of two steps; first, a parameter estimation step will be carried out, and second, the estimated parameters will be validated. The gains for the control law are shown in Table 4.7 along with those for (3.46)-(3.47) necessary for its implementation.

The tuning process for the experimental control gains followed that of the simulation and the tuning rules mentioned above in Section 4.3. In addition, the trial-and-error approach was also used to finish the tuning.

Table 4.7: Gains of the control law for experimentation.

| Gain | Value |
|---|---|
| $\boldsymbol{\Lambda}$ | $\mathrm{diag}\ \{25, 30, 55\}$ |
| $\boldsymbol{K}_\mathrm{e}$ | $\mathrm{diag}\ \{0.008, 0.012, 0.01\}$ |
| $\boldsymbol{K}_\mathrm{v}$ | $\mathrm{diag}\ \{0.008, 0.006, 0.004\}$ |
| $\boldsymbol{K}_\mathrm{p}$ | $\mathrm{diag}\ \{0.044, 0.026, 0.014\}$ |
| $\boldsymbol{\lambda}_\mathrm{s}$ | $[2 \quad 2 \quad 2]^\mathrm{T}$ |
| $\boldsymbol{\lambda}_\mathrm{e}$ | $[2.813 \quad 2.813 \quad 2.813]^\mathrm{T}$ |

The sample time was set to the smallest allowed by the hardware, which was $T = 0.002$ seconds. The input torque $\boldsymbol{\tau}$ is estimated by the device based on motor current, which is subject to high-frequency electrical noise and friction in cable transmission. This noise is treated in the first step of the MDREM algorithm as described above by applying the filter of equation (2.14). With respect to the joint position $\boldsymbol{q}$, this is measured. Since the resolution of the Geomagic Touch is relatively high, this noise is negligible for position, but becomes significant when the joint velocity $\dot{\boldsymbol{q}}$ is calculated by numerical differentiation. Hence, it is processed using a filter of the same form as in (2.13) and (2.14). Joint acceleration is not needed by the MDREM algorithm nor by the control law.

### 4.5.1 First scenario: parameter estimation without payload

The results of the parameter estimation are shown in Figure 4.7 and summarized in Table 4.8. It can be seen that the adaptation process begins about second 1 and that the parameters $\theta_5$ and $\theta_{10}$ quickly converge to some value, while others take a little bit longer or just the increased excitation ends and stops the adaptation. The estimated parameters for which the adaptation simply stops before they converge to some value might not be correctly estimated. This could be due to poor tuning or a difficulty in handling the uncertainties in modeling due to the simplifications. The behavior of the square determinant of the matrices $\boldsymbol{Y}_\mathrm{f}$ and $\boldsymbol{Y}_\mathrm{m}$, that is, $\phi^2$ and $\phi_\mathrm{m}^2$ are shown in Figure 4.8. Here, it can be clearly seen that when $\phi^2$ reaches $\eta_\mathrm{m} = 1.0 \times 10^{-9}$, it increases to $\phi_\mathrm{d}^2 = 0.04$ and that this occurs between $t = 1$ s and $t = 1.26$ s. Therefore, condition (3.1) holds for about 0.26 seconds.

Table 4.8: Estimated parameters for the experiment without payload.

| **Parameter** | **Expression** | **Value** |
|---|---|---|
| $\theta_1$ | $m_2\ell_{c2}^2 + m_3a_2^2 + I_2$ | $0.0088$ kg · m$^2$ |
| $\theta_2$ | $m_3a_2\ell_{c3}$ | $0.033599$ kg · m$^2$ |
| $\theta_3$ | $m_3\ell_{c3}^2 + I_{yy3}$ | $0.020699$ kg · m$^2$ |
| $\theta_4$ | $I_{xx3}$ | $0.0052$ kg · m$^2$ |
| $\theta_5$ | $m_3\ell_{c3}^2 + I_{zz3}$ | $0.0031$ kg · m$^2$ |
| $\theta_6$ | $c_{f1}$ | $0.0192$ kg/s |
| $\theta_7$ | $c_{f2}$ | $0.065498$ kg/s |
| $\theta_8$ | $c_{f3}$ | $0.008898$ kg/s |
| $\theta_9$ | $m_2\ell_{c2} + m_3a_2$ | $0.00783$ kg · m |
| $\theta_{10}$ | $m_3\ell_{c3}$ | $0.00702$ kg · m |

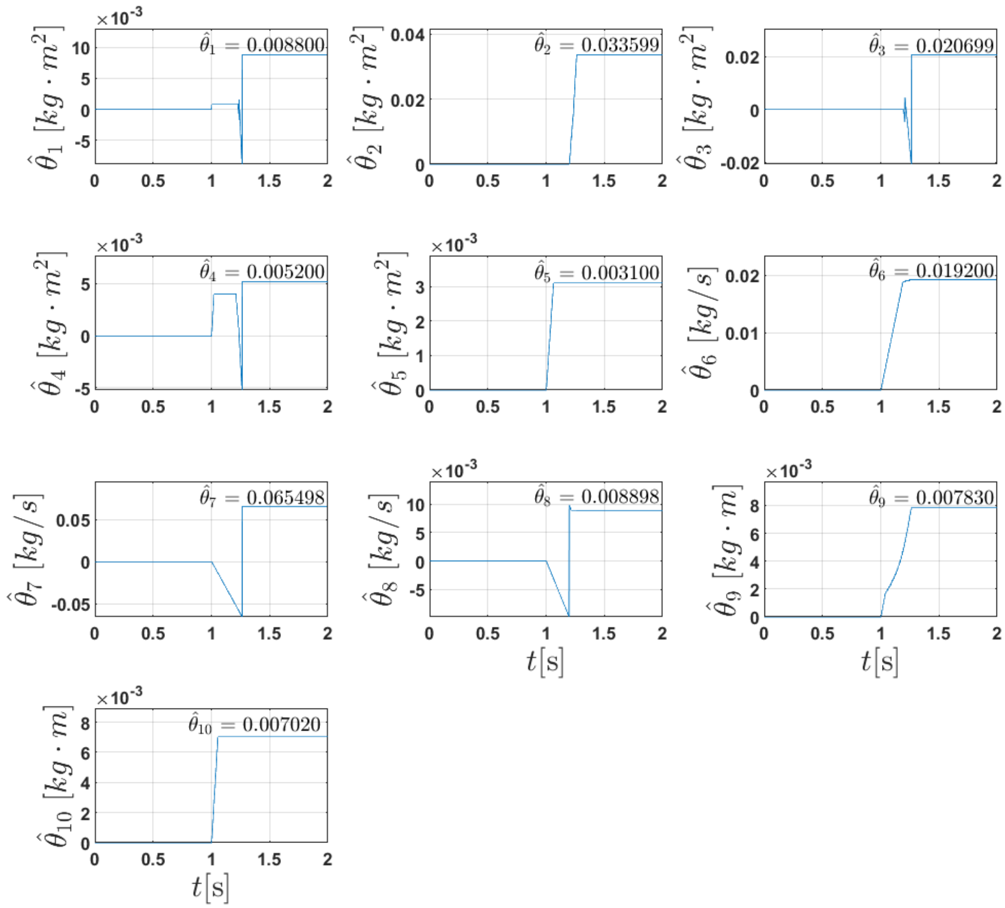


Figure 4.7: Estimated parameters for the experiment without payload.

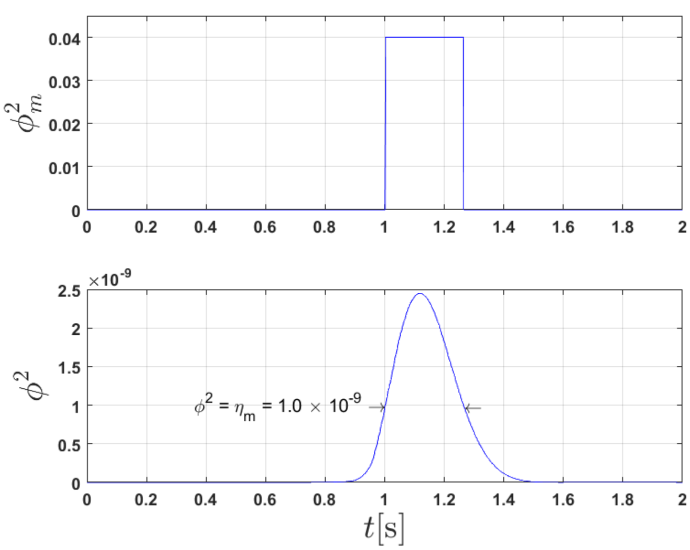


Figure 4.8: Experiment without payload. Square determinant $\phi_{\mathrm{m}}^2$ and $\phi^2$.

The estimated parameters shown in Table 4.8 should look similar to those of Table 4.1 since this experiment is carried out without payload. However, that is not the case for the estimated parameters $\hat{\theta}_1$ through $\hat{\theta}_8$, and as a result, this makes it difficult and not reliable to validate these estimated parameters. On the other hand, gravity compensation is performed to validate the parameters involved in gravity torques (3.43), which in this case are $\hat{\theta}_9$ and $\hat{\theta}_{10}$.

The joint positions for this experiment are shown in Figure 4.9. In the same run of the experiment, a validation procedure is performed. Hence, the first 2 seconds in Figure 4.9 are the MDREM algorithm phase, while the trajectory is executed in the first second. In second 2, the control law is switched to gravity compensation, and the validation phase begins. Right after second 2, it can be seen that the robot joints remain still. Then the operator manually moves the robot joints for a moment and stops the movement. It can be seen again that the joint positions remain fixed. Next, the operator repeats this process one more time to confirm that after stopping the movement, the joint positions remain where they should. Thus, the validation phase concludes by proving $\hat{\theta}_9$ and $\hat{\theta}_{10}$ to be good estimates.

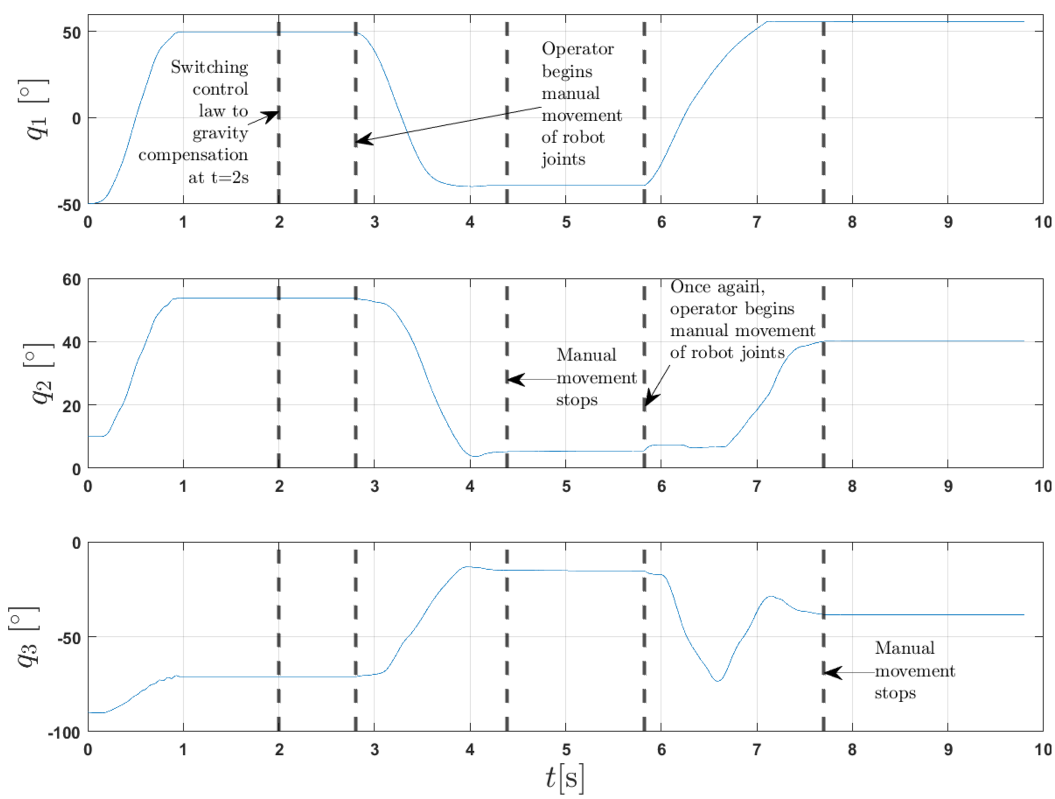


Figure 4.9: Experiment without payload. Joint positions.

### 4.5.2 Second scenario: parameter estimation with payload

The payload used in the experiment was a rectangular prism of dimensions $30 \times 30 \times 100$ mm. It was printed with a 3D printer, model ZORTRAX M200 PLUS. Figure 4.10 shows some pictures of the payload. In Figure 4.10a, it can be seen the actual printed payload. However, it can be seen that one face is not completely flat. It is curved to the top end of the face. This is due to an error at the end of the printing process. Regardless of this printing error, this piece was used in the experiment. The piece has a weight of 90 g. Figure 4.10b shows the adaptation made to the end of the third link to hold the payload. It was taken advantage of previous modifications made by former student fellows. The black piece is inserted into the plug connector on the basis of another similar piece. This piece also has a small rail on which the payload is mounted. Lastly, Figure 4.10c shows the payload mounted on the third link.

The results of the second experiment are shown in Figure 4.11 and the estimated parameters are put together in Table 4.9. It can be seen that they have an adaptation performance similar to that in the first experiment (see Figure 4.7). Also, with the exception of the parameters $\hat{\theta}_1$ and $\hat{\theta}_4$, an increase in the parameter values can be noticed; this is expected due to added payload.

Regardless of the estimation accuracy, the transient behavior of the parameters $\hat{\theta}_2$, $\hat{\theta}_5$, $\hat{\theta}_6$, $\hat{\theta}_9$, and $\hat{\theta}_{10}$ is straightforward, since they converge directly. Parameters $\hat{\theta}_5$ and $\hat{\theta}_{10}$ approach their final values rapidly. In contrast, $\hat{\theta}_2$, $\hat{\theta}_6$, and $\hat{\theta}_9$ adapt more slowly, appearing to reach their final values only when the high excitation of $\phi_\mathrm{m}$ ends; this suggests that the estimation was incomplete when the adaptation finished. Making gains $\gamma_2$, $\gamma_6$ and $\gamma_9$ larger could speed up convergence, but the likelihood of convergence to a relatively large value is present due to modeling uncertainties. In contrast, the remaining parameters behave differently. Although parameter values are known a priori to be positive, $\hat{\theta}_1$, $\hat{\theta}_3$, $\hat{\theta}_4$, $\hat{\theta}_7$, and $\hat{\theta}_8$ tend toward negative values. To partially address this, the algorithm enforces a constraint: if the error metric $e_{\theta i}$ in (3.12) falls below a threshold, the absolute value is applied to the estimated parameter. This ensures that when the estimation error becomes sufficiently small or when the $\phi_\mathrm{m}$ increased excitation phase ends, any negative estimate is rectified to a positive value. Consequently, $\hat{\theta}_1$ and $\hat{\theta}_4$ quickly converge to a positive value but then drift negative until the high-$\phi_\mathrm{m}$ phase ends, at which point they become positive. $\hat{\theta}_3$ tends to negative values repeatedly after every correction, until $\phi^2 < \eta_\mathrm{m}$ since applying the absolute value makes $e_{\theta i}$ large enough again to continue the

(a) Payload.

(b) End-efector adaptation for the payload.

(c) Mounted payload.

Figure 4.10: Payload used in experimentation.

adaptation. Finally, $\hat{\theta}_7$ and $\hat{\theta}_8$ drift negative without converging during the high-$\phi_{\mathrm{m}}$ phase. Increasing the adaptive gains $\gamma_7$ and $\gamma_8$ would likely induce behavior in $\hat{\theta}_7$ and $\hat{\theta}_8$ similar to that of $\hat{\theta}_3$. Generally, the drift toward negative values is likely caused by uncertainties from modeling simplifications.

In Figure 4.12 the performance of $\phi_{\mathrm{m}}^2$ and $\phi^2$ is shown. The behavior of $\phi_{\mathrm{m}}^2$ is very similar to that of the first experiment (see Figure 4.8). It increases at $t \approx 1$ s and decreases at $t = 1.28$ s, so condition (3.1) holds for about 0.28 seconds approximately.

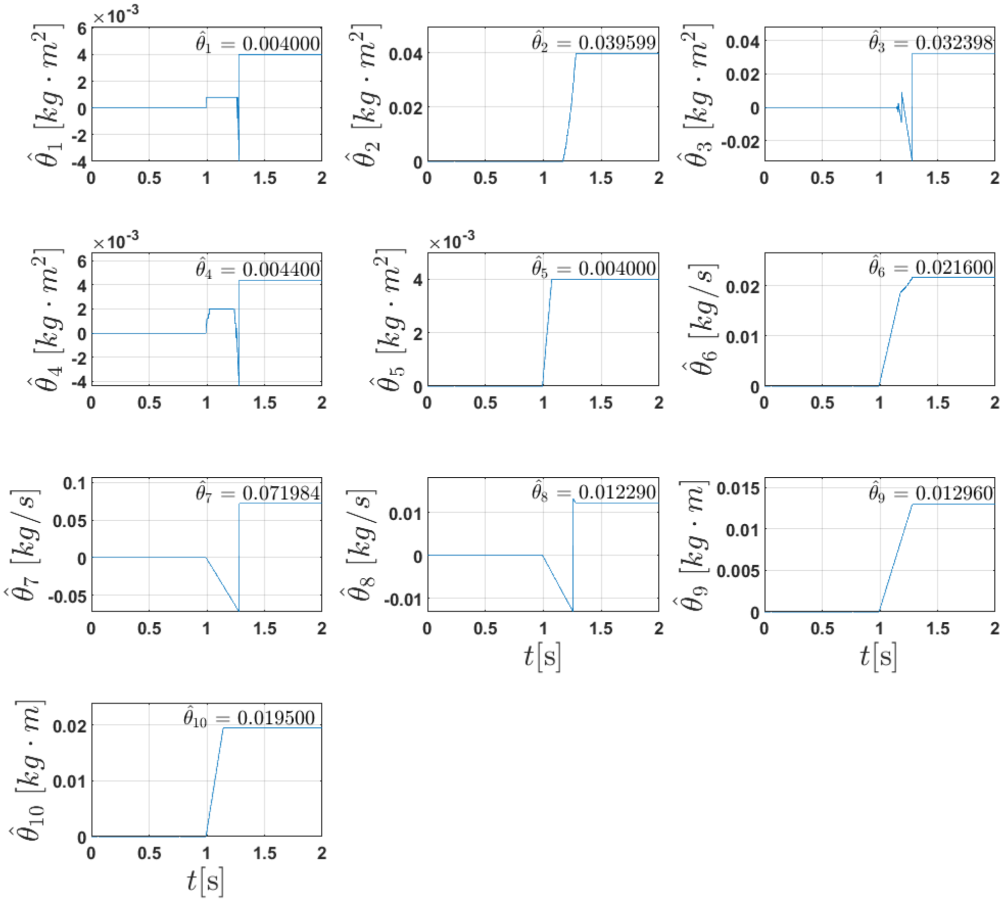


Figure 4.11: Estimated parameters for the experiment with payload.

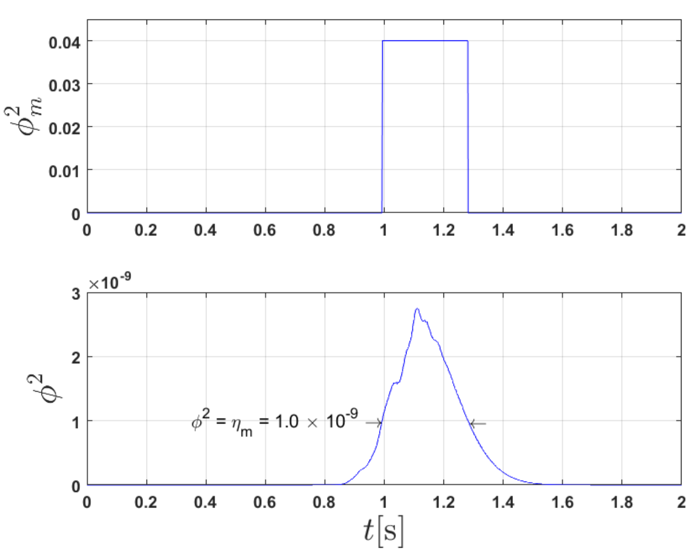


Figure 4.12: Experiment with payload. Square determinant $\phi_{\mathrm{m}}^2$ and $\phi^2$.

Table 4.9: Estimated parameters for the experiment with payload.

| Parameter | Expression | Value |
|---|---|---|
| $\theta_1$ | $m_2\ell_{c2}^2 + m_3a_2^2 + I_2$ | $0.004\ \mathrm{kg}\cdot\mathrm{m}^2$ |
| $\theta_2$ | $m_3a_2\ell_{c3}$ | $0.039599\ \mathrm{kg}\cdot\mathrm{m}^2$ |
| $\theta_3$ | $m_3\ell_{c3}^2 + I_{yy3}$ | $0.032398\ \mathrm{kg}\cdot\mathrm{m}^2$ |
| $\theta_4$ | $I_{xx3}$ | $0.0044\ \mathrm{kg}\cdot\mathrm{m}^2$ |
| $\theta_5$ | $m_3\ell_{c3}^2 + I_{zz3}$ | $0.004\ \mathrm{kg}\cdot\mathrm{m}^2$ |
| $\theta_6$ | $c_{f1}$ | 0.0216 kg/s |
| $\theta_7$ | $c_{f2}$ | 0.071984 kg/s |
| $\theta_8$ | $c_{f3}$ | 0.01229 kg/s |
| $\theta_9$ | $m_2\ell_{c2} + m_3a_2$ | $0.01296\ \mathrm{kg}\cdot\mathrm{m}$ |
| $\theta_{10}$ | $m_3\ell_{c3}$ | $0.0195\ \mathrm{kg}\cdot\mathrm{m}$ |

Recall that for its implementation, the control law (3.50) needs some parameters $\hat{\boldsymbol{\theta}}$. Updating these in real time, as in an adaptive control, could be dangerous in experimentation. Therefore, in both scenarios (without and with payload), the parameters used for the estimation phase were taken from Table 4.1. Joint positions are shown in Figure 4.13. In time, the figure is divided into two phases as in the first experiment. The first phase, the first 2 seconds, is a parameter estimation phase where the MDREM algorithm is applied. The second phase, from second 2 to the end, is a parameter validation phase. After the parameter estimation phase ends, the control law switches to gravity compensation using estimated parameters ($\hat{\theta}_9$ and $\hat{\theta}_{10}$) to validate them. Again, validation of the estimated parameters $\hat{\theta}_1$ through $\hat{\theta}_8$ was not possible due to the lack of reliable validation of these estimated parameters.

The trajectory for the estimation is executed from second 0 to 1 and it can be seen that the trajectories of joint $q_2$ and $q_3$ are a little irregular, which is expected due to the payload and that the control law uses an uncertain set of parameters for this scenario. However, thanks to the robustness of the control law, the joints reach the desired positions. Then, in second 2, the control law switches to gravity compensation, and the validation phase begins. In this phase, it is intended to validate the parameters involved with the gravity torques (3.43), these are $\hat{\theta}_9$ and $\hat{\theta}_{10}$. Right after Second 2, it can be seen that the joint positions remain motionless. After a short period of time, an operator manually moves the joint positions and then stops the movement; again, they remain fixed. This last process is repeated one more time and it is confirmed that the joint position remains unchanged, which concludes that $\hat{\theta}_9$ and $\hat{\theta}_{10}$ are good estimates.

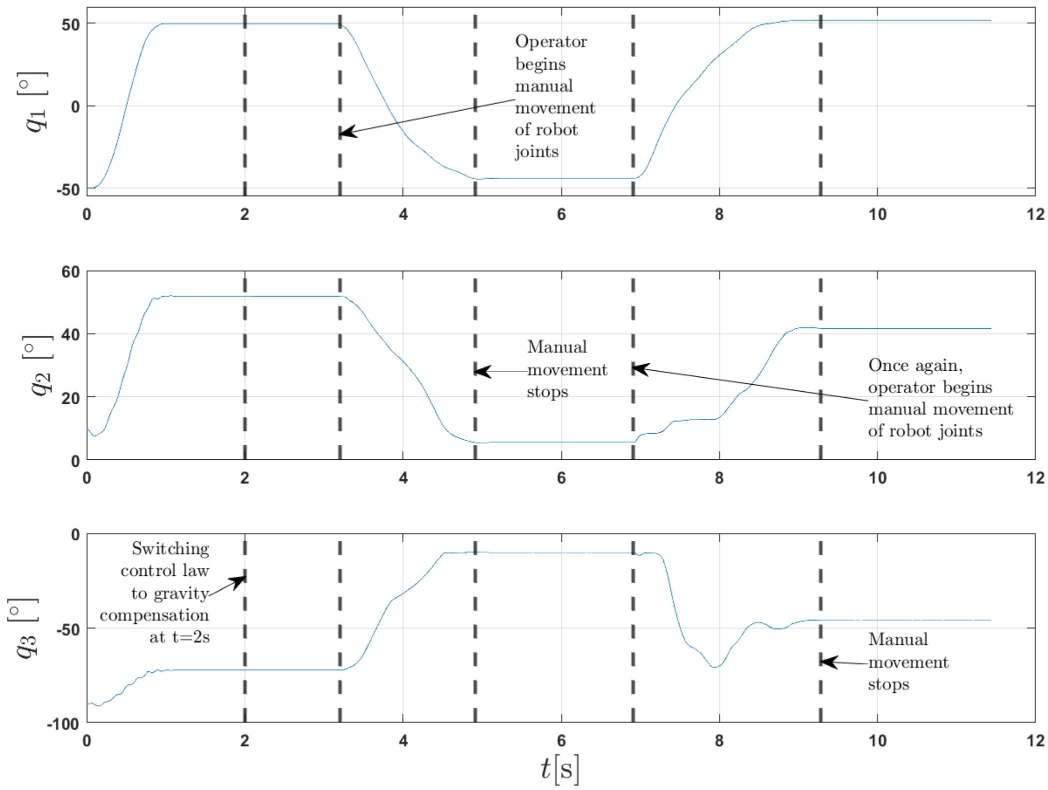


Figure 4.13: Experiment with payload. Joint positions.

A brief demonstration video of the previous experiments can be watched at the following YouTube link:

*https://youtube.com/shorts/MhCmMc3zWXQ?feature=share*

In summary, validation of the estimated parameters $\hat{\theta}_1$ through $\hat{\theta}_8$ was not possible due to a lack of time, a reliable estimate, and a suitable validation method. A common method is validation through the inverse dynamic model, where the input torques are calculated with the estimated parameters and compared with the measured ones. However, the assumptions made on the robot model make it difficult since they add uncertainty. The estimated parameters in the gravity torques remain relevant, since there are many medical applications where the robot manipulator needs to stay still and the physician/surgeon directly co-manipulate it. Lastly, it is worth noting that no external perturbations were added in the experiments, other than the inherent noise from the input torque (estimated from the motor current) and the joint velocity (obtained from numerical differentiation).

*Chapter 5*

# PAYLOAD PARAMETER ESTIMATION FOR THE FRANKA RESEARCH 3 (FR3) ROBOT

In this chapter, what was done during an internship at **École Centrale de Nantes** and the results obtained are reported. The MDREM algorithm was applied to the Franka Research 3 (FR3) robot (see Figure B.1) to estimate the inertial parameters of the payload. This robot is widely used in medical applications, where accurate inertial parameters of medical instruments (payload) are crucial. The contributions made in this chapter are based on part of the work reported in Guajardo Benavides (2025), so it is described as previous work.

Unlike in the previous chapter, where the entire set of robot parameters (including the robot's payload) was estimated, here only the inertial parameters of the payload are estimated. The set of inertial parameters for each robot link is assumed to be known. Furthermore, the only concern here is parameter estimation, so the control law is not involved. All experiments are carried out using one of the own robot controllers provided by the robot software. The one used is called "Joint position control", so the desired joint positions $\boldsymbol{q}_{\mathrm{d}}$ need to be given to the robot controller.

## 5.1 Preliminaries

The dynamic model of the FR3 robot is derived from the Newton-Euler (NE) formulation (see Appendix B for more details). Regardless of whether the Euler-Lagrange or Newton-Euler formulation is employed, recall that the dynamic model of a rigid-body n-DoF serial manipulator can be described as

$$\boldsymbol{\tau}_0 = \boldsymbol{H}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}), \tag{5.1}$$

Note that the matrix of viscous friction coefficients $\boldsymbol{D}$ was omitted on purpose because the FR3 robot model used does not take it into account. Again, because of Property 5, the right-hand side of the dynamic model (5.1) can be rewritten as

$$\boldsymbol{\tau}_0 = \boldsymbol{K}\boldsymbol{\theta}, \tag{5.2}$$

where, keeping in mind the NE formulation is used, $\boldsymbol{\tau}_0 \in \mathbb{R}^n$ is the input torque vector, $n$ is the number of DoF or generalized joint coordinates ($\boldsymbol{q}$); $\boldsymbol{K} \in \mathbb{R}^{n\times 10n}$ is the regressor matrix, and $\boldsymbol{\theta}^{10n}$ is the vector of constant parameters, which $10n$ means that there are 10 parameters for each link in this case. Note that $\boldsymbol{K} = \boldsymbol{K}(\ddot{\boldsymbol{q}}, \dot{\boldsymbol{q}}, \boldsymbol{q})$ to avoid writing the regressor arguments. The notation $\boldsymbol{K}$ has been used to distinguish it from the Lagrangian formulation regressor, and the abbreviation will be used hereafter.

If the robot payload is included, the dynamic model can be written as follows

$$\boldsymbol{\tau} = \boldsymbol{\tau}_0 + \boldsymbol{K}_{\mathrm{L}}\,\boldsymbol{\theta}_{\mathrm{L}} = \boldsymbol{K}\,\boldsymbol{\theta} + \boldsymbol{K}_{\mathrm{L}}\,\boldsymbol{\theta}_{\mathrm{L}} \tag{5.3}$$

where $\boldsymbol{\tau} \in \mathbb{R}^n$ is the input vector torque considering the payload, $\boldsymbol{K}_{\mathrm{L}} \in \mathbb{R}^{n\times 10}$ is the payload regressor matrix; $\boldsymbol{\theta}_{\mathrm{L}} \in \mathbb{R}^{10}$ is the payload vector of constant parameters. Again, note that $\boldsymbol{K}_{\mathrm{L}} = \boldsymbol{K}_{\mathrm{L}}(\ddot{\boldsymbol{q}}, \dot{\boldsymbol{q}}, \boldsymbol{q})$ has been used to avoid writing the regressor arguments.

### 5.1.1 Estimated inertial parameters

In the linear parameterization model (5.2) of the NE formulation, each link is associated with ten inertial parameters, as well as the payload. These ten parameters of each link or payload are represented by

$$\boldsymbol{\theta}_i = [m_i,\ c_{x_i}m_i,\ c_{y_i}m_i,\ c_{z_i}m_i,\ J_{xx_i},\ J_{xy_i},\ J_{xz_i},\ J_{yy_i},\ J_{yz_i},\ J_{zz_i}]^{\mathrm{T}}, \tag{5.4}$$

where $\boldsymbol{\theta}_i$ is the set of parameters of the $i$th link for $i = 1, \ldots, 7$ or $i = \mathrm{L}$ for the payload; $m_i$ is the mass, $c_{x_i}m_i$, $c_{y_i}m_i$ and $c_{z_i}m_i$ are mass moments, and $J_{xx_i}$, $J_{xy_i}$, $J_{xz_i}$, $J_{yy_i}$, $J_{yz_i}$, $J_{zz_i}$ are the elements of the inertia tensor with respect to the frame of the link $i$ for $i = 1, \ldots, 7$. The inertial parameters of the payload are estimated independently (not as part of link 7) but with respect to the frame of the $7^{th}$ link. Usually, the inertial parameters (5.4) are presented differently. Instead of the mass moment $c_{x_i}m_i$, $c_{y_i}m_i$, $c_{z_i}m_i$, the center of mass $\boldsymbol{c}_i = \{c_{x_i},\ c_{y_i},\ c_{z_i}\}$ is preferable, which is obtained by simply dividing by the estimated mass $m_i$. In addition, the inertia tensor with respect to the center of mass is preferred, which can be obtained by applying Steiner's theorem expressed as

$$\mathbf{I}_{Gi} = \mathbf{J}_i - m_i\left(\mathbf{c}_i^{\mathrm{T}}\mathbf{c}_i\mathbf{I} - \mathbf{c}_i\mathbf{c}_i^{\mathrm{T}}\right), \tag{5.5}$$

where $\mathbf{I}_{Gi}$ is the inertia tensor expressed at the center of mass of the link ( or payload), $\mathbf{J}_i$ is the inertia tensor expressed in the frame of the $i$th link, $\boldsymbol{c}_i$ is the vector from the origin of the frame of the $i$th link to the center of mass of the link (or payload), and $I$ is the identity matrix. Then

$$\boldsymbol{J}_i = \begin{bmatrix} J_{xx_i} & J_{xy_i} & J_{xz_i} \\ J_{xy_i} & J_{yy_i} & J_{yz_i} \\ J_{xz_i} & J_{yz_i} & J_{zz_i} \end{bmatrix}; \qquad \boldsymbol{I}_{Gi} = \begin{bmatrix} I_{xx_i} & I_{xy_i} & I_{xz_i} \\ I_{xy_i} & I_{yy_i} & I_{yz_i} \\ I_{xz_i} & I_{yz_i} & I_{zz_i} \end{bmatrix}$$

$$\boldsymbol{c}_i = [c_{x_i} \quad c_{x_i} \quad c_{x_i}]^{\mathrm{T}},$$

and the new set of parameters is expressed as

$$\overline{\boldsymbol{\theta}}_i = [m_i,\ c_{x_i},\ c_{y_i},\ c_{z_i},\ I_{xx_i},\ I_{xy_i},\ I_{xz_i},\ I_{yy_i},\ I_{yz_i},\ I_{zz_i}]^{\mathrm{T}}. \tag{5.6}$$

## 5.2 Previous work

Since the purpose is to estimate only the payload inertial parameters, equation (5.3) can be expressed as

$$\boldsymbol{u} = \boldsymbol{\tau} - \boldsymbol{\tau}_0 = \boldsymbol{K}_{\mathrm{L}}\,\boldsymbol{\theta}_{\mathrm{L}} \tag{5.7a}$$

$$= \boldsymbol{\tau} - \boldsymbol{K}\,\boldsymbol{\theta} = \boldsymbol{K}_{\mathrm{L}}\,\boldsymbol{\theta}_{\mathrm{L}}, \tag{5.7b}$$

where $\boldsymbol{u} \in \mathbb{R}^n$ is a new input torque vector whose effect corresponds to the payload. Typically, the torque $\boldsymbol{\tau}$ or $\boldsymbol{\tau}_0$ is measured. Therefore, using (5.7a) would imply measuring the input torque vector twice, with and without payload, and the variations in the trajectory would be neglected. An important assumption made is that the robot parameters are known, so (5.7b) is the one to which the MDREM algorithm is applied.

Unlike the Lagrangian dynamic model, where closed-form symbolic equations are available, the Newton-Euler model is usually obtained through a recursive numeric algorithm. This means that the filter (2.13) cannot be applied to the regressor $\boldsymbol{K}_{\mathrm{L}}$ to avoid using joint acceleration. For this reason, the first step is to compute an extended regressor $\boldsymbol{K}_{\mathrm{Lf}} \in \mathbb{R}^{10\times 10}$ and an extended input torque vector $\boldsymbol{u}_{\mathbf{f}} \in \mathbb{R}^{10}$ using $\boldsymbol{K}_{\mathrm{L}}$ and $\boldsymbol{u}$. These are given by

$$\boldsymbol{K}_{\mathrm{Lf}} = \begin{bmatrix} \boldsymbol{K}_{\mathrm{L}1} \\ \vdots \\ \boldsymbol{K}_{\mathrm{L}\tilde{n}} \end{bmatrix} \in \mathbb{R}^{10\times 10}, \qquad \boldsymbol{u}_{\mathrm{f}} = \begin{bmatrix} \boldsymbol{u}_1 \\ \vdots \\ \boldsymbol{u}_{\tilde{n}} \end{bmatrix} \in \mathbb{R}^{10}, \tag{5.8}$$

where $\boldsymbol{K}_{\mathrm{L}1} = \boldsymbol{K}_{\mathrm{L}}$ and $\boldsymbol{u}_1 = \boldsymbol{u}$, so the first $n$ rows of $\boldsymbol{K}_{\mathrm{Lf}}$ are the $n$ rows of the regressor $\boldsymbol{K}_{\mathrm{L}}$ and the first $n$ elements of $\boldsymbol{u}_{\mathrm{f}}$ are the $n$ elements of $\boldsymbol{u}$. The remaining elements of $\boldsymbol{K}_{\mathrm{Lf}}$ and $\boldsymbol{u}_{\mathrm{f}}$ are computed as usual following the procedure described in Section 2.4 by applying the $(\tilde{n}-1)$ linear $\mathcal{L}_\infty$ stable operators $G_j : \mathcal{L}_\infty \rightarrow \mathcal{L}_\infty$.

### 5.2.1 Singular Value Decomposition (SVD)

Later, it was found that no matter what happened, the determinant of $\boldsymbol{K}_{\mathrm{Lf}}$ never became different than zero. That is, $\phi \neq 0$ was never satisfied. The reason for this issue was that $\boldsymbol{K}_{\mathrm{L}}$ showed a liner dependency. Since it is a 7-by-10 matrix, it was expected that it had $rank = 7$, but it was indeed $rank = 6$. So, $\boldsymbol{K}_{\mathrm{Lf}}$ would always have a linear dependency as well. Hence, this problem was approached using a singular value decomposition (SVD) on $\boldsymbol{K}_{\mathrm{L}}$, which means that (5.7b) can be rewritten as

$$\begin{aligned} \boldsymbol{u} &= \boldsymbol{K}_{\mathrm{L}}\,\boldsymbol{\theta}_{\mathrm{L}} \\ &= \boldsymbol{U}\boldsymbol{S}\boldsymbol{V}^{\mathrm{T}}\boldsymbol{\theta}_{\mathrm{L}}, \end{aligned} \tag{5.9}$$

where the columns of $\boldsymbol{U} \in \mathbb{R}^{n\times n}$ and $\boldsymbol{V} \in \mathbb{R}^{10\times 10}$ are called left and right singular vectors, respectively; $\boldsymbol{S} \in \mathbb{R}^{n\times 10}$ contains the singular values on the diagonal and zeros off the diagonal. For this case $n < 10$, the matrix $\boldsymbol{S}$ has a maximum of $n$ non-zero singular values on the diagonal. In addition, $\boldsymbol{U}$ and $\boldsymbol{V}$ are unitary matrices, which means that $UU^{\mathrm{T}} = U^{\mathrm{T}}U = I$. It is possible to obtain an exact representation of the matrix $\boldsymbol{K}_{\mathrm{L}}$ by only taking the square block of the matrix $\boldsymbol{S}$ that contains the $r$ non-zero singular values and, in turn, taking only the first $r$ columns of $\boldsymbol{U}$ and the first $r$ rows of $\boldsymbol{V}^{\mathrm{T}}$. This is known as economy SVD and can be expressed as

$$\boldsymbol{u} = \boldsymbol{U}\boldsymbol{S}\boldsymbol{V}^{\mathrm{T}}\boldsymbol{\theta}_{\mathrm{L}} \tag{5.10a}$$

$$= \underbrace{\begin{bmatrix} \boldsymbol{U}_1 & \boldsymbol{U}_2 \end{bmatrix}}_{\boldsymbol{U}} \underbrace{\begin{bmatrix} \boldsymbol{\Sigma} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} \end{bmatrix}}_{\boldsymbol{S}} \underbrace{\begin{bmatrix} \boldsymbol{V}_1^{\mathrm{T}} \\ \boldsymbol{V}_2^{\mathrm{T}} \end{bmatrix}}_{\boldsymbol{V}^{\mathrm{T}}} \boldsymbol{\theta}_{\mathrm{L}} \tag{5.10b}$$

$$= \boldsymbol{U}_1\boldsymbol{\Sigma}\boldsymbol{V}_1^{\mathrm{T}}, \tag{5.10c}$$

where $\boldsymbol{U}_1 \in \mathbb{R}^{n\times r}$, $\Sigma \in \mathbb{R}^{r\times r}$ and $\boldsymbol{V}_1^{\mathrm{T}} \in \mathbb{R}^{r\times 10}$. For this particular case, it was seen that even though $\boldsymbol{S}$ could have at most 7 non-zero singular values, it actually had 6, which also explains that $\boldsymbol{K}_{\mathrm{L}}$ had $rank = 6$. It is worth mentioning that $\boldsymbol{U}_1^{\mathrm{T}}\boldsymbol{U}_1 = I$ and $\boldsymbol{V}_1^{\mathrm{T}}\boldsymbol{V}_1 = I$ while $\boldsymbol{U}_1\boldsymbol{U}_1^{\mathrm{T}} \neq I$ and $\boldsymbol{V}_1\boldsymbol{V}_1^{\mathrm{T}} \neq I$.

Next, both sides of (5.10c) are multiplied on the left by $\boldsymbol{U}_1^{\mathrm{T}}$ so that

$$\begin{aligned} \boldsymbol{U}_1^{\mathrm{T}}\boldsymbol{u} &= \boldsymbol{U}_1^{\mathrm{T}}\boldsymbol{U}_1\boldsymbol{\Sigma}\boldsymbol{V}_1^{\mathrm{T}}\boldsymbol{\theta}_{\mathrm{L}} && (5.11a)\\ &= \boldsymbol{\Sigma}\boldsymbol{V}_1^{\mathrm{T}}\boldsymbol{\theta}_{\mathrm{L}} && (5.11b)\\ \boldsymbol{\delta} &= \boldsymbol{\Psi}\boldsymbol{\theta}_{\mathrm{L}}, && (5.11c) \end{aligned}$$

where $\boldsymbol{U}_1^{\mathrm{T}}\boldsymbol{U}_1 = I$ has been used; $\boldsymbol{\delta} \in \mathbb{R}^{r}$ and $\boldsymbol{\Psi} \in \mathbb{R}^{r\times 10}$ are the input vector and regressor matrix of the new system, respectively, defined by

$$\boldsymbol{\delta} = \boldsymbol{U}_1^{\mathrm{T}}\boldsymbol{u} \tag{5.12}$$

and

$$\boldsymbol{\Psi} = \boldsymbol{\Sigma}\boldsymbol{V}_1^{\mathrm{T}}. \tag{5.13}$$

Now, from $\boldsymbol{\Psi}$ and $\boldsymbol{\delta}$ an extended regressor $\boldsymbol{\Psi}_{\mathrm{f}} \in \mathbb{R}^{10\times 10}$ and an extended input vector $\boldsymbol{\delta}_{\mathrm{f}} \in \mathbb{R}^{10}$ are computed exactly the same way as in (5.8). Thus, these are given by

$$\boldsymbol{\Psi}_{\mathrm{f}} = \begin{bmatrix} \boldsymbol{\Psi}_1 \\ \vdots \\ \boldsymbol{\Psi}_{\tilde{n}} \end{bmatrix} \in \mathbb{R}^{10\times 10}, \qquad \boldsymbol{\delta}_{\mathrm{f}} = \begin{bmatrix} \boldsymbol{\delta}_1 \\ \vdots \\ \boldsymbol{\delta}_{\tilde{n}} \end{bmatrix} \in \mathbb{R}^{10}, \tag{5.14}$$

where $\boldsymbol{\Psi}_1 = \boldsymbol{\Psi}$ and $\boldsymbol{\delta}_1 = \boldsymbol{\delta}$, so the first $r$ rows of $\boldsymbol{\Psi}_{\mathrm{f}}$ are the $r$ rows of the regressor $\boldsymbol{\Psi}$ and the first $r$ elements of $\boldsymbol{\delta}_{\mathrm{f}}$ are the $r$ elements of the input vector $\boldsymbol{\delta}$.

## 5.3 Contributions

The approach described previously relies on the precision of $\boldsymbol{u} = \boldsymbol{\tau} - \boldsymbol{K}\boldsymbol{\theta}$ as in (5.7b), which in turn relies on the accuracy of the set of robot parameters $\boldsymbol{\theta}$. For this reason, it is necessary to verify that the relationship (5.2) is valid in practice, which is $\boldsymbol{\tau}_0 = \boldsymbol{K}\boldsymbol{\theta}$. To do this, two things need to be reviewed. First, the dynamic model (5.1) does not consider any friction torque, so it is necessary to include a friction model. Second, there is no certainty of the accuracy of the available robot parameters $\boldsymbol{\theta}$. The estimation of these parameters is not within the scope of this work, and therefore research in the literature and comparison of them are performed to try to obtain a better set than the currently available one.

Before continuing, let us clarify why it is necessary to include friction torques. By including friction torques, we do not pretend to estimate the parameters related to a certain friction model but to obtain a more accurate torque vector $\boldsymbol{u}$. In equation (5.7b), that is,

$$\boldsymbol{u} = \boldsymbol{\tau} - \boldsymbol{K}\,\boldsymbol{\theta} = \boldsymbol{K}_{\mathrm{L}}\,\boldsymbol{\theta}_{\mathrm{L}},$$

$\boldsymbol{\tau}$ is measured and is subject to friction effects, while the second term $\boldsymbol{K}\,\boldsymbol{\theta}$ as in (5.2) is not measured and does not consider the effect of friction torque. Thus, what this would really yield is

$$\begin{aligned} \boldsymbol{u} &= \boldsymbol{\tau} - \boldsymbol{K}\,\boldsymbol{\theta} = \boldsymbol{K}_{\mathrm{L}}\,\boldsymbol{\theta}_{\mathrm{L}} + \boldsymbol{\tau}_{\boldsymbol{friction}} \\ \Rightarrow \boldsymbol{u} &= \boldsymbol{\tau} - \boldsymbol{K}\,\boldsymbol{\theta} - \boldsymbol{\tau}_{\boldsymbol{friction}} = \boldsymbol{K}_{\mathrm{L}}\,\boldsymbol{\theta}_{\mathrm{L}}, \end{aligned}$$

so, in order to obtain an input vector $\boldsymbol{u}$ that corresponds only to the effects of the payload, it is necessary to subtract the friction torque or otherwise include them in the robot model (5.1)-(5.2). Therefore, if the friction torque is included in the dynamic model, the following is obtained

$$\boldsymbol{\tau}_0 = \boldsymbol{H}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) + \boldsymbol{\tau}_{\boldsymbol{f}} = \boldsymbol{K}\boldsymbol{\theta} + \boldsymbol{\tau}_{\boldsymbol{f}}, \tag{5.15}$$

where $\boldsymbol{\tau}_{\boldsymbol{f}} \in \mathbb{R}^n$ is the friction torque vector.

As part of the inclusion of a friction model, three models are considered and then compared with their corresponding values (Soto et al. 2024). So, the three friction models presented here are the Lugre, Stribeck and Sigmoidal models.

The **Lugre** model is given by

$$\tau_{f,i} = \sigma_{0,i}\,\mathrm{sgn}\left(\dot{q}_i - v_{0,i}z_i\right) + \sigma_{2,i}\dot{q}_i S_i + \sigma_{3,i}\dot{q}_i\left(1 - S_i\right) \tag{5.16}$$

with $S_i = 1 - e^{-\frac{\left|\dot{q}_i - v_{1,i}\right|}{v_{2,i}}}$; and $z_i(t) = z_i(t-1) + \Delta t\left(\dot{q}_i(t) - \frac{\sigma_{0,i}}{\sigma_{1,i}} z_i(t-1)\right),$

where the subindex $i$ indicates the variable or parameter corresponding to the $i$-th joint for $i = 1, \dots, 7$; $\tau_{f,i}$ is the friction torque, $\sigma_{2,i}$ is the viscous friction force, $\sigma_{3,i}$ is the Stribeck effect, $v_{0,i}$ is the presiding displacement threshold, $z_i$ is the internal state variable of the friction model, $S_i$ is the Stribeck function, and $\Delta t$ is the time step.

The **Stribeck** model is provided:

$$\begin{aligned} \text{Stribeck term } &= F_s \cdot \exp\left(-\frac{|\dot{q}|}{v_s} \cdot k_s\right), \\ \tau_f &= \operatorname{sgn}(\dot{q}) \cdot (F_c + \text{ Striebeckterm }) + \dot{q} \cdot F_v, \end{aligned} \tag{5.17}$$

where $\tau_f$ is the friction torque, $F_c$ is the Coulomb friction coefficient, $F_v$ is the viscous friction coefficient, $v_s$ indicates the Stribeck velocity, $k_s$ is used to adjust the sensitivity of the stiction effect.

The **sigmoidal** model is given by:

$$\tau_{f,i}\left(\dot{q}_i\right) = \frac{\psi_{1,i}}{1 + e^{-\psi_{2,i}\left(\dot{q}_i + \psi_{3,i}\right)}} - \frac{\psi_{1,i}}{1 + e^{-\psi_{2,i}\psi_{3,i}}}, \tag{5.18}$$

where the subindex $i$ indicates the variable or parameter corresponding to the $i$-th joint for $i = 1, \dots, 7$; $\tau_{f,i}$ is the friction torque, $\psi_{1,i}$, $\psi_{2,i}$, and $\psi_{3,i}$ are the friction parameters.

The friction model parameters for the three models that will be compared were found in Soto et al. (2024), and are presented in Tables D.1 - D.3. In addition, another set of parameters for the Sigmoidal friction model (5.18) from Gaz et al. (2019) is also considered in the comparison and shown in Table D.4. As a starting point, the set of available robot parameters $\boldsymbol{\theta}$ is reported in the same paper mentioned above Gaz et al. (2019).

To determine the better friction model to use, the comparison will be performed using the estimated joint torque without any friction model (5.2), and the estimated joint torque for the different friction models using (5.15) to compare them with the measured torque. The experimental trajectory used is shown in Figure 5.1. It was generated using MoveIt, which is a motion planning, manipulation, and kinematics framework for the Robot Operating System (ROS). The initial positions are $\boldsymbol{q}(t_0) = [0^\circ \ -45^\circ \ 0^\circ \ -135^\circ \ 45^\circ \ 45^\circ \ 45^\circ]^{\mathrm{T}}$ and the desired final positions are $\boldsymbol{q}_{\mathrm{d}} = [0^\circ \ -45^\circ \ 0^\circ \ -135^\circ \ -45^\circ \ 90^\circ \ -45^\circ]^{\mathrm{T}}$. The trajectory was executed in approximately 1.4 seconds. It can be seen that theoretically the first 4 joints are kept fixed, while only the last 3 joints are moved. Nonetheless, the first 4 joints move slightly.

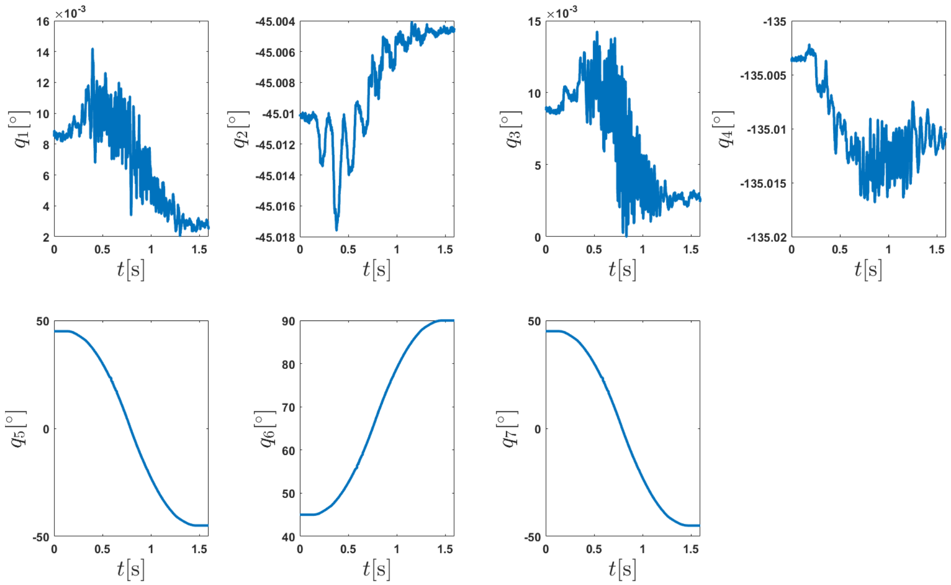


Figure 5.1: Experimental trajectory for the Franka Research 3 (FR3) robot.

In Figure 5.2 the joint torque comparison is presented. The figure shows the measured joint torque filtered using Matlab's functions *designfilt* and *filtfilt*. For the case of the Lugre friction model, (5.16) and the data from Table D.1 was used ; for the Stribeck model friction case, (5.17) and Table D.2; for the Sigmoidal 1 friction model case, (5.18) and Table D.3; for the Sigmoidal 2 friction model case, (5.18) and Table D.4. It can be seen in the figure that all the estimated torques of the first four joints are the same as they remain static and are not affected by motion friction. In the remaining three joints, it can be seen that by including the friction models, do not all the estimated joint torques necessarily get closer to the measured one, compared to the one without the friction model, except in joint 6.

To evaluate which friction model and its corresponding parameter values are better, the Root Mean Square Error (RMSE) (5.19) will be used as the extent of performance. Table 5.1 show the RMSE values of the estimated joint torque for the case without any friction model and the cases with the models presented above. It can be seen that the smallest RMSE values are from the sigmoidal friction model using the parameters in Table D.4 of Gaz et al. (2019). For this reason, this friction model and values will be used hereafter and next, a better set of robot inertial parameters $\boldsymbol{\theta}$ will be sought.

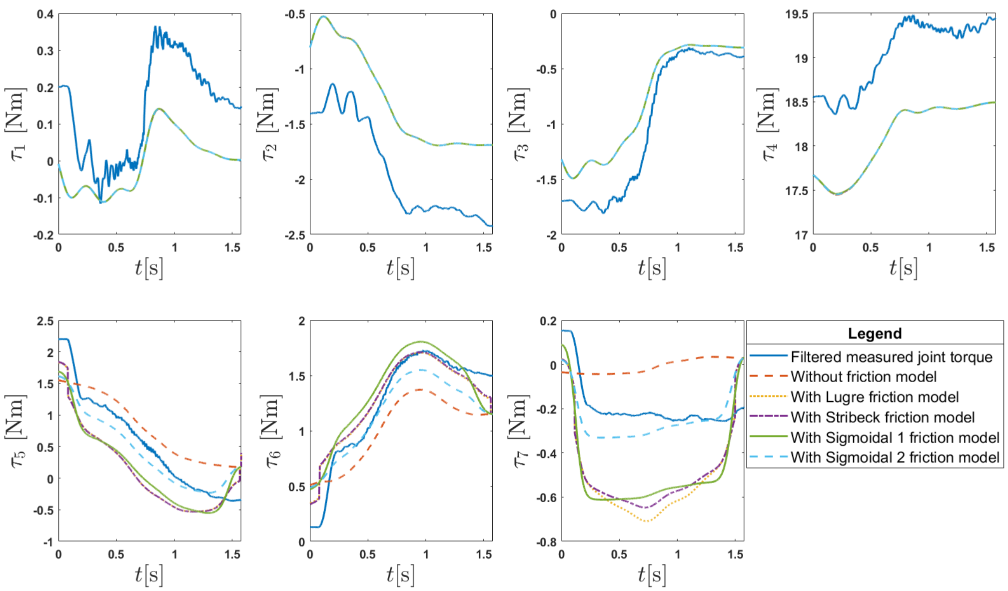


Figure 5.2: Comparison of joint torques for different friction models and values.

Table 5.1: Root Mean Square Error (RMSE) of the estimated joint torques for different friction models and values.

| **Joint** | **Without model** | **Lugre Model** | **Stribeck model** | **Sigmoidal model (1)** | **Sigmoidal model (2)** |
|---|---|---|---|---|---|
| **1** | 0.1627 | 0.1626 | 0.1626 | 0.1625 | 0.1626 |
| **2** | 0.6339 | 0.6336 | 0.6336 | 0.6335 | 0.6337 |
| **3** | 0.2808 | 0.2808 | 0.2808 | 0.2807 | 0.2808 |
| **4** | 0.9409 | 0.9403 | 0.9403 | 0.9396 | 0.9403 |
| **5** | 0.3783 | 0.5103 | 0.5117 | 0.4855 | 0.2847 |
| **6** | 0.3124 | 0.1543 | 0.1551 | 0.2003 | 0.1973 |
| **7** | 0.2219 | 0.3528 | 0.3236 | 0.3224 | 0.1084 |
| **Average RMSE** | 0.4187 | 0.4335 | 0.4297 | 0.4321 | 0.3725 |

$$RMSE = \sqrt{\frac{\sum_{i=1}^{N} (\text{ Estimated }_i - \text{ Measured }_i)^2}{N}} \tag{5.19}$$

Now, we will consider a few sets of inertial parameters of the robot to compare. It was already mentioned that the set reported in Gaz et al. (2019) had been used and is shown in Table D.5. Another set of parameters that will be studied is taken from the Github of the robot developer (Franka Robotics 2024) and presented in Table D.6. In the paper mentioned above Soto et al. (2024), along with the friction model parameters, a few set of robot parameters are reported; while all of them were examined, only the one with the best performance is compared here and is shown in Table D.7.

The estimated joint torque using the three different sets of inertial parameters and the measured filtered joint torque are shown in Figure 5.3. It is hard to tell which set has better results just by looking at the charts in the figure. Therefore, RMSE (5.19) is used again to evaluate which yields better performance. It is calculated for each set and joint and the results are presented in Table 5.2. In general, the set of inertial parameters from Franka Robotics (2024) has the smallest average RMSE. On the other hand, the parameters of Soto et al. (2024) have better results for individual joints. The reason for this discrepancy is due to the values of joint 4 where the latter has a relatively much higher value. Consequently, it was decided to continue the present work using the set of inertial parameters of Soto et al. (2024) shown in Table D.7.

It is necessary to make a distinction between the Franka Research 3 (FR3) robot (the one used in this work) and the Franka Emika Panda robot. Although they are structurally the same robot, some of their features vary. This distinction is made in order to note that the inertial parameters reported in Franka Robotics (2024) are for the FR3 robot, while those reported in Gaz et al. (2019) and Soto et al. (2024) are for the Franka Emika Panda robot; along with their corresponding friction model parameters.

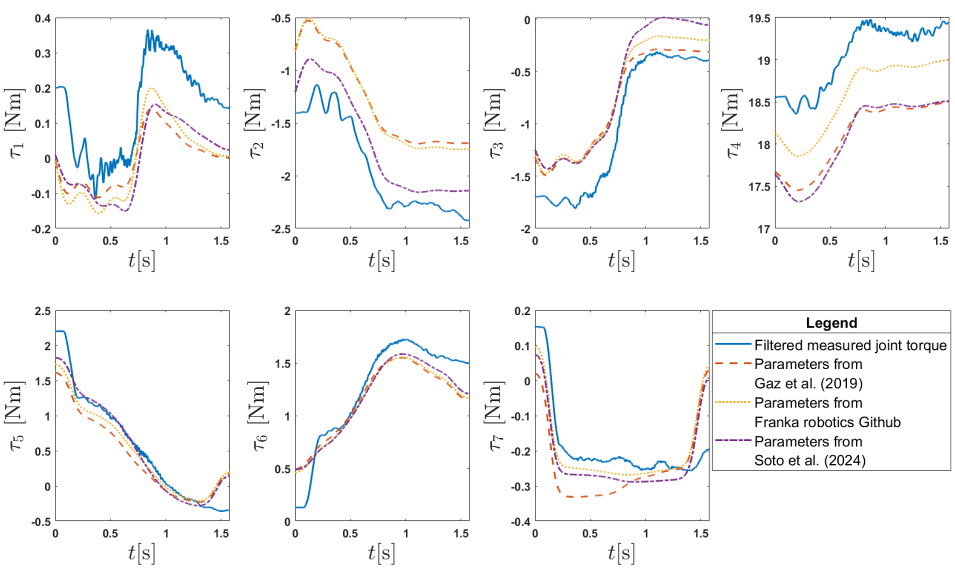


Figure 5.3: Joint torques for different sets of inertial parameters of the robot.

Table 5.2: Root Mean Square Error (RMSE) of the estimated joint torques for different set of inertial parameters of the robot.

| **Joint** | **Parameters from** Gaz et al. (2019) | **Parameters from** Franka Robotics (2024) | **Parameters from** Soto et al. (2024) |
|---|---|---|---|
| **1** | 0.1626 | 0.1572 | 0.1516 |
| **2** | 0.6337 | 0.6165 | 0.2453 |
| **3** | 0.2808 | 0.3086 | 0.3602 |
| **4** | 0.9403 | 0.4807 | 0.9741 |
| **5** | 0.2847 | 0.2266 | 0.1811 |
| **6** | 0.1973 | 0.1879 | 0.1717 |
| **7** | 0.1084 | 0.0731 | 0.0764 |
| **Average RMSE** | 0.3725 | 0.2930 | 0.3086 |

### 5.3.1 Filtering

It was mentioned earlier that the first step of the MDREM algorithm had been skipped in this chapter, that is, applying the filters (2.13) and (2.14) to the regressor $\boldsymbol{K}$ (and $\boldsymbol{K}_{\mathrm{L}}$) and the input vector $\boldsymbol{\tau}$, respectively. Leaving aside the goal of this of avoiding the joint acceleration (which in this case does not work), in simulation the result might not have any drawback. However, in practice where there are noisy sensors, not implementing these filters may have a negative effect. In Figure 5.4, it can be seen that the measured joint torque is subject to noise, while the implementation of the filter (2.14) with $\lambda = 25$ produces a cleaner signal. For that reason, the implementation of these filters is retaken, and thus from (5.7a), and taking into account the friction torque on the right-hand side of (5.15), this results in

$$\boldsymbol{u}_{\varphi} = \boldsymbol{\tau}_{\varphi} - (\boldsymbol{K}_{\varphi}\boldsymbol{\theta} + \boldsymbol{\tau}_{f\varphi}) = \boldsymbol{K}_{\mathrm{L}\varphi}\boldsymbol{\theta}_{\mathrm{L}}, \tag{5.20}$$

where $\boldsymbol{u}_{\varphi}$ is defined as

$$\boldsymbol{u}_{\varphi} = \boldsymbol{\tau}_{\varphi} - (\boldsymbol{K}_{\varphi}\boldsymbol{\theta} + \boldsymbol{\tau}_{f\varphi}), \tag{5.21}$$

$\boldsymbol{\tau}_{\varphi}$ and $\boldsymbol{\tau}_{f\varphi}$ are the filtered input vector and the filtered friction torque vector, respectively, obtained by applying the filter (2.14) to $\boldsymbol{\tau}$ and $\boldsymbol{\tau}_f$ ; $\boldsymbol{K}_{\varphi}$ and $\boldsymbol{K}_{\mathrm{L}\varphi}$ are the filtered robot and payload regressors, respectively, obtained by applying (2.13) to $\boldsymbol{K}$ and $\boldsymbol{K}_{\mathrm{L}}$.

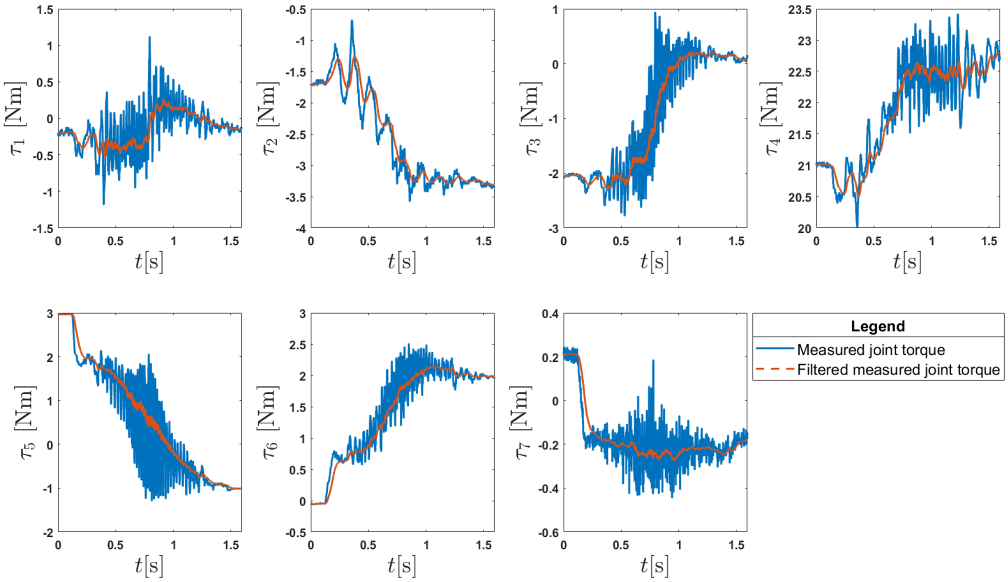


Figure 5.4: Raw measured joint torque vs filtered measured joint torque.

From this, it was surprising to find that the problem encountered earlier about the rank deficiency of the payload regressor $\boldsymbol{K}_{\mathrm{L}}$ disappeared. In other words, after filtering $\boldsymbol{K}_{\mathrm{L}}$, a full row-rank regressor $\boldsymbol{K}_{\mathrm{L}\varphi}$ is obtained. In Figure 5.5, the rank of these regressors is shown. The same trajectory as in Figure 5.1 was used. It can be seen that the rank of $\boldsymbol{K}_{\mathrm{L}}$ remains 6 for all time $t$, while the rank of $\boldsymbol{K}_{\mathrm{L}\varphi}$ quickly turns into 7 in the first instants and stays 7 for the remaining time.

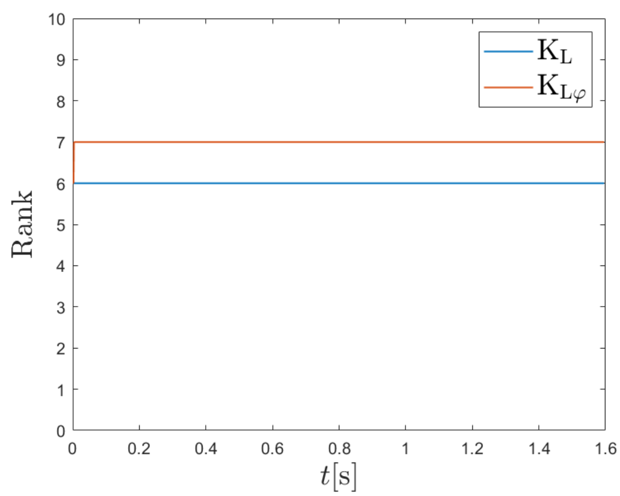


Figure 5.5: Rank of regressor $\boldsymbol{K}_{\mathrm{L}}$ and $\boldsymbol{K}_{\mathrm{L}\varphi}$

### 5.3.2 Matrix approximation using SVD

Despite the above, the SVD approach was maintained and even another modification was introduced. An approximation of $\boldsymbol{K}_{\mathrm{L}\varphi}$ can be obtained from the SVD (Brunton & Kutz 2019). First, SVD is applied to (5.20) as

$$\boldsymbol{u}_{\varphi} = \boldsymbol{\tau}_{\varphi} - (\boldsymbol{K}_{\varphi}\boldsymbol{\theta} + \boldsymbol{\tau}_{f\varphi}) = \boldsymbol{K}_{\mathrm{L}\varphi}\boldsymbol{\theta}_{\mathrm{L}} \tag{5.22a}$$

$$\boldsymbol{u}_{\varphi} = \boldsymbol{U}_{\varphi}\boldsymbol{S}_{\varphi}\boldsymbol{V}_{\varphi}^{\mathrm{T}}\boldsymbol{\theta}_{\mathrm{L}}, \tag{5.22b}$$

In the SVD, the singular values in the diagonal elements of the corresponding square sub-block are arranged from largest to smallest. That is, the larger the singular value, the more important and more information it saves. Hence, if only the leading $r$ singular values and their corresponding vectors are kept, and the rest are discarded, a rank-$r$ matrix approximation is obtained. Thus,

$$\boldsymbol{u}_{\varphi} = \underbrace{\begin{bmatrix} \tilde{\boldsymbol{U}}_{\varphi} & \boldsymbol{U}_{\varphi\,\mathrm{rem}} \end{bmatrix}}_{\boldsymbol{U}_{\varphi}} \underbrace{\begin{bmatrix} \tilde{\boldsymbol{\Sigma}}_{\varphi} & & \boldsymbol{0} \\ & \boldsymbol{\Sigma}_{\varphi\,\mathrm{rem}} & \boldsymbol{0} \end{bmatrix}}_{\boldsymbol{S}_{\varphi}} \underbrace{\begin{bmatrix} \tilde{\boldsymbol{V}}_{\varphi}^{\mathrm{T}} \\ \boldsymbol{V}_{\varphi\,\mathrm{rem}}^{\mathrm{T}} \\ \boldsymbol{V}_{\varphi\,0}^{\mathrm{T}} \end{bmatrix}}_{\boldsymbol{V}_{\varphi}{}^{\mathrm{T}}} \boldsymbol{\theta}_{\mathrm{L}} \tag{5.23a}$$

$$\boldsymbol{u}_{\varphi} \approx \tilde{\boldsymbol{U}}_{\varphi}\tilde{\boldsymbol{\Sigma}}_{\varphi}\tilde{\boldsymbol{V}}_{\varphi}^{\mathrm{T}}\,\boldsymbol{\theta}_{\mathrm{L}}, \tag{5.23b}$$

where $\tilde{\boldsymbol{U}}_{\varphi}\tilde{\boldsymbol{\Sigma}}_{\varphi}\tilde{\boldsymbol{V}}_{\varphi}^{\mathrm{T}} = \tilde{\boldsymbol{K}}_{\mathrm{L}\varphi}$ is an approximation of the payload regressor $\boldsymbol{K}_{\mathrm{L}\varphi}$; $\tilde{\boldsymbol{U}}_{\varphi} \in \mathbb{R}^{n\times r}$ and $\tilde{\boldsymbol{V}}_{\varphi}^{\mathrm{T}} \in \mathbb{R}^{r\times 10}$ consist of the first $r$ columns of $\boldsymbol{U}_{\varphi}$ and $\boldsymbol{V}_{\varphi}$, respectively, and $\tilde{\boldsymbol{\Sigma}}_{\varphi} \in \mathbb{R}^{r\times r}$ the first $r$ singular values of $\boldsymbol{S}_{\varphi}$. The subscript $_{\mathrm{rem}}$ means the remainder of $n - r$ columns or singular values that are discarded, and $\boldsymbol{V}_{\varphi\,0}^{\mathrm{T}}$ are the $10 - n$ columns of $\boldsymbol{V}_{\varphi}^{\mathrm{T}}$ that multiply the columns of $\boldsymbol{S}_{\varphi}$ that contain zeros that are also discarded. Thus, $r$ will be a variable that can be adjusted in the tuning stage of the MDREM algorithm.

The remainder procedure is the same as described in Section 5.2.1, but now it is continued with (5.23b). Then, both sides of (5.23b) are multiplied on the left by $\tilde{\boldsymbol{U}}_{\varphi}^{\mathrm{T}}$ so that

$$\tilde{\boldsymbol{U}}_{\varphi}^{\mathrm{T}}\boldsymbol{u}_{\varphi} \approx \tilde{\boldsymbol{U}}_{\varphi}^{\mathrm{T}}\tilde{\boldsymbol{U}}_{\varphi}\tilde{\boldsymbol{\Sigma}}_{\varphi}\tilde{\boldsymbol{V}}_{\varphi}^{\mathrm{T}}\boldsymbol{\theta}_{\mathrm{L}} \tag{5.24a}$$

$$\approx \tilde{\boldsymbol{\Sigma}}_{\varphi}\tilde{\boldsymbol{V}}_{\varphi}^{\mathrm{T}}\boldsymbol{\theta}_{\mathrm{L}} \tag{5.24b}$$

$$\boldsymbol{\delta}_{\varphi} \approx \boldsymbol{\Psi}_{\varphi}\boldsymbol{\theta}_{\mathrm{L}}, \tag{5.24c}$$

where $\tilde{\boldsymbol{U}}_{\varphi}^{\mathrm{T}}\tilde{\boldsymbol{U}}_{\varphi} = I$ has been used; $\boldsymbol{\delta}_{\varphi} \in \mathbb{R}^{r}$ and $\boldsymbol{\Psi}_{\varphi} \in \mathbb{R}^{r\times 10}$ are the input vector and regressor matrix of the new system, respectively, defined by

$$\boldsymbol{\delta}_{\varphi} = \tilde{\boldsymbol{U}}_{\varphi}^{\mathrm{T}}\boldsymbol{u}_{\varphi} \tag{5.25}$$

and

$$\boldsymbol{\Psi}_{\varphi} = \tilde{\boldsymbol{\Sigma}}_{\varphi}\tilde{\boldsymbol{V}}_{\varphi}^{\mathrm{T}}. \tag{5.26}$$

From $\boldsymbol{\Psi}_{\varphi}$ and $\boldsymbol{\delta}_{\varphi}$ an extended regressor $\bar{\boldsymbol{\Psi}}_{\mathrm{f}} \in \mathbb{R}^{10\times 10}$ and an extended input vector $\bar{\boldsymbol{\delta}}_{\mathrm{f}} \in \mathbb{R}^{10}$ are constructed precisely in the same way as in (5.8) and (5.14). Hence, these are represented by

$$\bar{\boldsymbol{\Psi}}_{\mathrm{f}} = \begin{bmatrix} \bar{\boldsymbol{\Psi}}_{1} \\ \vdots \\ \bar{\boldsymbol{\Psi}}_{\tilde{n}} \end{bmatrix} \in \mathbb{R}^{10\times 10}, \qquad \bar{\boldsymbol{\delta}}_{\mathrm{f}} = \begin{bmatrix} \bar{\boldsymbol{\delta}}_{1} \\ \vdots \\ \bar{\boldsymbol{\delta}}_{\tilde{n}} \end{bmatrix} \in \mathbb{R}^{10}, \tag{5.27}$$

where $\bar{\boldsymbol{\Psi}}_{1} = \boldsymbol{\Psi}_{\varphi}$ and $\bar{\boldsymbol{\delta}}_{1} = \boldsymbol{\delta}_{\varphi}$, so the first $r$ rows of $\bar{\boldsymbol{\Psi}}_{\mathrm{f}}$ are the $r$ rows of the regressor $\boldsymbol{\Psi}_{\varphi}$ and the first $r$ elements of $\bar{\boldsymbol{\delta}}_{\mathrm{f}}$ are the $r$ elements of the input vector $\boldsymbol{\delta}_{\varphi}$.

In other words, this process can be referred to as truncated SVD where the SVD is truncated at some value $r$. Subsequently, the rest of the MDREM algorithm is performed as usual.

## 5.4 Experimental results

Before estimating the inertial parameters of an unknown payload, it is recommended to try with a payload whose ten inertial parameters are known *a priori*. In this case, the payload will be the hand gripper of the robot shown in Figure 5.6. The corresponding payload parameters are presented in Table 5.3.

Figure 5.6: Hand gripper of the FR3 robot.

Table 5.3: Inertial parameters of the FR3 hand gripper (Sandoval & Laribi 2022).

| $\theta_{\mathrm{L}}$ | **Value** |
|---|---|
| $m$ (kg) | 0.730 |
| $c_x$ (m) | -0.010 |
| $c_y$ (m) | 0 |
| $c_z$ (m) | 0.030 |
| $I_{xx}$ $(kg \cdot m^2)$ | 0.001 |
| $I_{yy}$ $(kg \cdot m^2)$ | 0 |
| $I_{zz}$ $(kg \cdot m^2)$ | 0 |
| $I_{xy}$ $(kg \cdot m^2)$ | 0.0025 |
| $I_{xz}$ $(kg \cdot m^2)$ | 0 |
| $I_{yz}$ $(kg \cdot m^2)$ | 0.0017 |

### 5.4.1 Offline parameter estimation

Due to limited access to the actual robot and faster repetitiveness, the first experimental approach was to estimate the inertial parameters offline.

The experimental trajectory is the same as that shown in Figure 5.1. Since off-line estimation is performed, the trajectory was executed first, the data were saved and, finally, it was processed using Matlab afterwards. The FR3 robot provides joint positions $\boldsymbol{q}$, velocities $\dot{\boldsymbol{q}}$, and torque $\boldsymbol{\tau}$ through the libfranka interface. Joint acceleration was obtained by numerical differentiation and then filtered using Matlab's functions *designfilt* and *filtfilt*. The sample time was $T = 0.001$ s, which was the smallest possible.

The value $r$ at which the SVD is truncated as described in Section 5.3.2 is $r = 4$. The gains of the filters to compute the extended regressor, in this case $\bar{\boldsymbol{\Psi}}_{\mathrm{f}}$, are shown in Table 5.4 together with other MDREM tuning parameters. The corresponding gains of the adaptive law are shown in Table 5.5.

Table 5.4: MDREM gains for off-line experimentation with the FR3 robot.

| **Gain** | **Equation** | **Value** |
|---|---|---|
| $\lambda_\varphi$ | (2.13) | 25 |
| $a_2 = a_3$ | (2.19) | 0.19 |
| $b_2, b_3$ | (2.19) | $\{0.3, 0.8\}$ |
| $\phi_{\mathrm{d}}$ | (3.2) | 0.001 |
| $\eta_{\mathrm{m}}$ | (3.2) | $6.5 \times 10^{-9}$ |

Table 5.5: Gains of the adaptive law (3.10) for for the off-line experimentation with the FR3 robot.

| **Gain** | **Value** |
|---|---|
| $\boldsymbol{\Gamma}$ | diag $\{10.9,\ 0.49,\ 0.0,\ 0.308,\ 0.023,\ 0.0,\ 0.02,\ 0.046,\ 0.0,\ 0.034\}$ |
| $\boldsymbol{\lambda}_\theta$ | $[2\ \ 2\ \ 2\ \ 2\ \ 2\ \ 2\ \ 2\ \ 2\ \ 2\ \ 2\,]^{\mathrm{T}}$ |

The tuning process was not much different from that used before. The gains of Tables 5.4 and 5.5 were tuned once again, using the tuning rules introduced above in Section 4.3 and used for simulation and experimentation for the Geomagic Touch. In addition, the trial and error approach played a key role to minimize the parameter errors since there is uncertainty in the set of robot link parameters used, which adds some difficulty to the tuning process.

Figure 5.7 shows the results of the parameter estimation in terms of the errors since the inertial parameters are known *a priori*. The initial conditions of the estimated parameters $\hat{\boldsymbol{\theta}}_{\boldsymbol{L}}$ are set to zero. It is observed that six out of ten parameter estimation errors converge to zero, while other three ($\tilde{\theta}_{\mathrm{L}3}$, $\tilde{\theta}_{\mathrm{L}6}$ and $\tilde{\theta}_{\mathrm{L}9}$) remain zero for all time. This is done by taking advantage of the prior knowledge that $\theta_{\mathrm{L}3} = \theta_{\mathrm{L}6} = \theta_{\mathrm{L}9} = 0$ and setting the corresponding adaptive gains $\gamma_3 = \gamma_6 = \gamma_9 = 0$ in the adaptive law since the I.C. are zero. On the other hand, only $\tilde{\theta}_{\mathrm{L}7}$ converges to an incorrect value. In Figure 5.8 the square determinants $\phi^2$ and $\phi_{\mathrm{m}}^2$ are shown. From the figure, it is evident that $\phi_{\mathrm{m}}^2$ increases to the design value $\phi_{\mathrm{d}}^2 = 1 \times 10^{-6}$ when $\phi^2 \geq \eta_{\mathrm{m}} = 6.5 \times 10^{-9}$ and it lasts approximately 0.07 seconds. If we compare this with Figure 5.7, it can be seen that the estimated parameters begin to adapt as $\phi_{\mathrm{m}}^2$ increases due to the MDREM algorithm as expected.

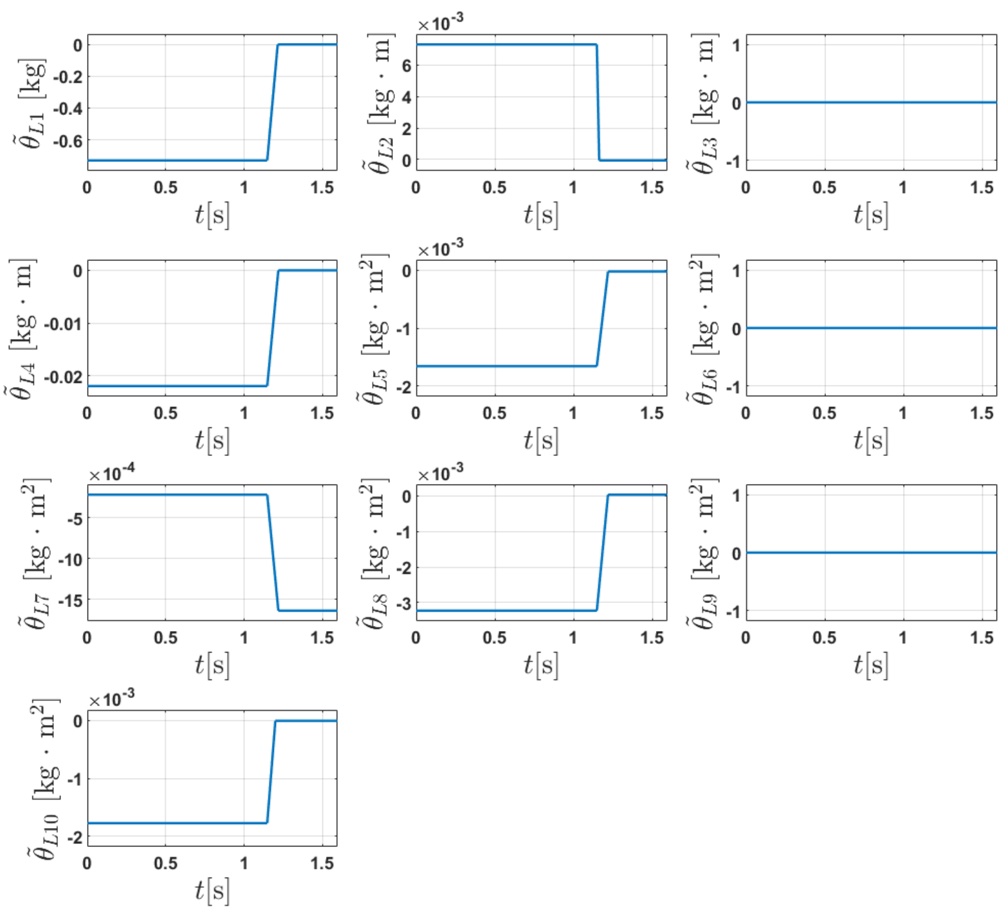


Figure 5.7: FR3 offline experiment. Parameter estimation errors. Initial conditions of $\hat{\boldsymbol{\theta}}_{\boldsymbol{L}}$ set to zero.

Since one of the characteristics of the DREM and MDREM algorithms is that an independent estimator can be designed and implemented for each parameter, and

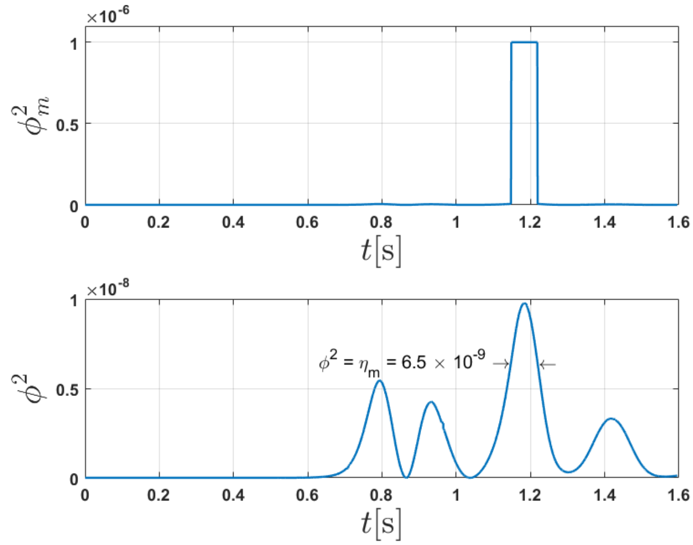


Figure 5.8: FR3 offline experiment. Square determinant $\phi^2$ and $\phi_{\mathrm{m}}^2$.

the fact that the parameter values are known beforehand, the gains $\gamma_3$, $\gamma_6$ and $\gamma_9$ in Table 5.5 are set to zero so that the estimation error remains always zero for $\tilde{\theta}_{\mathrm{L}3}$, $\tilde{\theta}_{\mathrm{L}6}$ and $\tilde{\theta}_{\mathrm{L}9}$, as shown in Figure 5.7. For this reason and to test the MDREM algorithm for these three parameters, the estimation process was repeated with the initial conditions of the estimated parameters set to -1 (except $\hat{\boldsymbol{\theta}}_{\mathrm{L}1}$). The gains of the adaptive law were subject to adjustment and are shown in Table 5.6.

Table 5.6: FR3 offline experiment. Gains of the adaptive law (3.10) with initial conditions of the estimated parameters set to -1.

| **Gain** | **Value** |
|---|---|
| $\boldsymbol{\Gamma}$ | diag $\{10.9,\ 13.98,\ 32.176,\ 14.39,\ 14.111,\ 14.087,\ 20.0,\ 14.132,\ 200.1,\ 18.937\}$ |
| $\boldsymbol{\lambda}_\theta$ | $[2 \quad 2 \quad 2 \quad 2 \quad 2 \quad 2 \quad 2 \quad 2 \quad 2 \quad 2\,]^{\mathrm{T}}$ |

The results of the parameter estimation for this second scenario are shown in Figure 5.9. As illustrated in the figure, eight out of ten estimated parameters achieve a good approximation with relatively small errors, while only $\tilde{\boldsymbol{\theta}}_{\mathrm{L}7}$ and $\tilde{\boldsymbol{\theta}}_{\mathrm{L}9}$ converge incorrectly. To better understand the results, Table 5.7 shows the known inertial parameters *a priori*, the estimated parameters, and the errors. Although the algorithm is designed to yield zero estimation error, the observed errors stem from discrepancies in the robot parameters. While the FR3 is structurally identical to the Franka Emika Panda, it is a newer generation and possesses slight physical variations. This parametric uncertainty in $\boldsymbol{\theta}$ directly affects the input $\boldsymbol{u}$ in (5.7b), which is a critical component of the MDREM algorithm. Consequently, as shown in (3.12), the adaptive law is driven by $\tau_{\epsilon i}$ (or rather the corresponding version of the torque input $u_i$), which inherits these uncertainties and prevents the error from converging to zero. Note that in Figures 5.7 and 5.9, as well as in Table 5.7, they present the mass moments instead of the center of mass, and the elements of the tensor of inertia with respect to the frame of link 7 rather than with respect to the center-of-mass frame of the payload, since the parameters are estimated this way.

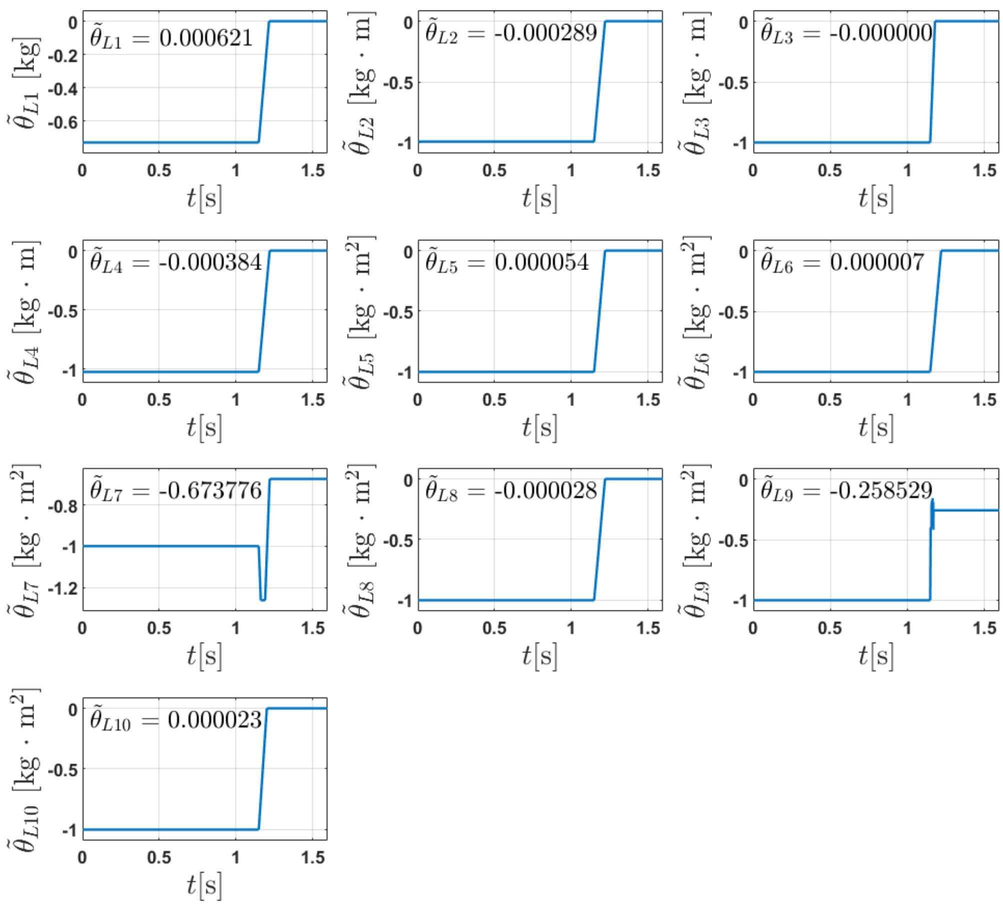


Figure 5.9: FR3 offline experiment. Parameter estimation errors with initial conditions of the estimated parameters set to -1.

Table 5.7: FR3 offline experiment. Inertial parameters of the FR3 hand gripper known *a priori*, estimated and their errors.

| $\boldsymbol{\theta}_{\mathrm{L}}$ | ***A priori*** | **Estimated** | **Error** |
|---|---|---|---|
| $m$ (kg) | 0.730 | 0.7306 | 0.0006 |
| $c_x m$ (kg · m) | -0.0073 | -0.0076 | -0.0003 |
| $c_y m$ (kg · m) | 0 | -0.0000 | -0.0000 |
| $c_z m$ (kg · m) | 0.0219 | 0.0215 | -0.0004 |
| $I_{xx}$ (kg · m$^2$) | 0.0017 | 0.0017 | 0.0001 |
| $I_{yy}$ (kg · m$^2$) | 0 | 0.0000 | 0.0000 |
| $I_{zz}$ (kg · m$^2$) | 0.0002 | -0.6736 | -0.6738 |
| $I_{xy}$ (kg · m$^2$) | 0.0032 | 0.0032 | -0.0000 |
| $I_{xz}$ (kg · m$^2$) | 0 | -0.2585 | -0.2585 |
| $I_{yz}$ (kg · m$^2$) | 0.0018 | 0.0018 | 0.0000 |

### 5.4.2 Online parameter estimation

After some acceptable results are achieved in the offline experiment, it is time to present the results of the online parameter estimation. The implementation of the MDREM algorithm was done in ROS2 for Linux Ubuntu, and the coding was done in the C++ programming language. A copy of the actual code used in the experiment can be found in

*https://github.com/Netul7/MDREM-inertial-parameters-estimation-for-FR3-robot*.

The path to the MDREM algorithm code is *\mdrem_algorithm\src* and the file is called *mdrem_node.cpp*. The code for the robot regressor $\boldsymbol{K}$ was converted from Matlab code to C++ using the tool called Matlab Coder, and their corresponding files can be found in the same path in the folder called *matlab_code*.

For the on-line parameter estimation, the trajectory was slightly changed from the off-line trajectory. It is shown in Figure 5.10. The initial position remains the same and is given by

$$\boldsymbol{q}(t_0) = [0^\circ \; -45^\circ \; 0^\circ \; -135^\circ \; 45^\circ \; 45^\circ \; 45^\circ]^{\mathrm{T}}, \tag{5.28}$$

while the desired final position is determined by

$$\boldsymbol{q}_{\mathrm{d}} = [0^\circ \; -45^\circ \; 0^\circ \; -135^\circ \; -90^\circ \; 115^\circ \; -40^\circ]^{\mathrm{T}}. \tag{5.29}$$

Recall that the joint acceleration is obtained through numerical differentiation, but, unlike in the offline experiment, now Matlab is not used, so that it is filtered by a

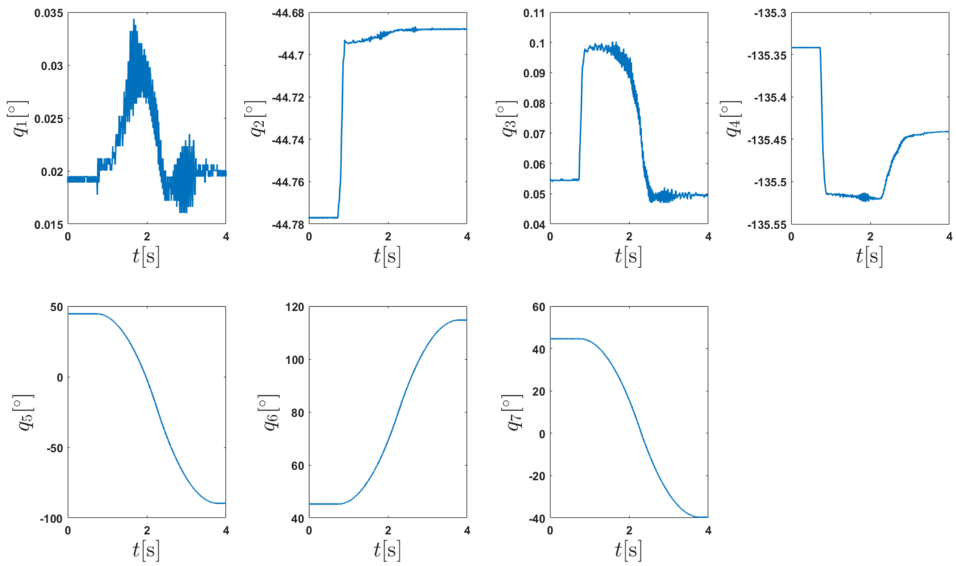


Figure 5.10: Trajectory for the online experiment.

filter of the same form as in (2.13) or (2.14). The sample time again is $T = 0.001$ s.

The truncation value $r$ for the SVD as described in Section 5.3.2 was chosen as $r = 4$. All the necessary gains of the MDREM algorithm for the online parameter estimation experiment are shown in Table 5.8. The equation in which each gain is used is also indicated in the table. Furthermore, the gains of the adaptation law are shown in Table 5.9.

Table 5.8: MDREM gains for online estimation experiment

| **Gain** | **Equation** | **Value** |
|---|---|---|
| $\lambda_\varphi$ | (2.13) | 15 |
| $a_2 = a_3$ | (2.19) | 0.1 |
| $b_2, b_3$ | (2.19) | $\{0.3, 0.7\}$ |
| $\phi_\mathrm{d}$ | (3.2) | 0.02 |
| $\eta_\mathrm{m}$ | (3.2) | $2.5 \times 10^{-8}$ |

Table 5.9: Gains of the adaptive law (3.10) for the online estimation experiment.

| **Gain** | **Value** |
|---|---|
| $\mathbf{\Gamma}$ | $\mathrm{diag}\,\{3.95,\ 0.0835,\ 0.0,\ 0.1178,\ 0.0073,\ 0.0,\ 0.01,\ 0.014,\ 0.0,\ 0.01\}$ |
| $\lambda_\theta$ | $[2\quad 2\quad 2\quad 2\quad 2\quad 2\quad 2\quad 2\quad 2\quad 2]^\mathrm{T}$ |

In Figure 5.11, the results of the online parameter estimation in terms of errors are presented. Again, taking advantage of knowledge *a priori* of the values of

$\theta_{L3} = \theta_{L6} = \theta_{L9} = 0$, gains $\gamma_3 = \gamma_6 = \gamma_9 = 0$ are set to zero. Consequently, their estimated value remains zero for all time because the initial conditions of $\hat{\boldsymbol{\theta}}_{\boldsymbol{L}}$ are also zero. Leaving aside the three estimation errors that remain zero, six of them have a good approximation, while only $\tilde{\theta}_{L7}$ does not converge to zero. Table 5.10 shows the estimated numerical value and their errors. The behavior of the square determinants $\phi^2$ and $\phi_m^2$ is shown in Figure 5.12 where it can be seen that the condition (3.1) holds for approximately 0.24 seconds.

Through many experiment executions, it was observed that the estimated parameter $\hat{\theta}_{L1}$ was the more consistent one and closer to the known value *a priori*. Even though the others get approximations, they show more variation from time to time. This could be explained by the uncertainty still present in the inertial parameters of the robot links. Again, recall that the ones used are from the Franka Emika Panda robot (Soto et al. 2024), while the actual robot used in the experimentation is the Franka Research 3, which is structurally the same robot.

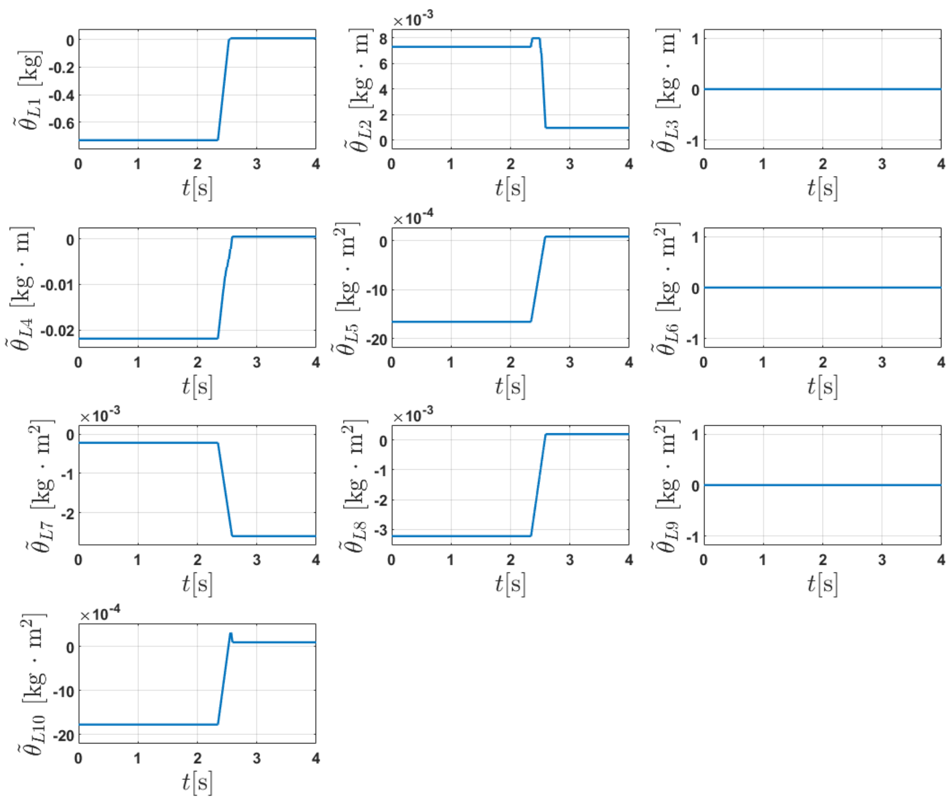


Figure 5.11: FR3 online experiment. Parameter estimation errors. Initial conditions of $\hat{\boldsymbol{\theta}}_{\boldsymbol{L}}$ set to zero.

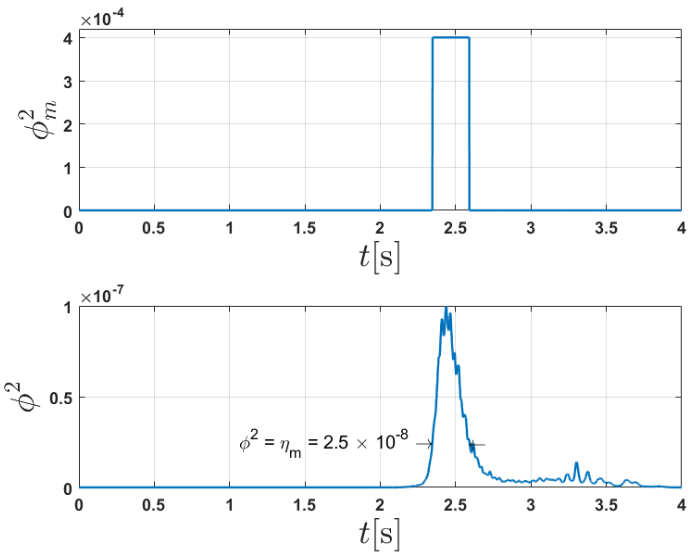


Figure 5.12: Square determinant $\phi^2$ and $\phi_{\mathrm{m}}^2$ of the online estimation experiment.

Table 5.10: Estimated inertial parameters of the hand gripper and their errors of the online experiment.

| $\boldsymbol{\theta}_{\mathrm{L}}$ | ***A priori*** | **Estimated** | **Error** |
|---|---|---|---|
| $m$ (kg) | 0.730 | 0.7387 | 0.0087 |
| $c_x m$ (kg · m) | -0.0073 | -0.0063 | 0.0010 |
| $c_y m$ (kg · m) | 0 | 0 | 0 |
| $c_z m$ (kg · m) | 0.0219 | 0.0224 | 0.0005 |
| $I_{xx}$ (kg · m$^2$) | 0.0017 | 0.0017 | 0.0001 |
| $I_{yy}$ (kg · m$^2$) | 0 | 0 | 0 |
| $I_{zz}$ (kg · m$^2$) | 0.0002 | -0.0024 | -0.0026 |
| $I_{xy}$ (kg · m$^2$) | 0.0032 | 0.0034 | 0.0002 |
| $I_{xz}$ (kg · m$^2$) | 0 | 0 | 0 |
| $I_{yz}$ (kg · m$^2$) | 0.0018 | 0.0019 | 0.0001 |

### 5.4.3 Experiment with hand gripper + 200g weight

In order to try to estimate a partially unknown payload, a weight of 200 grams was added and held by the hand gripper. The results of this experiment are shown in Figure 5.13 and summarized in Table 5.11. As expected, the estimated parameter $\hat{\theta}_{\mathrm{L}1}$, which is the mass, has an increase of slightly more than 200 g. Since the gains of the adaptive law were not changed, the estimated parameters $\hat{\theta}_{\mathrm{L}3}$, $\hat{\theta}_{\mathrm{L}6}$, and $\hat{\theta}_{\mathrm{L}9}$ remain zero, again because the I.C. of $\hat{\boldsymbol{\theta}}_{\boldsymbol{L}} = 0$ and the gains $\gamma_3 = \gamma_6 = \gamma_9 = 0$ in the adaptive law. This was technically a mistake because the actual parameters $\theta_{\mathrm{L}3}$, $\theta_{\mathrm{L}6}$ and $\theta_{\mathrm{L}9}$ are possibly no longer zero, but the purpose also was to keep the same gains from the experiment without the weight of 200g. The rest of the estimated parameters experience an increase in their value compared to the last experiment.

The estimation process begins in second 2 and if we look at Figure 5.14, it can be seen that it coincides when $\phi^2$ becomes greater than $\eta_\mathrm{m} = 2.5\times10^{-8}$ so that $\phi_\mathrm{m}^2$ increases. The transient behavior of the estimated parameters $\hat{\theta}_{\mathrm{L}2}$, $\hat{\theta}_{\mathrm{L}7}$, and $\hat{\theta}_{\mathrm{L}10}$ is obviously not as expected because the adaptation direction simply changes, which should not occur. This is explained by the uncertainties present in the robot parameters. The estimated parameters $\hat{\theta}_{\mathrm{L}1}$, $\hat{\theta}_{\mathrm{L}4}$, $\hat{\theta}_{\mathrm{L}5}$, and $\hat{\theta}_{\mathrm{L}8}$ adapt steadily as expected. In second 3, all parameters, excluding $\hat{\theta}_{\mathrm{L}3}$, $\hat{\theta}_{\mathrm{L}6}$ and $\hat{\theta}_{\mathrm{L}9}$, experienced a sudden small change in the estimated value. To understand this, we first need to look again at the behavior of $\phi_\mathrm{m}^2$ in Figure 5.14. It is seen that for an instant in second 3, $\phi^2$ becomes larger than $\eta_\mathrm{m}$, causing $\phi_\mathrm{m}^2$ to grow for a brief moment and start the adaptation again. Then, the small change in the estimated value is likely due, once again, to uncertainty in the robot parameters, the estimation was not completed the first time of the high-$\phi_\mathrm{m}$ phase, or both.

A video of the robot executing the trajectory can be found at the following link: *https://youtu.be/9pYngCh7uG8?si=ZL1O0ZlLWQ-FHZmU*. The hand gripper is seen holding the 200g weight with a rudimentary wrap to keep it tight.

Table 5.11: Estimated inertial parameters of the experiment with hand gripper + 200g weight.

| $\hat{\boldsymbol{\theta}}_\mathrm{L}$ | **Value** |
|---|---|
| $m$ (kg) | 0.9400 |
| $c_x m$ (kg · m) | -0.0111 |
| $c_y m$ (kg · m) | 0 |
| $c_z m$ (kg · m) | 0.0437 |
| $I_{xx}$ (kg · m$^2$) | 0.0019 |
| $I_{yy}$ (kg · m$^2$) | 0 |
| $I_{zz}$ (kg · m$^2$) | -0.0023 |
| $I_{xy}$ (kg · m$^2$) | 0.0049 |
| $I_{xz}$ (kg · m$^2$) | 0 |
| $I_{yz}$ (kg · m$^2$) | 0.0012 |

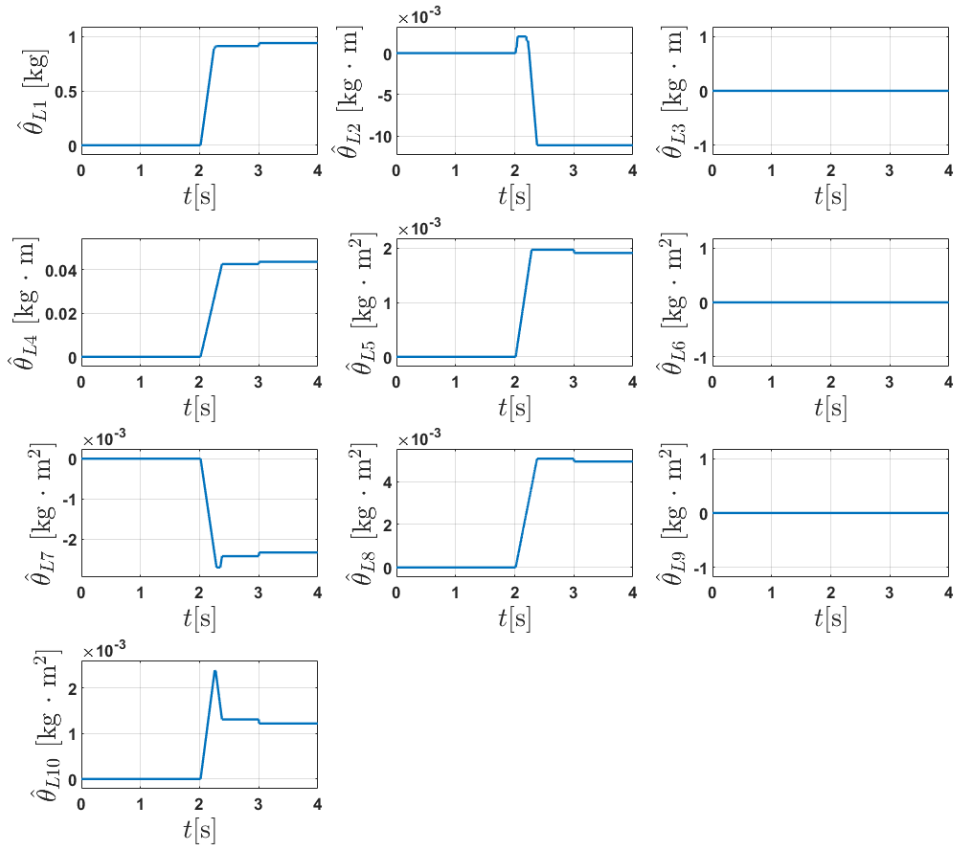


Figure 5.13: Estimated parameters. Experiment with hand gripper + 200g weight. Initial conditions of $\hat{\boldsymbol{\theta}}_{\boldsymbol{L}}$ set to zero.

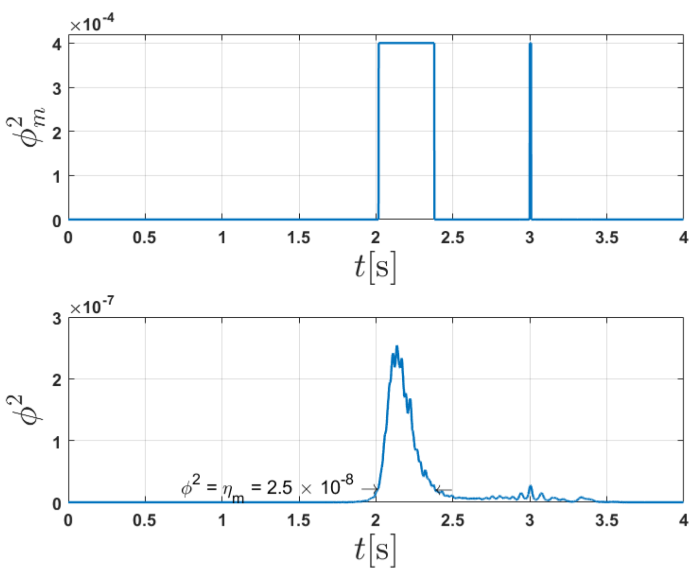


Figure 5.14: Square determinant $\phi^2$ and $\phi_{\mathrm{m}}^2$ of the online estimation experiment with hand gripper + 200g weight.

*Chapter 6*

# CONCLUSIONS

The main goal of this thesis was to estimate the inertial parameters of a robot manipulator payload using an adaptive control technique called *Modified Dynamic Regressor Extension and Mixing* (MDREM) algorithm. This approach has the advantage of achieving parameter estimation in a limited time period by increasing the little amount of excitation that a short trajectory could have. In other words, the PE condition is not necessarily to be satisfied; instead, if another more relaxed condition is fulfilled, theoretically, it is guaranteed convergence in fixed time.

The MDREM algorithm was implemented in simulation to a 3 DoF robot and then validated in experimentation with the *Geomagic touch* haptic device robot with 3 DoF. In addition, the implementation of the algorithm and experiments were also carried out with the Franka Research 3 robot with 7 DoF. The simulation results with the 3 DoF robot showed the effectiveness of the method to estimate the inertial parameters of the robot with and without payload; while the experimental validation results showed, as expected, an increase in the estimated parameters for the scenario with the payload attached. A validation test demonstrates the reliability of the estimated parameters involved in the compensation gravity. Meanwhile, experiments with the FR3 robot showed a good approximation of the estimated payload parameters compared to those known *a priori*. Through experiments, the estimated payload mass exhibits greater consistency of all estimated parameters, even when a 200g mass was added to the payload.

Although there is room for improvement, these results demonstrate that the MDREM algorithm could become a good solution to the parameter estimation problem, making the whole process easier and faster, and also offering a less restrictive alternative to the hard PE condition and the practical risks it imposes on the robot and their environment.

Obviously, there are some limitations to this parameter estimation method. It is designed in continuous time, while the implementation is done in discrete time, which could have a negative impact if the sample time is not small enough. Furthermore, the method is based on the accuracy of the dynamic model, and the one used in experimentation for the Geomagic touch robot was derived considering important simplifications, which can downgrade the certainty of the estimation. The validation of some estimated parameters could not be carried out; otherwise, it would have evidenced these model uncertainties. Concerning the FR3 robot, recall the assumption that the inertial parameters of the robot links are known, and that the actual parameters used were from the Franka Emika Panda robot, which is structurally the same robot, so robot parameters might be slightly different. This explain why, leaving aside the estimated mass, the rest of estimated inertial parameters showed more variation from experiment to experiment.

Additionally, among the future work that could be done is: study the perturbed case. Regarding the Geomagic touch robot, the employment of a dynamic model without considering the simplifications made, and validation of the rest of the estimated parameters with a feasible test. As to the FR3 robot, finding a more accurate set of inertial parameters of the robot links, identifying a totally unknown payload, and validating the results.

*Appendix A*

# DYNAMIC MODELING OF RIGID ROBOT MANIPULATORS

Recalling that $\mathcal{L} = \mathcal{K} - \mathcal{P}$, and from (2.2) and (2.4), the Lagrangian function is given by

$$\mathcal{L} = \frac{1}{2}\dot{\boldsymbol{q}}^{\mathrm{T}}\boldsymbol{H}(\boldsymbol{q})\dot{\boldsymbol{q}} + \sum_{i=1}^{n} m_i \overline{\boldsymbol{g}}^{\mathrm{T}\,0}\overline{\boldsymbol{r}}_i. \tag{A.1}$$

Equation (2.1) can be expressed in matrix form as

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial\mathcal{L}}{\partial\dot{\boldsymbol{q}}}\right) - \frac{\partial\mathcal{L}}{\partial\boldsymbol{q}} = \boldsymbol{\tau}, \tag{A.2}$$

where $\boldsymbol{\tau} \in \mathbb{R}^n$ is the generalized force/torque vector applied to the joints.

Now, in order to obtain the dynamic model of the manipulator, it is necessary to develop the left-hand side of (A.2). First, let us begin with $\frac{\partial\mathcal{L}}{\partial\boldsymbol{q}}$, so we get

$$-\frac{\partial\mathcal{L}}{\partial\boldsymbol{q}} = -\frac{\partial}{\partial\boldsymbol{q}}\left(\sum_{i=1}^{n} m_i \overline{\boldsymbol{g}}^{\mathrm{T}0}\overline{\boldsymbol{r}}_i\right) - \frac{1}{2}\frac{\partial}{\partial\boldsymbol{q}}\left(\dot{\boldsymbol{q}}^{\mathrm{T}}\boldsymbol{H}(\boldsymbol{q})\dot{\boldsymbol{q}}\right). \tag{A.3}$$

The first term on the right-hand side of (A.3) produces the gravity vector $\boldsymbol{g}(\boldsymbol{q})$ and can be written in a short way as

$$\boldsymbol{g}(\boldsymbol{q}) = \frac{\partial\mathcal{P}}{\partial\boldsymbol{q}} = -\frac{\partial}{\partial\boldsymbol{q}}\left(\sum_{i=1}^{n} m_i \overline{\boldsymbol{g}}^{\mathrm{T}0}\overline{\boldsymbol{r}}_i\right). \tag{A.4}$$

The second term on the right-hand side of (A.3) forms a part of the Coriolis and centrifugal torques vector and will be retaken later. In the meantime, the first term of (A.2) needs to be developed as well, thus

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial\mathcal{L}}{\partial\dot{\boldsymbol{q}}} = \frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{H}(\boldsymbol{q})\dot{\boldsymbol{q}} = \boldsymbol{H}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \dot{\boldsymbol{H}}(\boldsymbol{q})\dot{\boldsymbol{q}} \tag{A.5}$$

The first term on the right-hand side of (A.5) is ready as is to appear in the model. As for the second term, it forms the remaining part of Coriolis and centrifugal torque vector. Then, it is made up of this term and that second term on the right-hand side of (A.3). It is denoted as $\boldsymbol{h}_{\mathrm{c}}(\boldsymbol{q}, \dot{\boldsymbol{q}}) \in \mathbb{R}^n$ and as a result we have

$$\boldsymbol{h}_{\mathrm{c}}(\boldsymbol{q}, \dot{\boldsymbol{q}}) = \dot{\boldsymbol{H}}(\boldsymbol{q})\dot{\boldsymbol{q}} - \frac{1}{2}\frac{\partial}{\partial \boldsymbol{q}}\left(\dot{\boldsymbol{q}}^{\mathrm{T}}\boldsymbol{H}(\boldsymbol{q})\dot{\boldsymbol{q}}\right). \tag{A.6}$$

However, computing the Coriolis and centrifugal torque vector from (A.6) is not practical. Consequently, $\boldsymbol{h}_{\mathrm{c}}(\boldsymbol{q}, \dot{\boldsymbol{q}})$ can be expressed in a better way by

$$\boldsymbol{h}_{\mathrm{c}}(\boldsymbol{q}, \dot{\boldsymbol{q}}) = \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} \tag{A.7}$$

for a matrix $\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}}) \in \mathbb{R}^{n\times n}$. Here, we will not review how to compute the elements of matrix $\boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})$. For that purpose, see Spong et al. (2020) or Arteaga et al. (2022) Section 3.2.1.

Now we already have all the terms to form the dynamic model and thus it is given by

$$\boldsymbol{H}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{D}\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) = \boldsymbol{\tau} \tag{A.8}$$

where $\boldsymbol{D} \in \mathbb{R}^{n\times n}$ is a diagonal positive semidefinite matrix of viscous friction coefficients of the joint.

*Appendix B*

# DYNAMIC MODELS OF THE FRANKA RESEARCH 3 (FR3) ROBOT

Figure B.1: Franka Research 3 (FR3) robot.

This appendix chapter presents some information regarding the modeling of the 7 DoF Franka Research 3 robot (see Figure B.1). Table B.1 shows modified Denavit-Hartenberg (D-H) parameters of the robot, while Figure B.2 shows the visual D-H configuration. The modified D-H convention, in contrast to the original D-H convention, has the particularity of setting the $i$-th coordinate frame at the $i$-th joint.

The dynamic model of the FR3 robot is derived from the Newton-Euler (NE) formulation, which is the basis for the very efficient recursive Newton-Euler algorithm (RNEA) (Siciliano & Khatib 2008, Section 2.5.1), (Spong et al. 2020, Section 6.6). In the NE formulation, each link is represented by ten inertial parameters. For this reason, in the linear parameterization model, the regressor results in a 7-by-70 matrix (7-by-80 if the payload is included), which makes it difficult to show here. For more details on the linear parameterization model, it is suggested to see Siciliano & Khatib (2008) Section 14.3 and Sandoval & Laribi (2022).

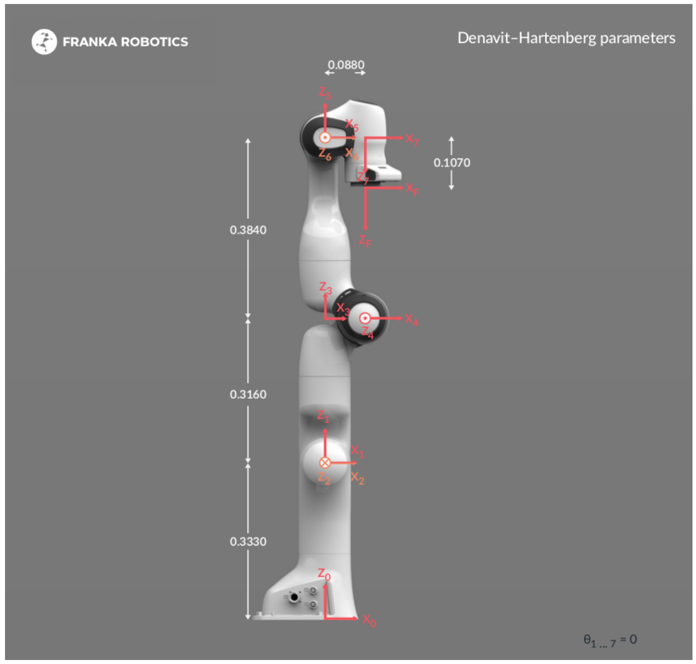


Figure B.2: Modified Denavit-Hartenberg configuration of FR3 robot.

Table B.1: Modified Denavit-Hartenberg parameters of the FR3 robot

| **Joint** | $a_i$ [m] | $d_i$ [m] | $\alpha_i$ [rad] | $q_i$ [rad] |
|---|---|---|---|---|
| **1** | 0 | 0.333 | 0 | ${q_1}^*$ |
| **2** | 0 | 0 | $-\frac{\pi}{2}$ | ${q_2}^*$ |
| **3** | 0 | 0.316 | $\frac{\pi}{2}$ | ${q_3}^*$ |
| **4** | 0.0825 | 0 | $\frac{\pi}{2}$ | ${q_4}^*$ |
| **5** | -0.0825 | 0.384 | $-\frac{\pi}{2}$ | ${q_5}^*$ |
| **6** | 0 | 0 | $\frac{\pi}{2}$ | ${q_6}^*$ |
| **7** | 0.088 | 0 | $\frac{\pi}{2}$ | ${q_7}^*$ |
| **Flange** | 0 | 0.107 | 0 | 0 |

*variable quantity

*Appendix C*

# COMPLEMENTARY THEORY

## C.1 Lemma 1.

*(Arteaga et al. 2025) Consider a positive definite continuous, radially unbounded function* $V(\boldsymbol{x}) : \mathbb{R}^n \to \mathbb{R}_+ \cup \{0\}$, *for* $\boldsymbol{x} \in \mathbb{R}^n$ *and with* $V(\boldsymbol{x}) = 0$ *only for* $\boldsymbol{x} \equiv \mathbf{0}$. *If*

$$\dot{V}(\boldsymbol{x}) \leq -\kappa_1 V^q(\boldsymbol{x}) \tag{C.1}$$

*for some constants* $\kappa_1 > 0$ *and* $q > 1$, *then* $V(\boldsymbol{x}) \leq 1$ *in a finite time no larger than* $t_0 + T_1$, *with* $T_1 = \frac{1}{\kappa_1(q-1)}$.

△

## C.2 Lemma 2.

*(Arteaga et al. 2025) Consider a positive definite continuous, radially unbounded function* $V(\boldsymbol{x}) : \mathbb{R}^n \to \mathbb{R}_+ \cup \{0\}$, *for* $\boldsymbol{x} \in \mathbb{R}^n$ *and with* $V(\boldsymbol{x}) = 0$ *only for* $\boldsymbol{x} \equiv \mathbf{0}$. *Assume that* $V(\boldsymbol{x}(t_0)) \leq 1$. *If*

$$\dot{V}(\boldsymbol{x}) \leq -\kappa_2 V^p(\boldsymbol{x}) \tag{C.2}$$

*for some positive constants* $\kappa_2$, $p$ *with* $p < 1$, *then* $V(\boldsymbol{x}) \equiv 0$ *in a finite time no larger than* $t_0 + T_2$, *with* $T_2 = \frac{1}{\kappa_2(1-p)}$.

△

## C.3 Swapping Lemma.

*(Sastry & Bodson 2011) Let* $\phi$, $w : \mathbb{R}_+ \to \mathbb{R}^n$ *and* $\phi$ *be differentiable. Let* $\hat{G}$ *be a proper rational transfer function. If* $\hat{G}$ *is stable, with a minimal realization*

$$\hat{G} = c^{\mathrm{T}}(sI - A)^{-1}b + d \tag{C.3}$$

*Then*

$$\hat{G}(w^{\mathrm{T}}\phi) - \hat{G}(w^{\mathrm{T}})\phi = \hat{G}_{\mathrm{c}}(\hat{G}_{\mathrm{b}}(w^{\mathrm{T}})\dot{\phi}) \tag{C.4}$$

*where*

$$\hat{G}_{\mathrm{b}} = (sI - A)^{-1}b \qquad \hat{G}_{\mathrm{c}} = -c^{\mathrm{T}}(sI - A)^{-1} \tag{C.5}$$

△

*Appendix D*

# FRICTION MODEL PARAMETERS & ROBOT INERTIAL PARAMETERS

Table D.1: **Lugre** friction model parameters (Soto et al. 2024).

| Joints | $\sigma_0$ (N·m) | $\sigma_1$ (N·m) | $\sigma_2$ (N·m) | $\sigma_3$ (N·m) | $v_0$ (rad/s) | $v_1$ (rad/s) | $v_2$ (rad/s) |
|---|---|---|---|---|---|---|---|
| 1 | 0.2390 | 0.0435 | 0.0316 | 0.4999 | 6.1102e-5 | 3.9028e-5 | 4.5659e-4 |
| 2 | 0.1360 | 0.1356 | 0.1795 | 0.4999 | 6.4280e-5 | 3.6342e-5 | 1.6242e-3 |
| 3 | 0.1764 | 0.0601 | 0.0663 | 0.4999 | 5.3066e-5 | 1.6280e-5 | 5.7174e-4 |
| 4 | 0.3662 | 0.1631 | 0.1772 | 0.4999 | 6.8089e-5 | 4.4244e-5 | 1.0923e-4 |
| 5 | 0.2152 | 0.0616 | 0.1082 | 0.5000 | 7.3521e-5 | 7.3274e-5 | 1.7022e-3 |
| 6 | 0.1398 | 0.0591 | 0.0192 | 0.5000 | 5.8370e-5 | 5.6923e-5 | 1.0196e-3 |
| 7 | 0.1837 | 0.0471 | 0.0997 | 0.5000 | 7.6521e-5 | 3.8782e-5 | 2.4456e-3 |

Table D.2: **Stribeck** friction model parameters (Soto et al. 2024).

| Joints | $F_C$ (Nm) | $F_V$ (Nm·s/rad) | $F_s$ (Nm) | $V_s$ (m/s) | $K_s$ |
|---|---|---|---|---|---|
| 1 | 0.2247 | 0.0497 | 6.5038e-22 | 2.1749 | 2.1748 |
| 2 | 0.1293 | 0.1708 | -8.6085e-28 | 2.1749 | 2.1748 |
| 3 | 0.1681 | 0.0774 | 1.6565e-20 | 2.1749 | 2.1748 |
| 4 | 0.3791 | 0.1792 | 5.8329e-24 | 2.1749 | 2.1748 |
| 5 | 0.2231 | 0.1038 | 3.7780e-25 | 1.5208 | 1.3656 |
| 6 | 0.1497 | 0.0129 | 7.4546e-22 | 1.5207 | 1.3656 |
| 7 | 0.1936 | 0.0655 | 2.7360e-21 | 1.5207 | 1.3656 |

Table D.3: **Sigmoidal** friction model parameters (Soto et al. 2024).

| Joints | $\psi_1$ (N·m) | $\psi_2$ (s/rad) | $\psi_3$ (rad/s) |
|---|---|---|---|
| 1 | 0.6183 | 10.8798 | 0.1120 |
| 2 | 0.5428 | 10.8708 | 0.0472 |
| 3 | 0.5048 | 10.8739 | -0.0111 |
| 4 | 0.9740 | 10.8846 | 0.0158 |
| 5 | 0.6200 | 13.0489 | 0.0053 |
| 6 | 0.3491 | 13.0484 | -0.0765 |
| 7 | 0.5572 | 13.0473 | 0.0010 |

Table D.4: **Sigmoidal** friction model parameters (Gaz et al. 2019).

| Joint | $\psi_1$ (N·m) | $\psi_2$ (s/rad) | $\psi_3$ (rad/s) |
|---|---|---|---|
| 1 | 5.4615e-1 | 5.1181 | 3.9533e-2 |
| 2 | 0.87224 | 9.0657 | 2.5882e-2 |
| 3 | 6.4068e-1 | 1.0136e+1 | -4.6070e-2 |
| 4 | 1.2794 | 5.5903 | 3.6194e-2 |
| 5 | 8.3904e-1 | 8.3469 | 2.6226e-2 |
| 6 | 3.0301e-1 | 1.7133e+1 | -2.1047e-2 |
| 7 | 5.6489e-1 | 1.0336e+1 | 3.5526e-3 |

Table D.5: Estimated parameters of the Franka Emika Panda robot (Gaz et al. 2019).

| $\theta$ | Link 1 | Link 2 | Link 3 | Link 4 | Link 5 | Link 6 | Link 7 |
|---|---|---|---|---|---|---|---|
| $m$ (kg) | 4.970684 | 0.646926 | 3.228604 | 3.587895 | 1.225946 | 1.666555 | 0.73552 |
| $c_x$ (m) | 3.875e-3 | -3.141e-3 | 2.7518e-2 | -5.317e-2 | -1.1953e-2 | 6.0149e-2 | 1.0517e-2 |
| $c_y$ (m) | 2.081e-3 | -2.872e-2 | 3.9252e-2 | 1.04419e-1 | 4.1065e-2 | -1.4117e-2 | -4.252e-3 |
| $c_z$ (m) | -0.0675818 | 3.495e-3 | -6.6502e-2 | 2.7454e-2 | -3.8437e-2 | -1.0517e-2 | 6.1597e-2 |
| $I_{xx}$ (kg·m$^2$) | 7.0337e-1 | 7.9620e-3 | 3.7242e-2 | 2.5853e-2 | 3.5549e-2 | 1.9640e-3 | 1.2516e-2 |
| $I_{yy}$ (kg·m$^2$) | 7.0661e-1 | 2.8110e-2 | 3.6155e-2 | 1.9552e-2 | 2.9474e-2 | 4.3540e-3 | 1.0027e-2 |
| $I_{zz}$ (kg·m$^2$) | 9.1170e-3 | 2.5995e-2 | 1.0830e-2 | 2.8323e-2 | 8.6270e-3 | 5.4330e-3 | 4.8150e-3 |
| $I_{xy}$ (kg·m$^2$) | -1.3900e-4 | -3.9250e-3 | -4.7610e-3 | 7.7960e-3 | -2.1170e-3 | 1.0900e-4 | -4.2800e-4 |
| $I_{xz}$ (kg·m$^2$) | 6.7720e-3 | 1.0254e-2 | -1.1396e-2 | -1.3320e-3 | -4.0370e-3 | -1.1580e-3 | -1.1960e-3 |
| $I_{yz}$ (kg·m$^2$) | 1.9169e-2 | 7.0400e-4 | -1.2805e-2 | 8.6410e-3 | 2.2900e-4 | 3.4100e-4 | -7.4100e-4 |

Table D.6: Inertial parameters of the Franka Research 3 robot (Franka Robotics 2024).

| $\theta$ | Link 1 | Link 2 | Link 3 | Link 4 | Link 5 | Link 6 | Link 7 |
|---|---|---|---|---|---|---|---|
| $m$ (kg) | 2.9275 | 2.9355 | 2.2449 | 2.6156 | 2.3271 | 1.8170 | 0.62714 |
| $c_x$ (m) | 4.1280e-7 | 3.1829e-3 | 4.0702e-2 | -4.5910e-2 | -1.6040e-3 | 5.9713e-2 | 4.5226e-3 |
| $c_y$ (m) | -1.8125e-2 | -7.4322e-2 | -4.8201e-3 | 6.3049e-2 | 2.9254e-2 | -4.1029e-2 | 8.6262e-3 |
| $c_z$ (m) | -3.8604e-2 | 8.8146e-3 | -2.8973e-2 | -8.5188e-3 | -9.7297e-2 | -1.0169e-2 | -1.6163e-2 |
| $I_{xx}$ $(kg \cdot m^2)$ | 2.3927e-2 | 4.1939e-2 | 2.4101e-2 | 3.4530e-2 | 5.1610e-2 | 5.4123e-3 | 2.1092e-4 |
| $I_{yy}$ $(kg \cdot m^2)$ | 2.2482e-2 | 2.5145e-2 | 1.9741e-2 | 2.8882e-2 | 4.7877e-2 | 1.4058e-2 | 1.7719e-4 |
| $I_{zz}$ $(kg \cdot m^2)$ | 6.3501e-3 | 6.1702e-2 | 1.9044e-2 | 4.1255e-2 | 1.6424e-2 | 1.6081e-2 | 5.9932e-5 |
| $I_{xy}$ $(kg \cdot m^2)$ | 1.3318e-5 | 2.0257e-4 | 2.4047e-3 | 1.3226e-2 | -5.7152e-3 | 6.1935e-3 | -2.4333e-5 |
| $I_{xz}$ $(kg \cdot m^2)$ | -1.1405e-4 | 4.0778e-3 | -2.8563e-3 | 1.0151e-2 | -3.5673e-3 | 1.4219e-3 | 4.5645e-5 |
| $I_{yz}$ $(kg \cdot m^2)$ | -1.9950e-3 | -4.2252e-3 | -2.1042e-3 | -9.7628e-4 | 1.0674e-2 | -1.3141e-3 | 8.7441e-5 |

Table D.7: Estimated parameters of the Franka Emika Panda robot (Soto et al. 2024).

| $\boldsymbol{\theta}$ | Link 1 | Link 2 | Link 3 | Link 4 | Link 5 | Link 6 | Link 7 |
|---|---|---|---|---|---|---|---|
| $m$ (kg) | 4.75 | 0.9918 | 3.2832 | 3.5858 | 1.3628 | 1.741 | 0.5158 |
| $c_x$ (m) | -3.48e-08 | 0.0396 | 0.0345 | -0.0474 | -0.0016 | 0.0733 | -0.0015 |
| $c_y$ (m) | -3.48e-08 | -0.1222 | 0.0351 | 0.1385 | 0.0307 | -0.0205 | 0.004 |
| $c_z$ (m) | -0.175 | -0.0096 | -0.0999 | 0.0327 | -0.125 | -0.0262 | 0.0638 |
| $I_{xx}$ (kg·m$^2$) | 0.5 | 0.0752 | 0.0528 | 0.0285 | 0.0669 | 0.0066 | 0.0005 |
| $I_{yy}$ (kg·m$^2$) | 0.5 | 0.0297 | 0.0102 | 0.0064 | 0.0534 | 0.0053 | 0.0019 |
| $I_{zz}$ (kg·m$^2$) | 0 | 0.016 | 0.0183 | 0.0486 | 0.0022 | 0.0006 | 0.0001 |
| $I_{xy}$ (kg·m$^2$) | 7.72e-14 | 0.0098 | -0.0195 | 0.0072 | -0.0039 | -0.0033 | -0.0009 |
| $I_{xz}$ (kg·m$^2$) | -8.56e-10 | -0.02 | -0.0298 | -0.0241 | -0.0112 | -0.0002 | -0.0002 |
| $I_{yz}$ (kg·m$^2$) | 8.55e-10 | -0.0201 | 0.0131 | 0.0053 | 0.0048 | 0.0015 | 0.0005 |

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